Galashin's Conjecture 4.3 via a convex gradient

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Abstract

For a loopless bounded affine permutation f, let Θ_f^{adm} be the admissible angle chamber and let $\Theta_f^{>0} \subseteq \Theta_f^{\mathrm{adm}}$ be Galashin's gauge slice, cut out by fixing one angle in each crossing component. Let Meas_f be the critical measurement map defined in [1, Def. 1.15, eq. (1.6)], and let $\mathrm{Crit}_f^{>0}$ be its image [1, Def. 1.8]. Conjecture 4.3 of [1] asserts that $\mathrm{Meas}_f \colon \Theta_f^{>0} \to \mathrm{Crit}_f^{>0}$ is a homeomorphism. We present an argument for the conjecture, covering every loopless f uniformly in (k, n), by exhibiting a convex-analytic structure. After the Muller–Speyer left twist [4] and passage to logarithmic Grassmann-necklace coordinates R_f, the measurement map becomes the *gradient* of an explicit potential:

$$R_f \circ \mathrm{Meas}_f \;=\; G_f \;=\; \nabla \Psi_f, \qquad \Psi_f(\theta) = \sum_e L\big(\delta_e(\theta)\big), \quad L'(u) = \log \sin u,$$

where the sum is over the directed edges $e = (p, q)$ of the crossing graph G_f^x and $\delta_e = \theta_{\widetilde{q}} - \theta_p \in (0, \pi)$ is the corresponding admissibility angle, read in Galashin's affine lift $\theta_{j+n} = \theta_j + \pi$ with $\widetilde{q}$ the forward lift of q past p (see (7)). Its Hessian is $H_f = B \,\mathrm{diag}(\cot \delta_e) B^{\mathsf{T}}$, with B the directed vertex–edge incidence matrix. Although an individual cotangent can be negative, the crossing edges partition into edge-disjoint *winding-one* cycles, and a cotangent cycle inequality makes H_f positive semidefinite with kernel exactly the crossing-component shifts. Convexity on a gauge slice then makes G_f an injective local diffeomorphism, and $\mathrm{Meas}_f^{-1} = G_f^{-1} \circ R_f$ is a constructive continuous inverse, characterized by a strictly convex minimization on the gauge slice. The critical cell is moreover a real-analytic submanifold, and Meas_f a real-analytic diffeomorphism onto it.

The gradient identity rests on an exact combinatorial fact: the cyclic row-difference of the twisted-necklace exponent matrix in [1, Prop. 5.12] is the incidence matrix B. We give a uniform half-open cyclic-arc argument for that identity, with no enumeration of cells, and two independent arguments for the winding-one partition, one by a finite window induction, which is machine checked, and one from a reduced alternating strand diagram. Quantitatively, if every crossing angle is at least η, the cycle decomposition gives the sharp single-cycle modulus $\kappa_m^\star(\eta) = [\cot \eta - (m-1)\cot((m-1)\eta)]/m$, a graph-Laplacian sandwich for H_f, and an inverse-Lipschitz estimate in logarithmic necklace-gradient coordinates.

As an application, the main theorem combined with Proposition 8.9 of [1] gives, in every rank, a constructive homeomorphism from $\mathrm{Crit}_f^{>0}$ to the dual critical variety of the shift $f^{\downarrow}$; for the electrical permutation of any matching τ, this restricts to a homeomorphism between the critical electrical and critical Ising loci of the same generalized region. The boundary behaviour of that shift, together with a partition-independent sharp modulus for the dense cyclic five-cell, is developed in the companion manuscript [2], which is in preparation; nothing in this paper depends on it.

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1 Introduction

Critical measurement maps and the main theorem

Let f be a loopless bounded affine permutation of rank k on $[n]$ in the sense of [1, §2.1, Def. 2.1] — the notion introduced by Knutson–Lam–Speyer [6], equivalent to Postnikov's decorated permutations [5, Def. 13.3, Thm. 17.1]. An f-admissible tuple is an angle tuple $\theta = (\theta_1, \dots, \theta_n)$ satisfying $\theta_p < \theta_q < \theta_p + \pi$ for every f-crossing $p < q$; these form the *ambient admissible chamber* $\Theta_f^{\mathrm{adm}} \subseteq \mathbb{R}^n$. Fixing one angle in each connected component of the crossing graph G_f^x cuts out Galashin's *gauge slice* $\Theta_f^{>0} \subseteq \Theta_f^{\mathrm{adm}}$ [1, §4, eq. (4.3)]; throughout, $\Theta_f^{>0}$ denotes that slice and never the ambient chamber. The curve construction of [1, Def. 1.15, eq. (1.6)] defines the critical measurement map Meas_f on the ambient chamber, and $\mathrm{Crit}_f^{>0} := \mathrm{Meas}_f(\Theta_f^{\mathrm{adm}})$ is its image [1, Def. 1.8]. That image is already attained on the slice:

$$\mathrm{Meas}_f(\Theta_f^{>0}) = \mathrm{Meas}_f(\Theta_f^{\mathrm{adm}}) = \mathrm{Crit}_f^{>0}. \tag{1}$$

This is not a definition. It is Galashin's gauge invariance: adding a common constant to the angles θ_p of the labels p in one connected component C of G_f^x leaves $\mathrm{Meas}_f(\theta)$ unchanged [1, Prop. 4.2 and the paragraph following it], and such component shifts move any admissible θ into the slice, so that $\mathrm{Meas}_f\colon \Theta_f^{>0} \to \mathrm{Crit}_f^{>0}$ is surjective [1, §4.1, the sentence after eq. (4.3)]. Component shifts also preserve admissibility, because the two labels of an f-crossing are by construction joined by an edge of G_f^x and hence lie in the same component, so every δ_e is unchanged; the shifted tuple is therefore again in Θ_f^{adm}. We use (1) once, in the proof of Theorem 1, and it is what supplies surjectivity there. Conjecture 4.3 of [1] states:

> $\mathrm{Meas}_f\colon \Theta_f^{>0} \to \mathrm{Crit}_f^{>0}$ is a homeomorphism.

The conjecture is immediate when the crossing graph G_f^x has no edges — every component is then a single label, the gauge fixes every angle, and $\Theta_f^{>0}$ is a point. Beyond that, Theorem 4.4 of [1] proves it for the top cells $f = f_{k,n}$, $1 \le k \le n$ [1, §9]; the general loopless case is the subject of Conjecture 4.3. Our main theorem addresses it for every loopless f, including coloops.

The restriction to the positive real chamber is essential. Example 5.7 of [1] gives two distinct complex parameter tuples with the same critical measurement for $f_{3,4}$; exactly one of the corresponding angle tuples lies in $\Theta_{f_{3,4}}^{>0}$. Thus there is no contradiction with the top-cell theorem or with Theorem 1: the convexity mechanism below is intrinsically real.

Theorem 1. *For every loopless bounded affine permutation f, the critical measurement map $\mathrm{Meas}_f\colon \Theta_f^{>0} \to \mathrm{Crit}_f^{>0}$ is a homeomorphism. Moreover the inverse is constructive: $\mathrm{Meas}_f^{-1} = G_f^{-1} \circ R_f$, where R_f reads off logarithmic twisted Grassmann-necklace coordinates (equivalently, logarithmic ratios of the ordinary necklace minors, Remark 13) and G_f^{-1} inverts the gradient of the explicit convex potential Ψ_f of (23) below. Equivalently, for every $X \in \mathrm{Crit}_f^{>0}$,*

$$\mathrm{Meas}_f^{-1}(X) = \operatorname*{argmin}_{\theta \in \Theta_f^{>0}} \{\Psi_f(\theta) - \langle R_f(X), \theta \rangle\}. \tag{2}$$

The minimizer exists and is unique for these targets.

The twist is needed to prove the gradient identity, not to invert the map. By Remark 13, R_f is read off the ordinary Grassmann-necklace minors, and Corollary 32 restates Theorem 1 in terms of them alone: the map

$$N_f\colon \mathrm{Crit}_f^{>0} \to \mathbb{P}_{>0}^{n-1}, \qquad X \mapsto [\Delta_{I_1}(X) : \cdots : \Delta_{I_n}(X)],$$

is a topological embedding, with $X = \mathrm{Meas}_f(G_f^{-1}(-P_f D \log N_f(X)))$. So the ratios of the n necklace minors of a critical point determine it, and recovering it needs no evaluation of the twist. Corollary 33 strengthens both statements: $\mathrm{Crit}_f^{>0}$ is a real-analytic submanifold, Meas_f is a real-analytic diffeomorphism onto it, and N_f is a real-analytic embedding.

Corollary 2 (All-rank critical Gr–OG shift)**.** *Let $f \in \mathcal{B}(k,n)$ be loopless, and let $f^{\downarrow}$ be Galashin's shift, $f^{\downarrow}(p) = f(p-1)$. It is looplessness of f that puts $f^{\downarrow}$ in $\mathcal{B}(k-1,n)$: $f^{\downarrow}(p) \geq p$ says exactly that $f(p-1) \neq p-1$, and the rank drops by one because $\sum_p (f(p-1)-p) = \sum_q (f(q)-q) - n$. There is a constructive homeomorphism*

$$\Phi_f : \mathrm{Crit}_f^{>0} \xrightarrow{\sim} \widehat{\mathrm{Crit}}_{f^{\downarrow}}^{>0}, \qquad \Phi_f(X) = \widehat{\mathrm{Meas}}_{f^{\downarrow}}(G_f^{-1}(R_f(X))).$$

If τ is any matching on $[2N]$, then $(f_\tau^{\mathrm{elec}})^{\downarrow} = f_\tau^{\mathrm{Ising}}$, and for every τ-admissible tuple θ,

$$\Phi_{f_\tau^{\mathrm{elec}}}(\phi^{\mathrm{elec}}(\Lambda_{\mathrm{elec}}^{(\tau,\theta)})) = \phi^{\mathrm{Ising}}(M_{\mathrm{Ising}}^{(\tau,\theta)}).$$

Equivalently, $\Phi_{f_\tau^{\mathrm{elec}}}$ restricts to a homeomorphism between the images of the τ-admissible tuples in the electrical and Ising critical loci, uniformly in N and τ; here the dual measurement in the definition of Φ is the regular projective map, including at common zeros of graph coordinates [1, Remark 5.3]. Here $\widehat{\mathrm{Crit}}_{f^{\downarrow}}^{>0}$ and $\widehat{\mathrm{Meas}}_{f^{\downarrow}}$ are Galashin's dual critical cell and dual measurement map [1, §8], and f_τ^{elec}, f_τ^{Ising} his electrical and Ising permutations [1, §7]; §6 recalls them.

The idea: the measurement map is a convex gradient

The proof does not manipulate Meas_f directly. Two coordinate changes reveal a variational structure. First, the Muller–Speyer left twist $\widetilde{\tau}$ ([1, §5.4], [4]) is an automorphism of the open positroid variety, and Galashin's boundary-measurement formula ([1, Prop. 5.12]) makes the twisted Grassmann-necklace Plücker coordinates explicit monomials in the edge factors. Taking logarithms of their absolute values, and using the hinge identity $|[\![t_q, t_p]\!]_+| = 2\sin\delta_e$ of Lemma 3 — proved in §2.2 from Galashin's bracket $[\![x,y]\!] = x/y - y/x$ and the specialization $t_p = e^{i\theta_p}$, and equal to the chord length $|e^{2i\theta_q} - e^{2i\theta_p}|$ of [1, Rem. 1.2] — defines a continuous map R_f from $\mathrm{Crit}_f^{>0}$ to a Euclidean space with

$$R_f \circ \mathrm{Meas}_f \;=\; G_f \;=\; \nabla \Psi_f, \qquad \Psi_f(\theta) = \sum_e L(\delta_e(\theta)), \quad L'(u) = \log\sin u. \tag{3}$$

Thus, in these coordinates, the measurement map is the gradient of the potential Ψ_f. Its Hessian is

$$H_f(\theta) = \mathrm{Hess}\,\Psi_f = B\,\mathrm{diag}(\cot\delta_e(\theta))B^{\mathsf{T}}, \tag{4}$$

where B is the directed vertex–edge incidence matrix of G_f^x (column $e_q - e_p$ for $e = (p,q)$). A gradient map is injective on any convex domain where its potential is strictly convex; and an injective gradient map whose Hessian is nonsingular is, by the inverse function theorem, a homeomorphism onto its open image, with a continuous inverse. The content of the theorem is therefore reduced to a single convexity statement:

$$\Psi_f \text{ is convex on } \Theta_f^{\mathrm{adm}}, \text{ with } H_f \succeq 0 \text{ and } \ker H_f \text{ exactly the crossing-component shifts.}$$

This is not obvious — $\cot\delta_e < 0$ when $\delta_e > \pi/2$, so Ψ_f is *not* convex edge by edge. The resolution (§4) is combinatorial: the crossing edge set partitions into edge-disjoint *winding-one* cycles, and a cotangent inequality shows each cycle contributes a positive semidefinite form. Grouped this way, the negative cotangents are dominated exactly.

Contributions and proof architecture

The new results are organized around the following chain, which Figure 1 summarizes.

1. The twisted logarithmic necklace coordinates turn the critical measurement map into the explicit gradient (3). The combinatorial identity behind it is $P_f D M_N = B$: cyclic row differences of the necklace exponents in [1, Prop. 5.12] give the crossing-graph incidence matrix (Proposition 10).

2. The crossing edges of every loopless f admit an edge-disjoint partition into winding-one directed cycles (Proposition 15). Together with the cotangent cycle inequality, this proves positive semidefiniteness of (4), with kernel exactly the component shifts (Proposition 26).

3. Strict convexity on the gauge slice makes G_f injective and locally invertible. Invariance of domain and the continuous coordinate map R_f then give the constructive inverse in Theorem 1, including its variational form (2); inverting G_f at any point of its image is that same minimization (Corollary 31).

4. The inverse needs only the ordinary necklace minors: $X \mapsto [\Delta_{I_1}(X) : \cdots : \Delta_{I_n}(X)]$ embeds $\mathrm{Crit}_f^{>0}$ in $\mathbb{P}^{n-1}_{>0}$, and the twist enters only the proof of the gradient identity (Remark 13 and Corollary 32). The critical cell is a real-analytic submanifold, and the measurement map and this embedding are real-analytic (Corollary 33).

5. The same Hessian decomposition yields the sharp single-cycle modulus and a fixed-partition graph-Laplacian sandwich (Theorem 28 and Corollary 29).

6. Combining Theorem 1 with [1, Prop. 8.9] produces the all-rank critical Gr–OG shift and the critical electrical–Ising correspondence of Corollary 2; Remark 35 says what it does not give.

The definitions of the critical curve, admissibility chamber, measurement map, and necklace exponent formula are taken from [1]; the left twist is the automorphism of open positroid varieties constructed by Muller–Speyer [4]. Sections 3–5 prove the convex-gradient homeomorphism, and Section 6 derives the shift corollary.

Scope, and what is deferred to the companion manuscript

Everything in this paper is a symbolic proof: no step depends on a finite computation or a numerical certificate. Three further developments of the same machinery are correspondingly separated into the companion manuscript [2], which is in preparation. It treats the boundary compactification of the shift and its stratified descent; the exact finite cross-stratum theorems at $n = 5$ and on the $n = 6$ one-skeleton, which are computer-assisted; and the partition-independent sharp full-Hessian modulus for the dense cyclic five-cell, which is an exact computer-assisted certificate. None of the three is used here. Decorated and looped cells lie outside the loopless scope of Theorem 1, as does Problem 5.2 of [1]; see Remark 8 for what a looped extension would involve, which we do not claim.

The figures illustrate and are used in no proof. Most draw the worked example of Appendix B, $f = (3, 5, 6, 7, 9)$, at the admissible angles $\theta/\pi = (0, 0.1, 0.7, 0.3, 0.8)$. Their values were computed in Lean: combinatorial values by evaluating the formal definitions of Appendix A or by direct count, real values in floating point. Each caption says which.

Relation to hyperbolic volume and to earlier variational principles

The immediate statistical-mechanical antecedent is Kenyon's critical dimer model [8]. Galashin's introduction constructs critical cells from this model on isoradial plabic tilings [1, Introduction and §1.2]. Kenyon's critical dimer weight is $2\sin\alpha$, where α is a rhombus half-angle [8, §2.3]; the same sine factor appears in Lemma 3, whereas Galashin's interior-edge normalization is $\sin(\theta_q-\theta_p)$. The train-track criterion for rhombic embeddings is due to Kenyon–Schlenker [9, Thm. 3.1]. Kenyon's earlier variational argument [8, §8, Prop. 8.1 and Thm. 8.2] parametrizes isoradial embeddings by a convex space of rhombus angles and obtains strict concavity by restricting a sum of per-rhombus functions to a linear subspace. This is a close antecedent of the mechanism here in the embedded regime, though his normalized log-determinant includes an energy term and is not identified with $-\Psi_f$.

The distinction relevant here is between positive individual cotangents and positivity after cycle assembly. When every $\delta_e<\pi/2$, H_f is a graph Laplacian with positive weights, so its positivity and kernel follow directly from the incidence matrix. The admissible chamber also permits $\delta_e>\pi/2$, where individual weights are negative; Proposition 15 supplies the cycles that preserve positivity there. The local rhombus comparison requires an orientation convention: for ordered unit sides $u_p=e^{-2i\theta_p}$ and $u_q=e^{-2i\theta_q}$, the counterclockwise signed area is $\det(u_p,u_q)=-\sin(2\delta_e)$. Thus the *clockwise* signed area is $\sin(2\delta_e)=2\sin^2(\delta_e)\cot\delta_e$ and has the sign of the Hessian weight. This local calculation does not assert that every admissible tuple gives a global rhombic embedding, or identify the crossing-graph Laplacian with the electrical network of [1, §1.6].

Folded rhombi also occur in the flat and minimal isoradial immersions of Boutillier–Cimasoni–de Tilière [10, Thms. 15 and 23; §3.4]. Their minimal-immersion theorem is stated for graphs without unbounded faces; the disk boundary and local angle-map conditions require a separate comparison before one could identify our admissible chamber with their space Y_G [10, §4 and Lem. 26]. No such identification is used here. Proposition 15 supplies the winding-one decomposition that extends our convexity argument across the folds.

The potential of (23) is not a new function. Normalise the antiderivative by $L(u)=\int_0^u\log\sin t\,dt$ (so $L(0)=0$; (23) fixes only L'), and let $\Lambda(x)=-\int_0^x\log|2\sin t|\,dt$ be Milnor's Lobachevsky function. Then $L(u)=-\Lambda(u)-u\log 2$ identically on $(0,\pi)$, and by (10) together with Proposition 15 the quantity $\sum_e\delta_e(\theta)$ is a constant — π times the number of cycles in any winding-one partition — on the whole chamber. Hence

$$\Psi_f(\theta)=-\sum_e\Lambda(\delta_e(\theta))+\text{const},\tag{5}$$

with the same gradient and the same Hessian as the right-hand sum: Ψ_f *is minus a sum of Lobachevsky functions.* Under Galashin's identification $v_p=\exp(2i\theta_p)$ [1, Rem. 1.2] a winding-one cycle of G_f^x is an ideal polygon inscribed in a circle in $\partial\mathbb{H}^3$, whose edges are the chords of length $2\sin\delta_e$ of Remark 4, and $\sum_{e\in Z}\Lambda(\delta_e)$ is the hyperbolic volume of the ideal polyhedron P_Z obtained by coning that polygon to the ideal point ∞. For a 3-cycle, P_Z is an ideal tetrahedron and this is Milnor's formula for its volume in terms of its dihedral angles [11, Appendix, Lem. 2], in the form $\mathrm{vol}(T)=\Lambda(\alpha)+\Lambda(\beta)+\Lambda(\gamma)$ recorded in [14, Lem. 5.1].[1] For $m\ge 4$ it follows from that case by fan triangulation. Cutting the polygon from v_{p_1} decomposes P_Z into the ideal tetrahedra $\mathrm{conv}(v_{p_1},v_{p_j},v_{p_{j+1}},\infty)$, $2\le j\le m-1$; the Euclidean base triangle of the j-th is inscribed in the same circle, so by the inscribed-angle theorem its angles are δ_{e_j}, $\pi-A_j$ and A_{j-1}, where $A_j=\delta_{e_1}+\cdots+\delta_{e_j}$ (these sum to π, as they must, because $A_m=\pi$ by (10)). Since Λ is odd and π-periodic [11, Appendix, Lem. 1], $\Lambda(\pi-A_j)=-\Lambda(A_j)$, so summing Milnor's formula over j

[1] No result of this paper depends on any reference of this subsection: (5), the constancy of $\sum_e\delta_e$ and the fan triangulation below are internal, Lemma 25 is proved in §4, and the $m=3$ formula is used only to read (5).

telescopes to $\Lambda(A_1) - \Lambda(A_{m-1}) + \sum_{j=2}^{m-1} \Lambda(\delta_{e_j}) = \sum_{e\in Z} \Lambda(\delta_e)$, using $A_1 = \delta_{e_1}$ and $A_{m-1} = \pi - \delta_{e_m}$. So Proposition 26 says that a total hyperbolic volume is concave in these angles — the analytic mechanism of the Casson–Rivin programme, in which strict concavity of the volume functional on the space of angle structures of an ideal triangulation is the key input [12, 13], [14, Thm. 1.2 and Lem. 5.3]. The same functional underlies Bobenko and Springborn's variational principles for circle patterns, whose functionals are built from $\operatorname{Im}\operatorname{Li}_2$, i.e. from Clausen's Cl_2 with $\Lambda(x) = \frac{1}{2}\mathrm{Cl}_2(2x)$ [15], and Lemma 25 for $m = 3$ is the classical cotangent formula for the Dirichlet energy of a piecewise-linear function on a Euclidean triangle [16], [17, Lem. 5]; for $m \geq 4$ the lemma reduces to $m = 3$ by telescoping the partial sums $A_j = \alpha_1 + \cdots + \alpha_j$, $D_j = d_1 + \cdots + d_j$, so it may be read as a classical fact rather than a new one. Figure 2 draws Λ at the seven gaps of the worked example of Appendix B.

Kenyon also relates the Lobachevsky sum to dimer entropy: in the periodic setting of [8, §10.4, Thm. 10.3], volume per site equals 2π times entropy per site. This is a related variational interpretation, not an entropy formula for our finite boundary problem. In particular, no normalization of a finite dimer partition function is identified with Ψ_f here.

What is new here is not the functional but the setting, and the decomposition that produces it. The domain is not the angle-structure polytope of an ideal triangulation: there is no 3-manifold and no edge condition. The variables are the n circle positions θ_p; the "angles" are their differences along the crossing graph; and the relation $\sum_{e\in Z} \delta_e = \pi$ is an *identity* in θ, equation (10), not a constraint cutting out the domain — so the nonemptiness of the angle-structure space, which is the hard part in the Casson–Rivin setting, does not arise, while the image of $\theta \mapsto (\delta_e)_e$ is in general a proper affine subspace of the product of the cycle simplices. Convexity restricts to a subspace, so the positivity in Proposition 26 is inherited from the classical concavity *once the cycles are available*, and the new ingredient is Proposition 15: the crossing graph of any loopless bounded affine permutation decomposes into inscribed polygons each winding exactly once around the circle. We are not aware of a counterpart to that statement in the variational literature.

Two things this reading does not give. It does not identify Meas_f with a map studied there: Rivin's uniqueness of a critical point is a different assertion and is used nowhere below. And it does not supply boundary control. Λ is finite at 0 and at π, so Ψ_f extends continuously to the closed chamber; each $\ell_e = \log\sin\delta_e$ is singular there, but the singularities can cancel in $G_f = B\ell$. For the directed triangle of Figure 8 and $\theta(t) = (0, t, 2t)$, $\delta = (t, t, \pi - 2t)$ and $G_f(\theta(t)) = (\log(2\cos t), 0, -\log(2\cos t)) \to (\log 2, 0, -\log 2)$ as $t \downarrow 0$, although $\theta(t)$ tends to a vertex of the chamber. So the gradient need not diverge at the boundary, let alone with the uniform coercivity a compactification argument would need. Nor is the gradient image convex: for the same triangle $(\log 2, 0, -\log 2)$ is the midpoint of the images of the triangles with side ratios $(1/4, 1, 1)$ and $(1, 1/4, 1)$, but it would need the degenerate ratio $(1 : 1 : 2)$.

2 Setup

We follow Galashin's conventions [1, §1,§4,§5]. Let f be loopless of rank k on $[n]$, with induced permutation $\bar{f}$ and Grassmann necklace $(I_r)_{r\in[n]}$, $I_{r+1} = (I_r \setminus \{r\}) \cup \{f(r)\}$. The *crossing graph* G_f^x has vertex set all of $[n]$ [1, eq. (4.1)], with an edge $\{p, q\}$ whenever p, q form an f-crossing; we orient it $p \to q$ when the ε-resolved boundary endpoints occur clockwise as $b_{s_p}^+, b_{s_q}^+, b_p^-, b_q^-$, where $s_x = \bar{f}^{-1}(x)$ ([1, §5.4]). A label fixed by $\bar{f}$ is an isolated vertex, not an omitted one. Its connected components partition $[n]$ as $C^{(1)} \sqcup \cdots \sqcup C^{(c_f)}$ [1, §4.4]; the component-shift vectors $\{\mathbf{1}_C : C \text{ a component}\}$ span the gauge directions, so $\operatorname{rank} B = n - c_f$.

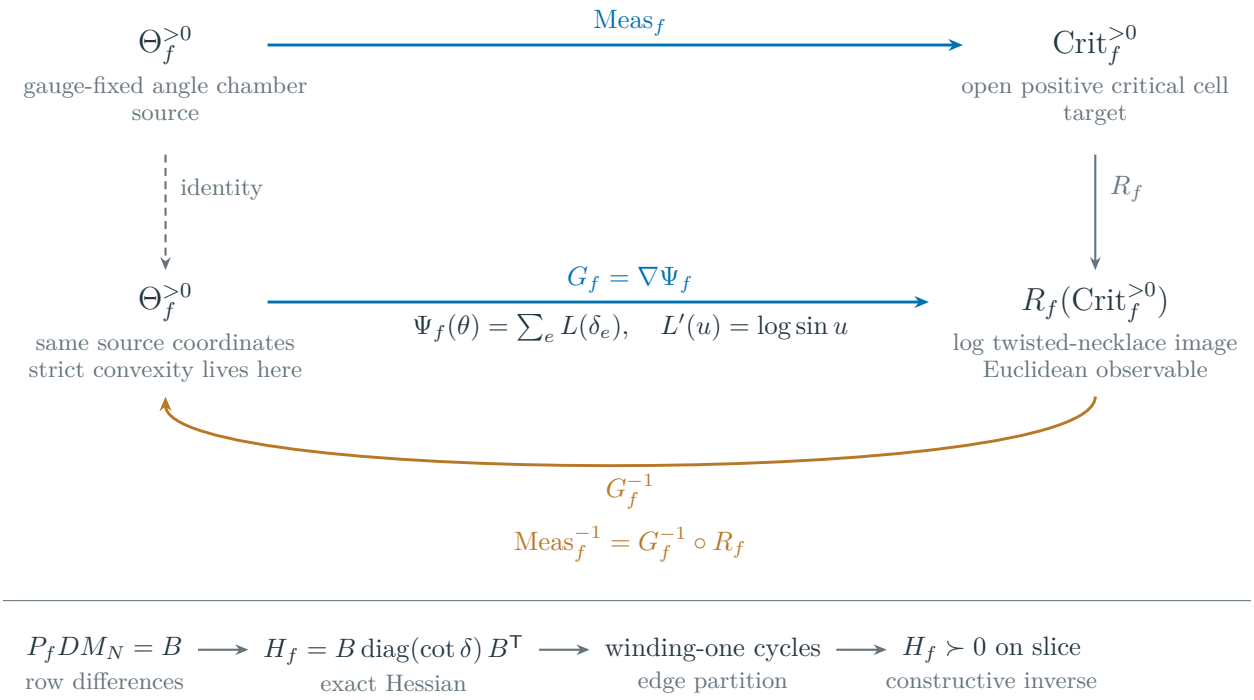


Figure 1: The proof architecture after the Muller–Speyer twist and logarithmic necklace coordinate change. The map becomes the gradient of an explicit potential; the row-difference and winding-cycle identities make its Hessian positive definite on the gauge slice, exposing the constructive inverse.

2.1 The admissibility angles δ_e

Throughout we use Galashin's affine convention: a tuple $\theta \in \mathbb{R}^n$ is extended to a sequence indexed by $\mathbb{Z}$ by

$$\theta_{j+n} = \theta_j + \pi \qquad (j \in \mathbb{Z}) \tag{6}$$

[1, eq. (3.1)]. For a directed crossing edge $e = (p, q)$ of G_f^x let $\widetilde{q}$ be the *forward lift* of q past p, that is $\widetilde{q} = q$ if $p < q$ and $\widetilde{q} = q + n$ if $p > q$, and set

$$\delta_e(\theta) := \theta_{\widetilde{q}} - \theta_p \;=\; \theta_q - \theta_p + \pi \cdot \mathbf{1}[p > q]. \tag{7}$$

An edge with $p > q$ is called *wrapped*. We use the lift to keep the potential real-valued: the orientation of G_f^x is given by the endpoint rule above, so on a wrapped edge the raw difference $\theta_q - \theta_p$ lies in $(-\pi, 0)$ and $\log\sin(\theta_q - \theta_p)$ is undefined over $\mathbb{R}$.

With (7), admissibility is exactly positivity of the angles:

$$\theta \in \Theta_f^{\mathrm{adm}} \iff 0 < \delta_e(\theta) < \pi \;\text{ for every directed crossing edge } e. \tag{8}$$

Indeed, [1, Def. 1.6] imposes $\theta_p < \theta_q < \theta_p + \pi$ for every f-crossing $p < q$. If $e = (p, q)$ with $p < q$ this reads $\delta_e = \theta_q - \theta_p \in (0, \pi)$ directly; if $e = (p, q)$ with $p > q$, the same inequality applied to the pair $q < p$ gives $\theta_p - \theta_q \in (0, \pi)$, hence $\delta_e = \pi - (\theta_p - \theta_q) \in (0, \pi)$. In particular $\sin\delta_e > 0$ and $\ell_e := \log\sin\delta_e$ is defined on all of Θ_f^{adm}. Figure 3 shows Θ_f^{adm} on two planes through a point of the worked example of Appendix B.

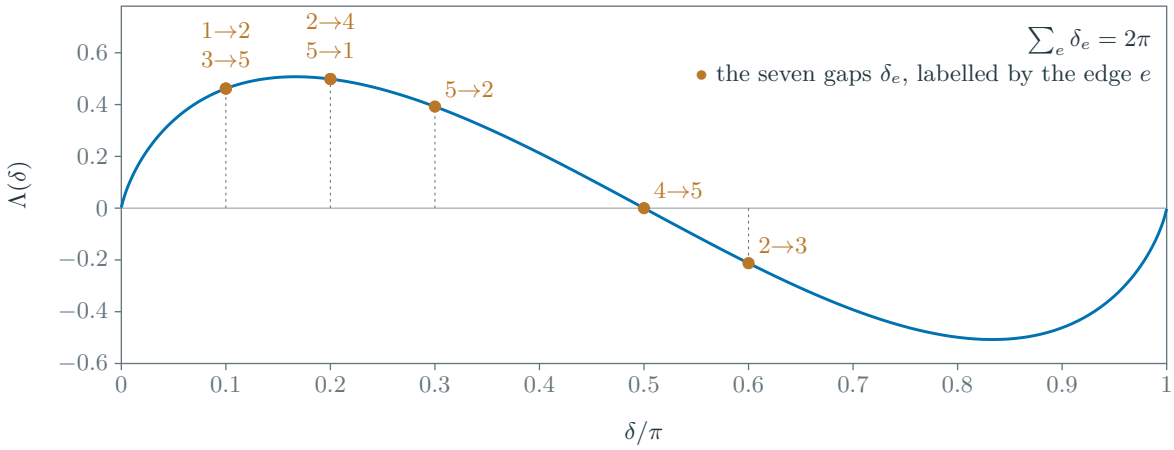


Figure 2: The Lobachevsky function $\Lambda(\delta) = -\int_0^\delta \log|2\sin t|\,dt$ on $(0,\pi)$, with the seven gaps δ_e of $\bar{f} = (3,5,1,2,4)$ and $\theta/\pi = (0, 0.1, 0.7, 0.3, 0.8)$ marked and labelled by their edges e. They sum to 2π, so by (5) $\Psi_f = -\sum_e \Lambda(\delta_e) + \text{const}$. Numerical values, in floating point.

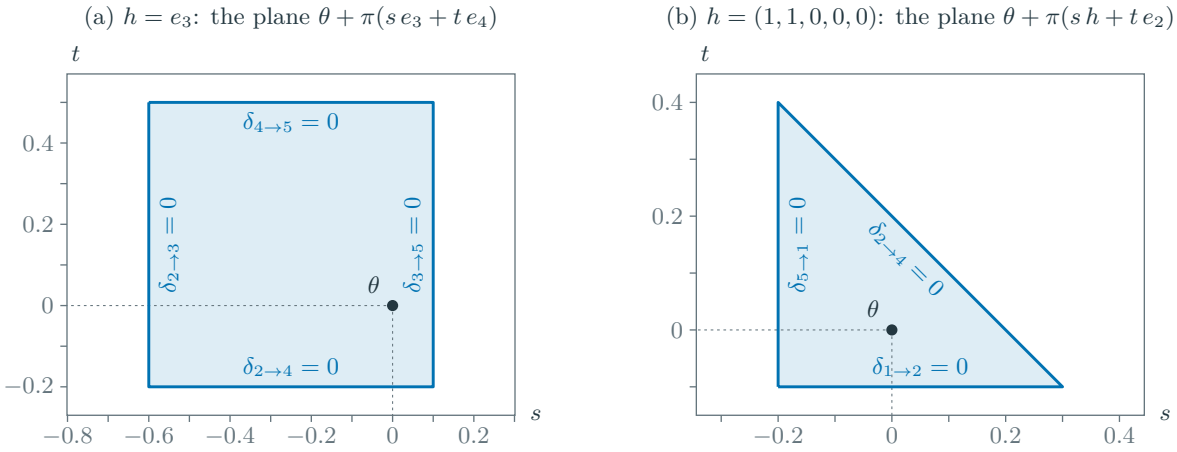


Figure 3: The admissible set Θ_f^{adm} on two planes through θ, for $\bar{f} = (3,5,1,2,4)$ and $\theta/\pi = (0, 0.1, 0.7, 0.3, 0.8)$, in the coordinates (s,t) of the point $\theta + \pi(s\,h + t\,v)$. (a) $h = e_3$, $v = e_4$: a rectangle; the plane lies in the affine space of the gauge slice $(h_1 = v_1 = 0)$. (b) $h = (1,1,0,0,0)$, $v = e_2$: a triangle. Each side lies on one hyperplane $\delta_e = 0$ of (8). Numerical values, in floating point.

Two consequences of (7) are used repeatedly. First, the oriented increment is unaffected by the lift,

$$\partial_\theta \delta_e = e_q - e_p, \tag{9}$$

so δ_e is an *affine* (not linear) function of θ, constant along component shifts. Second, passing from θ_q to its lift $\theta_q + \pi$ leaves $|\sin \delta_e|$ and $\cot \delta_e$ unchanged but *reverses the sign* of $\sin \delta_e$; the quantity the potential needs is $\sin \delta_e > 0$, and that is what (8) delivers. With the edge orientations, and hence B, fixed, writing $\ell_e = \log|\sin(\theta_q - \theta_p)|$ gives the same gradient values; the π-periodicity of cotangent likewise leaves the Hessian unchanged. The lift supplies the real domain of $L' = \log \sin$ and the positive cycle angles whose sum is π, as used in the convexity argument below. This invariance under adding π does not authorize changing edge orientations. Figure 4 shows two shifts of θ on the worked example: one constant on the crossing component, which changes no δ_e, and one that is not.

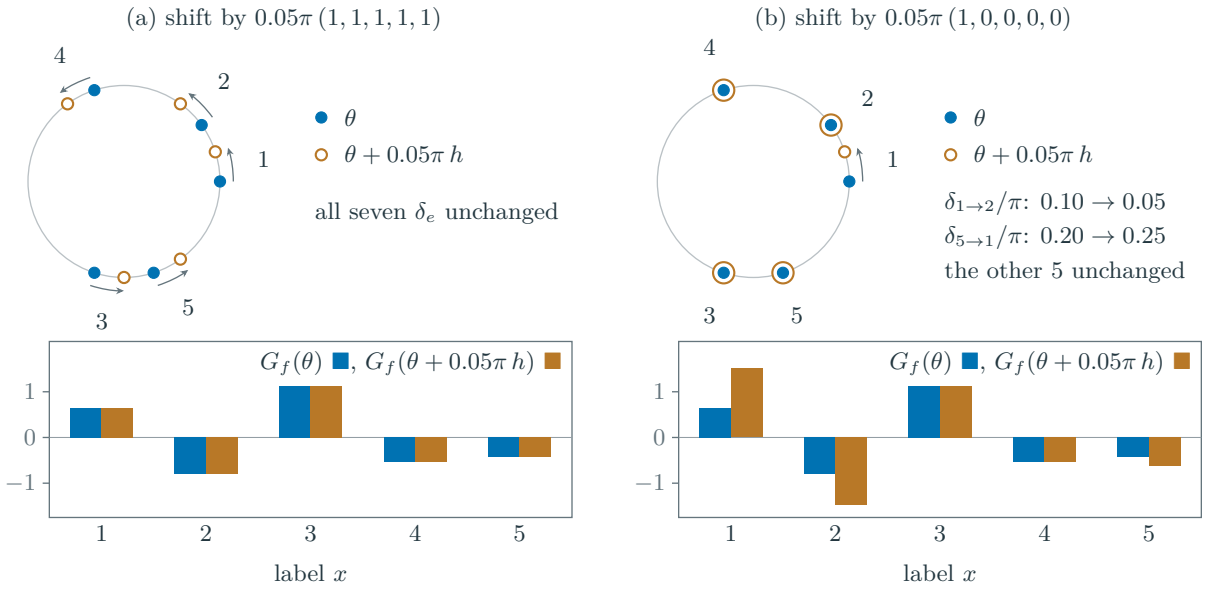


Figure 4: The shift $\theta \mapsto \theta + 0.05\pi\,h$ for $\bar{f} = (3,5,1,2,4)$ and $\theta/\pi = (0, 0.1, 0.7, 0.3, 0.8)$. Points are $e^{2i\theta_p}$ before (filled) and after (open); bars give $G_f = B\ell$ at each label. (a) $h = (1,1,1,1,1)$, constant on the crossing component: every δ_e and G_f are unchanged. (b) $h = e_1$, which moves θ_1 alone: $\delta_{1\to2}$ and $\delta_{5\to1}$ change, and G_f with them. Numerical values, in floating point.

Finally we record the cycle identity that Lemma 25 will consume. A directed cycle $Z = (p_1 \to p_2 \to \cdots \to p_m \to p_1)$ of crossing edges has winding one — total exit-advance n — if and only if it has exactly one descent in the cyclic label order, i.e. exactly one wrapped edge (see the closing paragraph of §4.2 — a forward reference; nothing in §4.2 depends on this section, so there is no circularity). Lifting each successive exit forward, the sum of (7) along Z telescopes to $\theta_{p_1+n} - \theta_{p_1}$, so by (6)

$$\sum_{e \in Z} \delta_e(\theta) = \pi \qquad \text{for every winding-one cycle } Z \text{ and every } \theta. \tag{10}$$

This is an identity in θ on all of $\mathbb{R}^n$, not a constraint cutting out Θ_f^{adm}.

2.2 Galashin's complex parameters, the bracket, and the hinge identity

The exponent formula we import is stated in the complex parameters of [1, §1.4], so we fix that notation here rather than inside the proof that uses it. Let $\mathcal{G}_{\mathrm{red}}(f)$ be the set of reduced planar bipartite graphs in the disk with strand permutation $\bar{f}$, and for a weight function wt on $E(G)$ let $\mathrm{Meas}_G(\mathrm{wt}) := (\Delta_I(G,\mathrm{wt}))_{I\in\binom{[n]}{k}}$ be the vector of dimer partition functions, viewed in $\mathrm{Gr}(k,n)$ when it is nonzero [1, §1.1,§5.1]. Let $\widetilde{\tau}$ denote the Muller–Speyer *left twist* automorphism of the open positroid variety Π_f° ([1, §5.4, eq. (5.6)], [4]).

For $x, y \in \mathbb{C}^*$ put

$$[\![x,y]\!] := \frac{x}{y} - \frac{y}{x}, \tag{11}$$

and for $t = (t_1,\dots,t_n) \in (\mathbb{C}^*)^n$ let wt_t be the weight function which gives an interior edge labelled $\{p,q\}$, $p<q$, the weight $[\![t_q,t_p]\!]$ and every boundary edge the weight 1 [1, Def. 1.12, eq. (1.4)]. Following [1, §5.4] we abbreviate

$$[\![t_p,t_q]\!]_+ = [\![t_q,t_p]\!]_+ := [\![t_q,t_p]\!] \qquad (1\le p<q\le n), \tag{12}$$

which is the symmetric form in which the brackets appear in (22). The tuple t is f-admissible when $t_p \neq \pm t_q$ for every f-crossing $\{p,q\}$, and

$$\Theta_f^\circ := \{t\in(\mathbb{C}^*)^n : \ t \text{ is } f\text{-admissible and } t_{p_1} = \cdots = t_{p_{c_f}} = 1\} \tag{13}$$

[1, eq. (5.2)] is the complex analogue of the gauge slice $\Theta_f^{>0}$, with $p_1,\dots,p_{c_f}$ the same component representatives that cut out $\Theta_f^{>0}$ in [1, eq. (4.3)].

Angles enter through the specialization of [1, eq. (5.1)],

$$t_p := \exp(i\theta_p) \qquad (p\in[n]), \tag{14}$$

under which $\mathrm{wt}_\theta = \frac{1}{2i}\,\mathrm{wt}_t$ on interior edges. The two weight functions are gauge-equivalent, so [1, Lem. 5.13] gives, for every $G\in\mathcal{G}_{\mathrm{red}}(f)$ and every f-admissible θ,

$$\mathrm{Meas}_G(\mathrm{wt}_t) = \mathrm{Meas}_G(\mathrm{wt}_\theta) = \mathrm{Meas}_f(\theta) \quad \text{in } \mathrm{Gr}(k,n), \tag{15}$$

the last equality being the square-move invariance by which $\mathrm{Meas}_f(\theta)$ is defined [1, §1.2]. Thus (22) below, although written in t, is a statement about $\mathrm{Meas}_f(\theta)$.

Lemma 3 (Hinge identity)**.** *Under* (14)*, for every directed crossing edge* $e=(p,q)$ *and every* $\theta\in\Theta_f^{\mathrm{adm}}$*,*

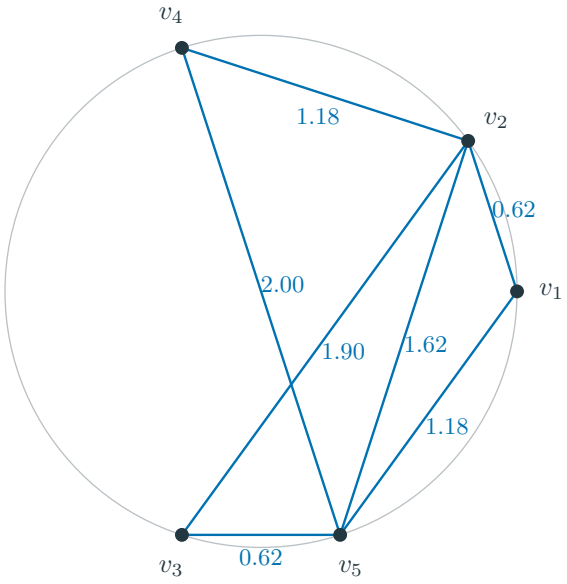
$$\big|[\![t_q,t_p]\!]_+\big| = 2\sin\delta_e(\theta). \tag{16}$$

Proof. By (11) and (14), for any $a,b\in[n]$, $[\![t_a,t_b]\!] = e^{i(\theta_a-\theta_b)} - e^{-i(\theta_a-\theta_b)} = 2i\sin(\theta_a-\theta_b)$, so $|[\![t_a,t_b]\!]| = 2|\sin(\theta_a-\theta_b)|$. Both orientations of the edge are now covered. If $p<q$, (12) gives $[\![t_q,t_p]\!]_+ = [\![t_q,t_p]\!]$ and $\delta_e = \theta_q-\theta_p \in (0,\pi)$, so both sides equal $2\sin(\theta_q-\theta_p)$. If $p>q$ — the wrapped case — then $[\![t_q,t_p]\!]_+ = [\![t_p,t_q]\!]$ because $q<p$, whose modulus is $2|\sin(\theta_p-\theta_q)| = 2\sin(\theta_p-\theta_q)$ by admissibility, while $\delta_e = \pi-(\theta_p-\theta_q)$ gives $\sin\delta_e = \sin(\theta_p-\theta_q)$. The two sides agree again. □

Remark 4 (the hinge is a chord length)**.** Setting $v_r := \exp(2i\theta_r)$, Galashin observes that $\sin(\theta_q-\theta_p) = \frac{1}{2}|v_q-v_p|$ [1, Rem. 1.2]. So (16) says that the edge factor of a crossing is the distance between two points v_p, v_q on a circle, and the argument of the sine is pinned by that reading: $|[\![t_q,t_p]\!]_+| = |v_q-v_p|$, not $c\sin(\delta_e/2)$ or $c\sin(2\delta_e)$. For $k=3$, $n=4$, [1, Ex. 1.9] reads the resulting Plücker coordinates as the side lengths of a convex inscribed quadrilateral. Admissibility does not

put all n points in cyclic order: for $f = (3, 5, 6, 7, 9)$ and $\theta/\pi = (0, 0.1, 0.7, 0.3, 0.8)$, the worked example of Appendix B, every crossing angle lies in $(0, \pi)$, yet labels 3 and 4 are out of order. The vertices of each winding-one cycle are in cyclic order, since their angles $2\delta_e$ are positive and sum to 2π; that is all the volume reading of §1 uses.

Figure 5 draws the points v_r and the chords $|v_q - v_p|$ for the worked example of Appendix B.



e	δ_e/π	$2\sin\delta_e$
$1 \to 2$	0.1	0.62
$2 \to 3$	0.6	1.90
$2 \to 4$	0.2	1.18
$3 \to 5$	0.1	0.62
$4 \to 5$	0.5	2.00
$5 \to 1$	0.2	1.18
$5 \to 2$	0.3	1.62

$|v_q - v_p| = 2\sin\delta_e$ for $e = (p, q)$

Figure 5: The points $v_r = e^{2i\theta_r}$ on the unit circle for $f = (3, 5, 6, 7, 9)$ and $\theta/\pi = (0, 0.1, 0.7, 0.3, 0.8)$. Each directed crossing edge $e = (p, q)$ is the chord from v_p to v_q, labelled by its length $|v_q - v_p| = 2\sin\delta_e$ as in (16); the table lists δ_e/π and $2\sin\delta_e$. Numerical values, in floating point.

2.3 The necklace exponent matrix

The *necklace exponent matrix* M_N has rows indexed by the necklace positions $r \in [n]$ and columns by unordered pairs $\{p, q\}$; by Galashin's Proposition 5.12 (eq. (5.8), displayed as (22) below) the twisted Plücker coordinate $\Delta_{I_r}(\widetilde{\tau}(\cdot))$ is a monomial whose $\{p, q\}$-exponent is determined by the pair's misalignment/crossing class together with the membership of p, q in I_r. We index B by the same set of unordered pairs: since the crossing relation is antisymmetric (again the closing paragraph of §4.2, a forward reference of the same harmless kind), a pair $\{p, q\}$ carries at most one directed crossing edge, and we let its B-column be $e_q - e_p$ when $p \to q$ is that edge and the zero column when $\{p, q\}$ carries no crossing. With this convention M_N and B have the same column set, the vector $\ell = (\ell_e)$ is understood as extended by an arbitrary value on non-crossing pairs, and every such value is annihilated by the corresponding zero column of B. All identities below are proved column by column, so the extension is immaterial.

2.4 Coloops, and the component reduction

A label fixed by $\bar{f}$ — a coloop, $f(p) = p + n$ — is an isolated vertex of G_f^x and carries no crossing (proved in the "Coloops" step of Proposition 10), so it contributes nothing to Ψ_f, to B, or to

H_f. The fixed-point-free case is therefore the essential one. Two independent devices make this precise, and we record both because they discharge different worries: Proposition 10 is proved below *with coloops allowed*, so no statement in this paper carries a fixed-point-free hypothesis; and the component reduction of this subsection independently factors the whole problem over the components of G_f^x.

Lemma 5 (Component reduction)**.** *Let f be loopless and let $[n] = C^{(1)} \sqcup \cdots \sqcup C^{(c_f)}$ be the connected components of G_f^x. Then:*

1. *for each r, erasing from the strand diagram D_f every strand S_p with $p \notin C^{(r)}$ and relabeling the surviving strands and boundary vertices by $[n_r]$, $n_r = |C^{(r)}|$, yields a strand diagram in the sense of [1, Def. 4.5]; its strand permutation is a loopless bounded affine permutation $f^{(r)} := f|_{C^{(r)}} \in \mathcal{B}(k_r, n_r)$ [1, §4.4];*

2. *the restriction map $\rho : \theta \mapsto (\theta|_{C^{(1)}}, \dots, \theta|_{C^{(c_f)}})$ is a homeomorphism $\Theta_f^{>0} \to \Theta_{f^{(1)}}^{>0} \times \cdots \times \Theta_{f^{(c_f)}}^{>0}$, and there is a homeomorphism $\Sigma : \mathrm{Crit}_f^{>0} \xrightarrow{\sim} \mathrm{Crit}_{f^{(1)}}^{>0} \times \cdots \times \mathrm{Crit}_{f^{(c_f)}}^{>0}$ [1, Prop. 4.10] which* intertwines the measurement maps*: the square*

$$\begin{array}{ccc} \Theta_f^{>0} & \xrightarrow{\ \rho\ } & \prod_r \Theta_{f^{(r)}}^{>0} \\ \downarrow \mathrm{Meas}_f & & \downarrow \prod_r \mathrm{Meas}_{f^{(r)}} \\ \mathrm{Crit}_f^{>0} & \xrightarrow{\ \Sigma\ } & \prod_r \mathrm{Crit}_{f^{(r)}}^{>0} \end{array} \tag{17}$$

   commutes, i.e. $\Sigma(\mathrm{Meas}_f(\theta)) = \big(\mathrm{Meas}_{f^{(r)}}(\theta|_{C^{(r)}})\big)_r$;

3. *if $C^{(r)} = \{p\}$ is a singleton then S_p crosses nothing, $\Theta_{f^{(r)}}^{>0}$ is a point, and the factor is a point;*

4. *every non-singleton $C^{(r)}$ has $\overline{f^{(r)}}$ fixed-point-free.*

Proof. Part (1) is [1, §4.4]: after the erasure the directions of the surviving strands still alternate around the boundary of the disk, so the relabeled diagram is a strand diagram in the sense of [1, Def. 4.5].

For (2), the commutation in (17) is not an extra hypothesis: it is how Proposition 4.10 of [1] is proved. There the disk is split at a boundary label whose two ε-resolved endpoints lie in different components, producing two cyclic intervals P, Q, each a union of components of G_f^x, with restrictions $g = f|_P$ and $h = f|_Q$. Galashin first records that θ is f-admissible if and only if $\theta|_P$ is g-admissible and $\theta|_Q$ is h-admissible, and then establishes the homeomorphism by the explicit minor identities (4.4)–(4.6) between $X = \mathrm{Meas}_f(\theta)$ and the pair $(\mathrm{Meas}_g(\theta|_P), \mathrm{Meas}_h(\theta|_Q))$, showing that each determines the other. In other words the map exhibited there is $\mathrm{Meas}_f(\theta) \mapsto (\mathrm{Meas}_g(\theta|_P), \mathrm{Meas}_h(\theta|_Q))$, which is exactly the commuting square for a two-factor splitting; iterating the splitting over the c_f components gives (17). The top arrow ρ is a homeomorphism because admissibility is a conjunction of conditions each involving a single component (both labels of an f-crossing lie in one component), and because the gauge slice fixes exactly one angle in each component, so ρ is the restriction of a linear isomorphism between the ambient affine slices.

For (3), an isolated vertex of G_f^x is a strand crossing no other, so the erased diagram has a single strand and no admissibility constraint; the gauge fixes its angle. For (4), a fixed point of $\overline{f^{(r)}}$ is a strand of $D_f|_{C^{(r)}}$ running between adjacent ε-resolved endpoints, which therefore crosses no other surviving strand and is isolated in the crossing graph of $f^{(r)}$; since that graph is connected and $n_r \geq 2$, there is no such strand. □

Remark 6 (why this is not a restriction of $\bar{f}$)**.** Part (1) is an erasure-and-relabeling, not the restriction of $\bar{f}$ to an invariant subset, and the distinction is not cosmetic: a component of G_f^x need not be $\bar{f}$-stable. Galashin flags the mechanism — for $p \in [n]$ the endpoints b_p^- and b_p^+ may lie in different components of D_f, so the boundary vertex b_p appears to belong to two components at once [1, §4.4]. The smallest instance is $n = 2$, $\bar{f} = (2\,1)$: both chords join adjacent ε-resolved endpoints, so G_f^x has no edges and $\{1\}$ is a component, yet $\bar{f}(1) = 2$. Relabeling after erasure is what makes $f^{(r)}$ well defined regardless.

Because (17) commutes and its horizontal arrows are homeomorphisms, Meas_f is a homeomorphism $\Theta_f^{>0} \to \mathrm{Crit}_f^{>0}$ if and only if every $\mathrm{Meas}_{f^{(r)}}$ is one on its factor; and a finite product of homeomorphisms is a homeomorphism. So Theorem 1 for all loopless f follows from Theorem 1 for those f with $\bar{f}$ fixed-point-free.

Remark 7 (no standing fixed-point-free hypothesis)**.** The proof of Proposition 10 covers coloops directly (the "Coloops" step below), and Proposition 15, Proposition 26 and Theorem 1 are stated and proved for every loopless f. No statement in §3–§6 assumes $\bar{f}$ fixed-point-free. Lemma 5 is therefore not needed for the general loopless case; we keep it because it identifies the product structure of $\mathrm{Crit}_f^{>0}$ and gives a second, independent route to it.

Remark 8 (loops: a proof candidate for the extension, not claimed here)**.** Galashin states Conjecture 4.3 for loopless f only, and Theorem 1 is stated on exactly that domain. Reading [1, Def. 1.8] verbatim for an arbitrary $f \in \mathcal{B}(k,n)$ — a reading that is ours, not Galashin's, since his Section 4 fixes f loopless — a loop label p (that is, $f(p) = p$) is a lollipop in every reduced graph [1, §2.2]: its boundary vertex is white and adjacent to a black interior leaf, so the pendant edge lies in every almost perfect matching, carries weight 1, and forces p out of the index set of every nonvanishing minor. Hence the p-th column of $\mathrm{Meas}_f(\theta)$ vanishes, p is an isolated vertex of G_f^x, $\Theta_f^{\mathrm{adm}} = \Theta_{f'}^{\mathrm{adm}} \times \mathbb{R}$, and deleting the lollipop leaves a graph with strand permutation $f' = f \setminus \{p\}$, a bounded affine permutation of period $n-1$ and rank k. That graph is again reduced: deleting a pendant edge with a free end keeps the $k(n-k)+1-\ell(f)$ faces of a reduced graph [1, §2.2], and $\ell(f) = \ell(f')+k$ (stated in the proof of [18, Thm. 7.12]) makes this the count $k(n-1-k)+1-\ell(f')$ that characterizes the reduced graphs for f'; independently, the four conditions of [5, Thm. 13.2] pass to the deleted graph, the lollipop's edge lying on no other trip. The combinatorial facts are Lean theorems (`TP.deleteLoop_bdd`, `TP.affRank_deleteLoop`, `TP.affLength_deleteLoop`). So $\mathrm{Meas}_f(\theta)$ is the image of $\mathrm{Meas}_{f'}(\theta')$, θ' the other angles, under the zero-column embedding $\mathrm{Gr}(k,n-1) \hookrightarrow \mathrm{Gr}(k,n)$, and the looped statement, that Meas_f is a homeomorphism from the gauge slice onto $\mathrm{Crit}_f^{>0}$, follows from Theorem 1 for f with its loops deleted. This is a proof candidate under the reading above; *none of it is claimed in this paper*, whose scope remains loopless.

Figure 6 shows the two constructions of this subsection: the components of the crossing graph (Lemma 5) and the deletion of a loop (Remark 8).

3 The measurement map is a gradient

3.1 The row-difference identity

The engine of (3) is an exact integer-matrix identity between Galashin's necklace exponents and the crossing incidence. Let D be the cyclic forward row-difference $(Dz)_r = z_{r+1} - z_r$ and let P_f be the necklace relabeling $(P_f z)_x = z_{s_x}$.

The proof of that identity rests on one bookkeeping fact about the ε-resolved boundary, which we isolate first. Split each boundary position $j \in [n]$ into an exit point j^- followed clockwise by an

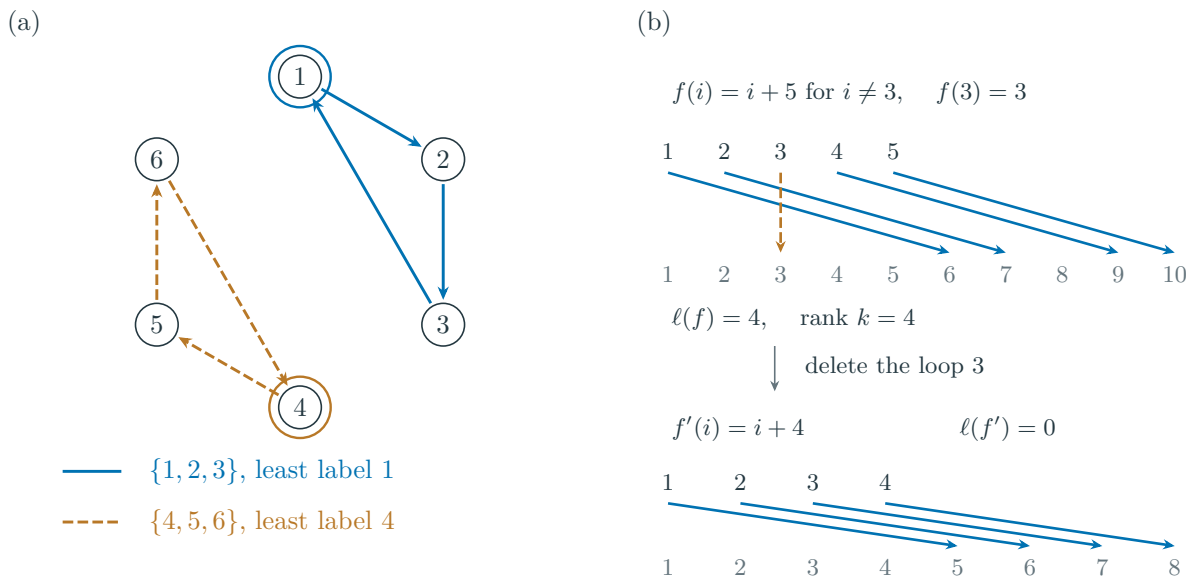


Figure 6: (a) The crossing graph of the loopless f with $\bar{f} = (3, 1, 2, 6, 4, 5)$ has two connected components, $\{1, 2, 3\}$ (blue, solid) and $\{4, 5, 6\}$ (ochre, dashed); the ringed vertex is the least label of its component. (b) Deleting a loop: $n = 5$, $f(i) = i + 5$ except $f(3) = 3$; here $\ell(f) = 4$ and $k = 4$, and after the deletion $\ell(f') = 0$. Panel (a) is evaluated by the formal definitions of Appendix A; the lengths and the rank in panel (b) are counted directly.

entry point j^+, Galashin's b_j^- and b_j^+ [1, Def. 1.5], so that the $2n$ refined points occur clockwise as $1^-, 1^+, 2^-, 2^+, \ldots, n^-, n^+$. For an entry point u^+ and an exit point v^-, let $J(u^+, v^-)$ be the set of refined points met when moving clockwise from u^+, excluding u^+ and including v^-. For $u, v \in [n]$ write $(u, v]$ for the cyclic half-open integer arc $\{u + 1, u + 2, \ldots, v\}$, indices modulo n, with $(u, u] := [n]$.

Lemma 9 (Half-open reading of the resolved boundary)**.**

(a) *For all $u, v \in [n]$, $\{r \in [n] : r^- \in J(u^+, v^-)\} = (u, v]$.*

(b) *If j^- and j^+ both occur in a list of distinct refined points written in clockwise cyclic order, then j^+ immediately follows j^- in that list.*

(c) *Let a^+, c^+ be entry points and b^-, d^- exit points, the four refined points being distinct. Then*

$$a^+, b^-, c^+, d^- \text{ clockwise} \implies (a, b] \cap (c, d] = \varnothing; \tag{18}$$

$$a^+, b^-, c^+, d^- \text{ counterclockwise} \implies (a, b] \cup (c, d] = [n]; \tag{19}$$

$$a^+, c^+, b^-, d^- \text{ clockwise} \implies (a, b] \setminus (c, d] = (a, c]. \tag{20}$$

Proof. (a) Moving clockwise from u^+ one meets $(u + 1)^-, (u + 1)^+, (u + 2)^-, \ldots$, so the exit points in $J(u^+, v^-)$ are the r^- with $r = u + 1, u + 2, \ldots, v$, read modulo n. If $u = v$, the walk reaches u^-, the last refined point before u^+, only after every other exit point.

(b) No refined point lies strictly between j^- and j^+ on the clockwise circle, so in any cyclic list containing both, the successor of j^- is j^+.

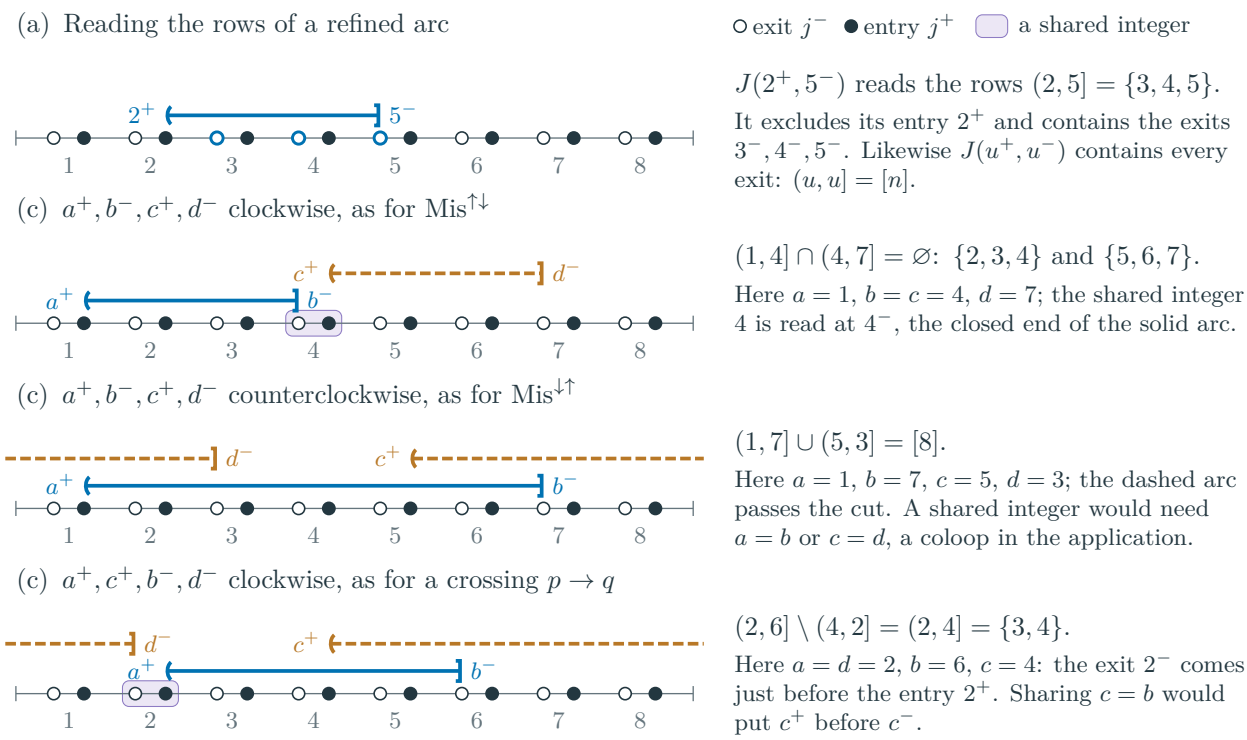


Figure 7: Lemma 9 on the resolved boundary, cut open for $n = 8$; open points are exits j^- and filled points entries j^+. A row r is read at its exit point r^-, so the refined arc $J(u^+, v^-)$, drawn from its open end u^+ to its closed end v^-, reads the half-open arc $(u, v]$. In part (c), with $(a, b, c, d) = (s_p, p, s_q, q)$ as in the proof of Proposition 10, the solid arc $J(a^+, b^-)$ reads A_p and the dashed arc $J(c^+, d^-)$ reads A_q. In the first and third patterns an integer is shared, and its exit comes immediately before its entry. Schematic; the positions are illustrative.

(c) Write the four points in clockwise order, compare refined arcs, and read rows with (a). In (18) the arcs $J(a^+, b^-)$ and $J(c^+, d^-)$ are the clockwise stretches from a^+ to b^- and from c^+ to d^- of the order a^+, b^-, c^+, d^-; they are disjoint, so no exit point lies in both. In (19) the clockwise order is a^+, d^-, c^+, b^-, and the complements of the two arcs are the stretches from b^- to a^+ and from d^- to c^+ (each without its first point and with its last), which are disjoint; so every exit point lies in $J(a^+, b^-)$ or in $J(c^+, d^-)$. In (20) the clockwise order is a^+, c^+, b^-, d^-, and $J(a^+, b^-) \setminus J(c^+, d^-)$ is the stretch from a^+ (excluded) to c^+ (included). Its exit points are those of $J(a^+, c^-)$, because c^- is the refined point just before c^+ and is not a^+; by (a) they are the rows of $(a, c]$. □

Part (b) is the exit-before-entry rule at a shared label, and it decides which integer coincidences the three patterns of (c) admit. Two of the four points can share an integer only if one is an exit and the other an entry, and then only where, in the pattern, that exit is immediately followed by that entry. In (18) this allows $b = c$ and $d = a$; in (19), whose clockwise order is a^+, d^-, c^+, b^-, only $c = d$ and $a = b$; in (20) only $d = a$, while $c = b$ is excluded because c^+ would be followed by c^-. Part (c) needs no other case distinction: it assumes only that the four refined points are distinct. Figure 7 draws the reading (a) and the three patterns, each with a shared label where one is allowed.

Proposition 10. *For every loopless f,*

$$P_f \, D \, M_N \;=\; B,$$

where M_N is the necklace-row restriction of Galashin's Proposition 5.12 exponent matrix and B is the directed incidence matrix of G_f^x, both indexed by unordered pairs as in §2.3.

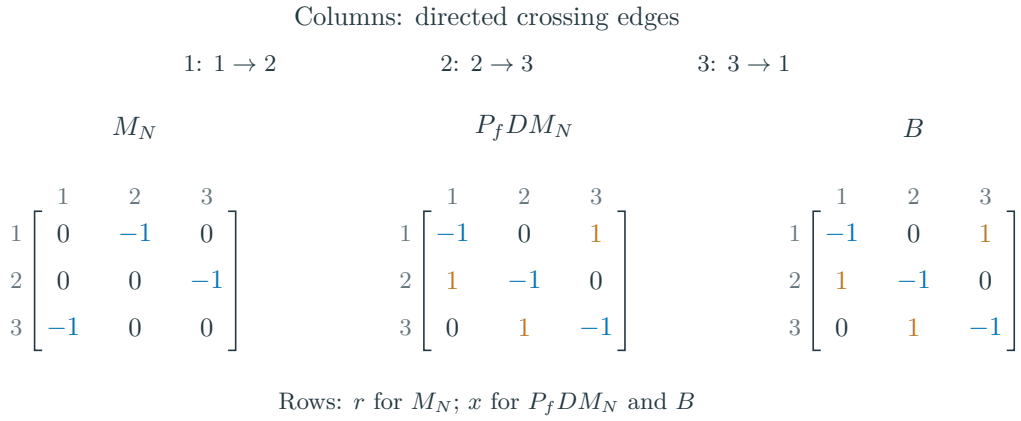


Figure 8: The row-difference identity for $\bar{f} = (3, 1, 2)$. The columns are the three crossings $1 \to 2$, $2 \to 3$, $3 \to 1$, each with $s_p = q$. Rows of M_N are necklace positions; rows of $P_f DM_N$ and B are exit labels. Thus the negative arc indicators become incidence columns, with -1 at the tail and $+1$ at the head.

Proof. The identity is the boundary of a half-open arc, read column by column. We show below that the columns of M_N are as follows: a pair carrying no directed crossing, in either direction, has a zero column, which covers every pair meeting a coloop; and for a crossing $p \to q$ the column is

$$m_r = -\mathbf{1}_{r \in (s_p, s_q]},$$

minus the indicator of the cyclic half-open arc of rows from s_p to s_q. Granting this, a pair carrying no crossing has a zero column in $P_f DM_N$ and in B, which is the incidence matrix of the crossings. For a crossing $p \to q$ the indicator of $(s_p, s_q]$ changes only from row s_p to $s_p + 1$, where it enters the arc, and from row s_q to $s_q + 1$, where it leaves (these differ, $\bar{f}$ being injective), so

$$(Dm)_r = m_{r+1} - m_r = \mathbf{1}_{r=s_q} - \mathbf{1}_{r=s_p}, \qquad Dm = e_{s_q} - e_{s_p},$$

and relabeling by P_f gives $P_f Dm = e_q - e_p$, the incidence column $B_{p \to q}$. Hence $P_f DM_N = B$. It remains to compute the columns.

The columns of M_N. By Galashin's affine formula $I_r = \{f(a) \bmod n : a < r \le f(a)\}$, a label x lies in I_r iff, for the lift s of s_x with the paired lift x' of x obeying $0 < x' - s \le n$, one has $s < r \le x'$. Reducing modulo n,

$$x \in I_r \iff r \in A_x := (s_x, x] \text{ (cyclically)}. \tag{21}$$

The inequality $x' - s \le n$ is strict exactly when x is not a coloop; for a coloop ($\bar{f}(x) = x$, so $s_x = x$ and $f(x) = x + n$) one has $x' - s = n$, and A_x is then the whole row set $[n]$ — x lies in *every* necklace row. Both readings are used below. In the language of Lemma 9(a), A_x is read off the refined arc $J(s_x^+, x^-)$ swept by the strand S_x, and the three half-open arc facts (18)–(20) are that lemma's part (c). They hold for the ε-resolved order with no generic-position assumption, because the four refined endpoints of two distinct labels are always distinct.

We now recall the exponent formula itself, since the argument turns on exactly which pairs contribute and with which sign. For $p \neq q$ write $s = s_p$ and $t = s_q$; following [1, §5.4], one says $\{p, q\} \in \mathrm{Mis}_f^{\uparrow\downarrow}$ when $b_s^+, b_p^-, b_t^+, b_q^-$ occur clockwise, and $\{p, q\} \in \mathrm{Mis}_f^{\downarrow\uparrow}$ when they occur counterclockwise; the two-element sets $\{s, t\}$ and $\{p, q\}$ are not required to be disjoint, so these definitions explicitly admit coincidences. Figure 9 illustrates these classes together with a crossing and an alignment.

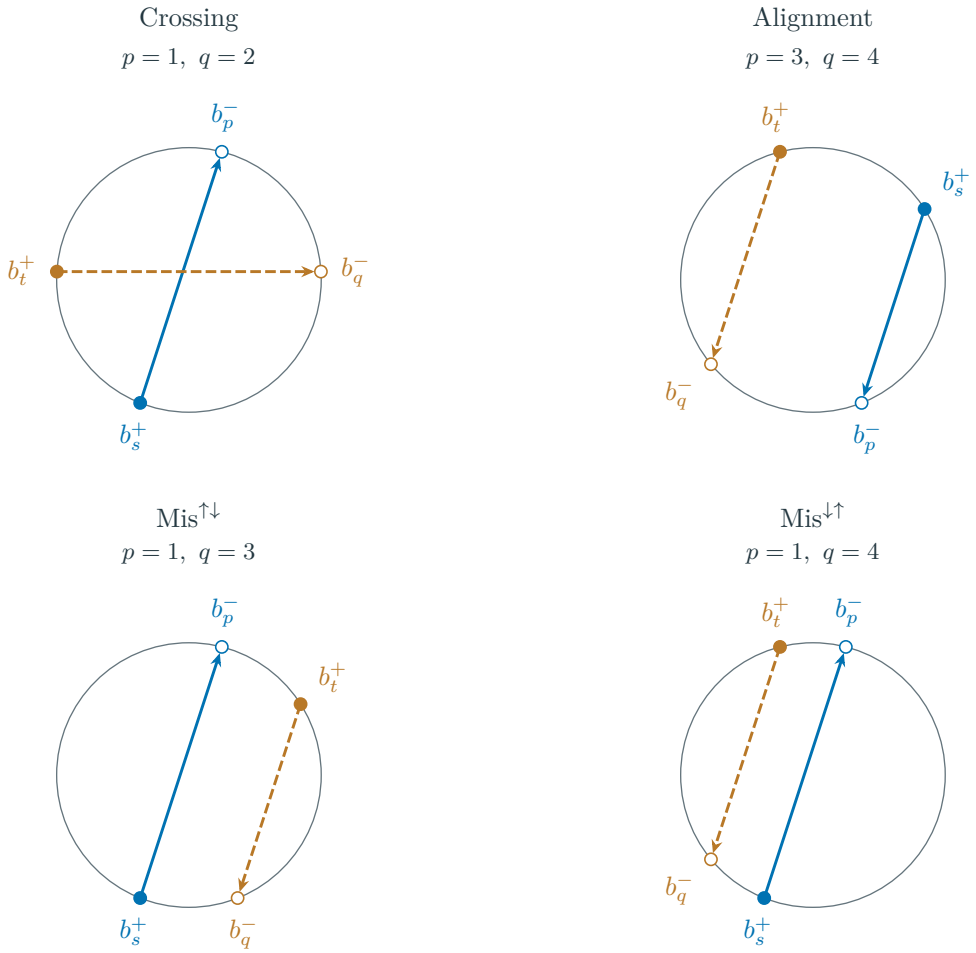


Figure 9: The four pair classes for $\bar{f} = (3, 5, 1, 2, 4)$, with $s = s_p$ and $t = s_q$. Each circle is read clockwise. The blue solid strand runs from b_s^+ to b_p^-; the ochre dashed strand runs from b_t^+ to b_q^-. Filled points are entries and open points are exits. The crossing has order $b_s^+, b_t^+, b_p^-, b_q^-$; the alignment has order $b_s^+, b_p^-, b_q^-, b_t^+$. The two misalignments read $b_s^+, b_p^-, b_t^+, b_q^-$ clockwise and counterclockwise, respectively.

With $[\![t_p, t_q]\!]_+ = [\![t_q, t_p]\!]_+ := [\![t_q, t_p]\!]$ for $p < q$, Proposition 5.12 of [1] states that, for $G \in \mathcal{G}_{\mathrm{red}}(f)$

and $t \in \Theta_f^\circ$ at which Meas_G is defined, after multiplication by a common scalar

$$\Delta_I(\widetilde{\tau}(\mathrm{Meas}_G(\mathrm{wt}_t))) = \prod_{\substack{\{p,q\}\in \mathrm{Mis}_f^{\uparrow\downarrow} \\ p,q\in I}} [\![t_q, t_p]\!]_+ \cdot \prod_{\substack{\{p,q\}\in \mathrm{Mis}_f^{\downarrow\uparrow} \\ p,q\notin I}} [\![t_q, t_p]\!]_+ \cdot \left(\prod_{\substack{(p,q)\in E(\vec{G}_f^{\,\times}) \\ p\in I,\ q\notin I}} [\![t_q, t_p]\!]_+ \right)^{-1} \tag{22}$$

for every face label I of G [1, eq. (5.8)]. We use it only at the necklace sets $I = I_r$, which label the boundary faces of every $G \in \mathcal{G}_{\mathrm{red}}(f)$ (see the proof of [1, Prop. 5.1] in [1, §5.5]), so the restriction to face labels costs nothing here. Three features are all we use. The product is supported on the two Mis classes and the directed-crossing class: a pair lying in none of them is absent from (22) outright, so its exponent is zero. Membership in a class is not by itself enough — a $\mathrm{Mis}^{\uparrow\downarrow}$ pair contributes only on rows I containing *both* of p, q, a $\mathrm{Mis}^{\downarrow\uparrow}$ pair only on rows containing *neither*, and a directed crossing $p \to q$ only on rows containing p but not q. And the crossing exponent is -1, the other two $+1$.

Coloops. Suppose first that one of the two labels, say p, is a coloop, so $s_p = p$ and $b^+_{s_p} = b^+_p$. The ε-convention places the exit b^-_p immediately before the entry b^+_p, so *no* refined endpoint lies strictly inside the clockwise arc from b^-_p to $b^+_{s_p}$. Three of the four patterns are therefore impossible for the pair $\{p, q\}$, whatever q is: $\mathrm{Mis}^{\uparrow\downarrow}$, whose clockwise order $b^+_{s_p}, b^-_p, b^+_{s_q}, b^-_q$ returns to $b^+_{s_p}$ only after $b^+_{s_q}$ and b^-_q, would put both of those inside that arc; the crossing $p \to q$, of clockwise order $b^+_{s_p}, b^+_{s_q}, b^-_p, b^-_q$, would put b^-_q inside it; and the crossing $q \to p$, of clockwise order $b^+_{s_q}, b^+_{s_p}, b^-_q, b^-_p$, would put $b^+_{s_q}$ inside it. Only $\mathrm{Mis}^{\downarrow\uparrow}$ survives, and it contributes nothing: by (21) a coloop lies in every necklace row, so no necklace row omits both p and q, and the second product of (22) is empty on every row. Hence the $\{p, q\}$ column of M_N is identically zero; and the matching column of B is zero too, since p lies in no crossing, i.e. a coloop is an isolated vertex of G^x_f. Both sides of the asserted identity vanish on every pair meeting a coloop, so the identity holds there.

The generic pair. Assume from here that neither p nor q is a coloop, so $s_p \neq p$ and $s_q \neq q$. Taking $I = I_r$ and reading off the $\{p, q\}$ column of M_N, the three cases are governed by the endpoint orders:

- $\mathrm{Mis}^{\uparrow\downarrow}$: order s^+_p, p^-, s^+_q, q^- clockwise; by (18) and (21), $A_p \cap A_q = \varnothing$, i.e. *no* necklace row contains both p and q. Since the first product of (22) contributes only on rows containing both, the (p, q) exponent vanishes on every necklace row.
- $\mathrm{Mis}^{\downarrow\uparrow}$: same points counterclockwise; by (19), $A_p \cup A_q$ is the whole row set, so no necklace row omits both. The second product contributes only on rows containing neither, so the exponent again vanishes.
- Directed crossing $p \to q$: order s^+_p, s^+_q, p^-, q^- clockwise; by (20), $\{r : p \in I_r,\ q \notin I_r\} = A_p \setminus A_q = (s_p, s_q]$. Proposition 5.12 assigns exponent -1 here, so the column is $m_r = -\mathbf{1}_{r\in(s_p,s_q]}$.

The coincidence pool is closed: $p \neq q$, $s_p \neq s_q$ ($\bar{f}$ injective), and $s_p \neq p$, $s_q \neq q$ because neither label is a coloop, so the only possible coincidences are $s_p = q$ and $s_q = p$, always an entry and an exit sharing an integer. The four refined endpoints remain distinct, so Lemma 9(c) applies in every case, and by the paragraph after that lemma a $\mathrm{Mis}^{\uparrow\downarrow}$ pair admits both coincidences, a $\mathrm{Mis}^{\downarrow\uparrow}$ pair of non-coloops admits neither, and a crossing $p \to q$ admits only $s_p = q$. We write out both cases for the crossing $p \to q$, whose pattern s^+_p, s^+_q, p^-, q^- is (20) with $(a, b, c, d) = (s_p, p, s_q, q)$.

- $s_p = q$. The pattern is q^+, s^+_q, p^-, q^-; read from q^- it is q^-, q^+, s^+_q, p^-, so the exit q^- is immediately followed by the entry q^+, as Lemma 9(b) requires. Then $A_p \setminus A_q = (q, s_q]$, and the

shared integer q itself is a row of $A_q = (s_q, q]$, at its closed end, and not of $A_p = (q, p]$, at its open end, in agreement with the exponent $-\mathbf{1}_{r\in(s_p,s_q]}$. This case is common: for $\bar{f} = (3, 1, 2)$, the top cell of $\mathrm{Gr}(2, 3)$, each of the three crossings $1 \to 2 \to 3 \to 1$ has $s_p = q$.

- $s_q = p$. The pattern would be s_p^+, p^+, p^-, q^-, in which the entry p^+ is immediately followed by the exit p^-. Lemma 9(b) excludes this: on the refined circle every point other than $p^\pm$ lies on the clockwise way from p^+ to p^-, in particular s_p^+ and q^-. So $s_q = p$ never occurs for a crossing $p \to q$. It does occur for the reversed crossing $q \to p$, where it is the previous case with p and q exchanged, and for $\mathrm{Mis}^{\uparrow\downarrow}$ pairs; in both, Lemma 9(c) applies unchanged.

Hence the column $m_r = -\mathbf{1}_{r\in(s_p,s_q]}$ of the third bullet is correct for every crossing, with or without a coincidence. All non-crossing pair columns are zero, by the coloop step for pairs meeting a coloop and by the first two bullets otherwise. □

Remark 11 (Finite corroboration, not a premise)**.** Independent reimplementations of the necklace rows and pair classes found no row-difference mismatch on any fixed-point-free cell with $3 \le n \le 8$, nor on any loopless cell with a coloop and $n \le 7$; Appendix A (*Finite checks*) records them. They are corroboration, not premises.

3.2 The gradient identity

Write $\ell_e(\theta) = \log \sin \delta_e(\theta)$ and $G_f(\theta) = B\,\ell(\theta)$.

Proposition 12. *The map*

$$R_f(X) \;:=\; P_f D\Big(\log|\Delta_{I_r}(\widetilde{\tau}(X))|\Big)_{r=1}^{n}, \qquad R_f\colon \mathrm{Crit}_f^{>0} \to \mathrm{im}(B),$$

is continuous, with $R_f \circ \mathrm{Meas}_f = G_f$*, and*

$$G_f = \nabla \Psi_f, \qquad \Psi_f(\theta) = \sum_e L(\delta_e(\theta)), \quad L'(u) = \log \sin u, \tag{23}$$

with Hessian $H_f = B\,\mathrm{diag}(\cot \delta_e) B^{\mathsf{T}}$.

Figure 10 draws L and Ψ_f along a line, for the worked example of Appendix B.

Proof. From a point $X = \mathrm{Meas}_f(\theta) \in \mathrm{Crit}_f^{>0}$, form the left twist $\widetilde{\tau}(X)$ (regular on the open positroid cell, [4]), read its Grassmann-necklace Plücker coordinates $\Delta_{I_r}(\widetilde{\tau}(X))$ (nonzero on the open cell), and take the vector of logarithms of their absolute values; the projective scale is removed by passing to differences, i.e. to $\mathrm{im}(B)$. This is $R_f(X)$ as displayed in the statement. Take θ first in the gauge slice, and let t be the complex parameter of (14), so that $\widetilde{\tau}(X) = \widetilde{\tau}(\mathrm{Meas}_G(\mathrm{wt}_t))$ by (15) and (22) applies verbatim (the last paragraph of this proof says why, collisions included). Taking $\log|\cdot|$ of (22) turns the monomial into a linear form,

$$\log|\Delta_{I_r}(\widetilde{\tau}(X))| = \big(M_N\ \log|[\![t_q, t_p]\!]_+|\big)_r$$

up to the projective constant, and by Lemma 3 $\log|[\![t_q, t_p]\!]_+| = \ell_e + \log 2$ on every crossing pair (the non-crossing columns of M_N being zero by Proposition 10, the value assigned there is irrelevant, cf. §2.3). The cyclic row-difference and P_f-relabeling then give, via Proposition 10, $R_f \circ \mathrm{Meas}_f = P_f D M_N\,(\ell + \log 2 \cdot \mathbf{1}) = B\ell = G_f$; the per-edge constant contributes $(\log 2)\,P_f D M_N \mathbf{1} = (\log 2)\,B\mathbf{1} = 0$, because in-degree equals out-degree at every vertex of G_f^x by Proposition 16 — the third and last forward reference from §§2–3 into §4.2, which is proved there directly from the refined

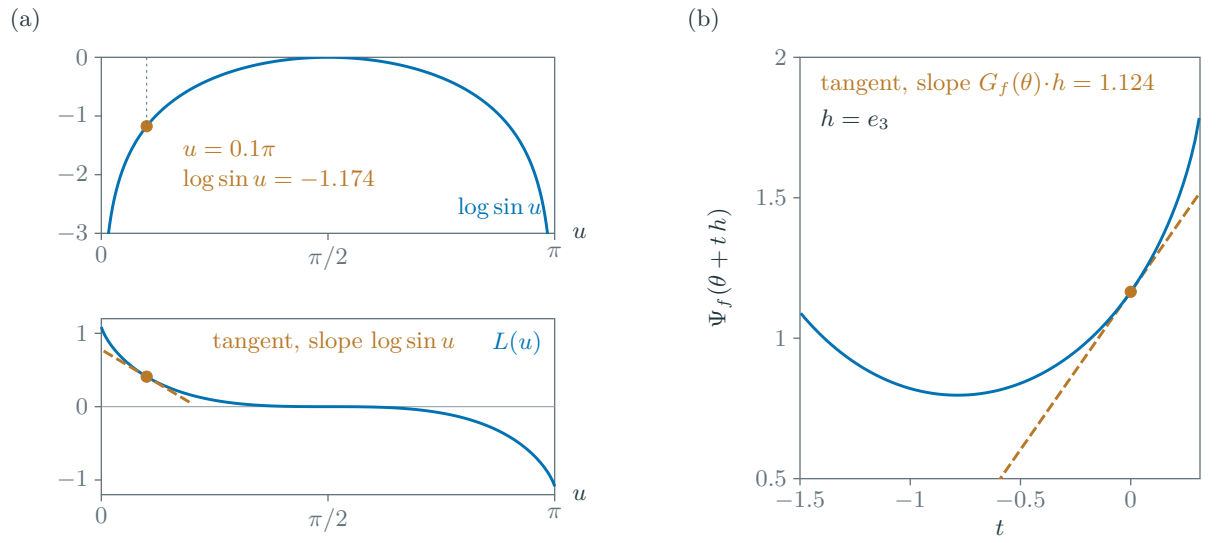
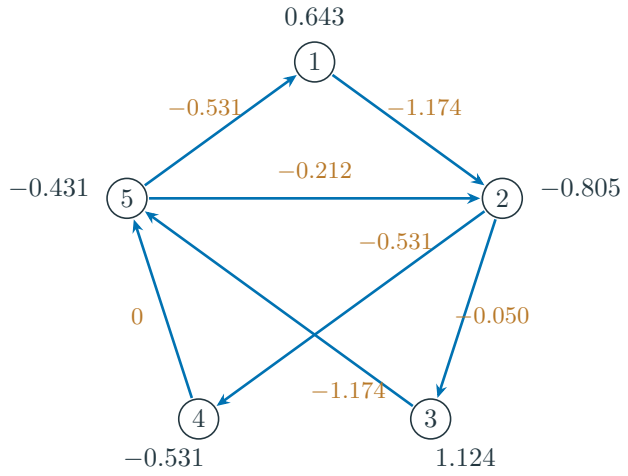


Figure 10: (a) $\log\sin u$ on $(0,\pi)$ and its primitive $L(u) = \int_{\pi/2}^{u} \log\sin t\, dt$, with the tangent of L at $u = 0.1\pi$, of slope $\log\sin u$. (b) $\Psi_f(\theta + t\,h)$, with L normalised as in (a), for $h = e_3$, $\bar{f} = (3,5,1,2,4)$ and $\theta/\pi = (0, 0.1, 0.7, 0.3, 0.8)$, over the t for which every gap stays in $(0,\pi)$. The tangent at $t = 0$ has slope $G_f(\theta)\cdot h$; the curve lies above it. Numerical values, in floating point.

endpoint order and is independent of everything in this section. Finally $\partial_\theta \ell_e = \cot\delta_e\, \partial_\theta \delta_e$ and $\partial_\theta \delta_e = e_q - e_p$ (equation (9), unaffected by the affine lift) give $G_f = B\ell = \nabla\Psi_f$ with Ψ_f as in (23), and $\operatorname{Hess}\Psi_f = B\,\operatorname{diag}(\cot\delta_e)B^{\mathsf{T}}$. Figure 11 shows the identity $G_f = B\ell$ on the worked example, label by label.

Where the identity holds. Galashin states Proposition 5.12 for every $t \in \Theta_f^\circ$ at which Meas_G is well defined. For θ in the gauge slice, $t = e^{i\theta}$ lies in Θ_f°: the slice fixes the component representatives at angle 0, so $t_{p_1} = \cdots = t_{p_{c_f}} = 1$, and admissibility of θ gives $t_p \neq \pm t_q$ on every crossing pair. And Meas_G is well defined at t by [1, Lem. 5.13]. So the computation above holds at every point of the gauge slice, including points where two noncrossing labels collide ($t_p = \pm t_q$); no density or continuity argument is needed. At such a point the vanishing bracket does not occur in the necklace expression, because colliding labels cannot form a crossing pair and Proposition 10 shows that every noncrossing-pair exponent is zero on every necklace row. And $\mathrm{Meas}_f(\theta)$ stays in the open positroid cell, where the twist and the nonzero necklace coordinates, hence R_f, are defined and continuous: all interior edge weights are strictly positive at an admissible tuple [1, Prop. 4.2], so $\mathrm{Meas}_f(\theta) \in \Pi_f^{>0}$ [1, Def. 1.8 and eq. (2.6)]. On the ambient chamber the identity follows by a component shift: it moves any $\theta \in \Theta_f^{\mathrm{adm}}$ into the slice without changing $\mathrm{Meas}_f(\theta)$ [1, Prop. 4.2 and the paragraph following it] or any δ_e, so neither side of $R_f \circ \mathrm{Meas}_f = G_f$ changes. (A density argument on the locus of pairwise distinct parameters, which is dense in Θ_f^{adm}, would not serve on the slice itself: when $c_f \geq 2$ the slice fixes all component representatives at the same angle, so that locus does not meet it.) □

Remark 13 (recovery from the ordinary necklace minors)**.** The twist is needed to apply (22), not to evaluate R_f. For A in the open positroid cell, $\Delta_{I_r}(A) = 1/\Delta_{I_r}(\widetilde{\tau}(A))$ for every r [1, proof of Prop. 5.4], which cites [4, proof of Prop. 6.6]; a change of representative rescales the necklace minors



$$G_f(\theta)_x = \sum_{e=(p,x)} \ell_e - \sum_{e=(x,q)} \ell_e = (B\ell)_x$$

$$\begin{aligned} G_1 &= \ell_{5\to1} - \ell_{1\to2} \\ &= -0.531 + 1.174 = 0.643 \end{aligned}$$

$$\begin{aligned} G_2 &= \ell_{1\to2} + \ell_{5\to2} - \ell_{2\to3} - \ell_{2\to4} \\ &= -1.174 - 0.212 + 0.050 + 0.531 = -0.805 \end{aligned}$$

$$\begin{aligned} G_3 &= \ell_{2\to3} - \ell_{3\to5} \\ &= -0.050 + 1.174 = 1.124 \end{aligned}$$

$$\begin{aligned} G_4 &= \ell_{2\to4} - \ell_{4\to5} \\ &= -0.531 + 0 = -0.531 \end{aligned}$$

$$\begin{aligned} G_5 &= \ell_{3\to5} + \ell_{4\to5} - \ell_{5\to1} - \ell_{5\to2} \\ &= -1.174 + 0 + 0.531 + 0.212 = -0.431 \end{aligned}$$

Figure 11: The crossing graph of $\bar{f} = (3,5,1,2,4)$ at $\theta/\pi = (0, 0.1, 0.7, 0.3, 0.8)$: the weight $\ell_e = \log\sin\delta_e$ on each edge (ochre) and, at each label x, the weights into x minus the weights out of x (black), which is $G_f(\theta)_x = (B\ell)_x$. Numerical values, in floating point.

of X by one common factor, which D annihilates. Since $(Dz)_r = z_{r+1} - z_r$ and $(P_f z)_x = z_{s_x}$,

$$R_f(X) = -P_f D\Big(\log|\Delta_{I_r}(X)|\Big)_{r=1}^{n}, \qquad \big(R_f(X)\big)_x = \log\left|\frac{\Delta_{I_{s_x}}(X)}{\Delta_{I_{s_x+1}}(X)}\right|, \tag{24}$$

with the necklace position $s_x + 1$ read cyclically. So the angles are recovered from ratios of the ordinary Grassmann-necklace minors of X, and the twist need not be computed; Corollary 32 records the consequence.

4 The potential is convex

4.1 Existence of winding-one partitions

Both proofs of the partition below use one counting fact about advances, which we state once. For distinct $p, q \in [n]$ put $\mathrm{adv}(p,q) = (q-p) \bmod n \in \{1, \dots, n-1\}$.

Lemma 14 (Closed walks of advance n are simple)**.** *Let $v_0 \to v_1 \to \cdots \to v_{m-1} \to v_0$ be a closed walk of labels in $[n]$ (or in $\mathbb{Z}_n$, as in §4.2) with $m \geq 2$ and consecutive vertices distinct. Its total advance $\sum_i \mathrm{adv}(v_i, v_{i+1})$ (indices modulo m) is a positive multiple of n. If it equals n, the vertices $v_0, \dots, v_{m-1}$ are pairwise distinct.*

Proof. Each advance is at least 1, and modulo n the advances telescope: $\sum_i \mathrm{adv}(v_i, v_{i+1}) \equiv \sum_i (v_{i+1} - v_i) = 0$. If $v_i = v_j$ with $0 \leq i < j \leq m-1$, the walk splits at that vertex into the closed walks $v_i \to \cdots \to v_j$ and $v_j \to \cdots \to v_{m-1} \to v_0 \to \cdots \to v_i$, with $j - i \geq 1$ and $m - (j-i) \geq 1$ edges. Each has a total advance that is a positive multiple of n, so the whole walk has advance at least $2n$. □

Proposition 15 (W-P). *For every loopless f, the directed crossing-edge set E_f of G_f^x admits a partition $\mathcal{Z}$ into edge-disjoint, vertex-simple directed cycles, each of total exit-advance n (*winding one*) and length ≥ 3. Every such partition has the same number of cycles:*

$$|\mathcal{Z}| = \frac{1}{n} \sum_{(p,q)\in E_f} \mathrm{adv}(p,q) = \#\{(p,q) \in E_f : p > q\}, \qquad \mathrm{adv}(p,q) = (q-p) \bmod n \in \{1,\dots,n-1\}. \tag{25}$$

The partition itself need not be unique.

Proposition 15 is proved in §4.2, by a finite window induction from the refined endpoint order alone; that is the proof Lean checks. A second, geometric proof from a reduced alternating strand diagram follows in §4.3. Nothing later in the paper depends on which of the two is used.

Choice of partition and scope of the combinatorics. For example, the loopless window $f = (3,5,6,7,9)$ has directed crossing edges

$$E_f = \{(1,2),(2,3),(2,4),(3,5),(4,5),(5,1),(5,2)\}.$$

Two distinct winding-one partitions are

$$\{(1,2,3,5),(2,4,5)\} \quad \text{and} \quad \{(1,2,4,5),(2,3,5)\},$$

where each tuple denotes a cyclic vertex list. Each partition has two cycles, matching the two wrapped edges $(5,1)$ and $(5,2)$. The Hessian decomposition below may use any fixed winding-one partition. Figure 12 draws both choices; Appendix B gives the necklace rows and matrices for the same window.

Figure 13 counts the cycles of (25) for the two partitions of Figure 12.

The refined-endpoint crossing rule depends only on the induced permutation $\bar{f}$, not on the decoration of its fixed points. Every permutation has a loopless bounded affine lift, obtained by choosing $p < f(p) \leq p + n$; a fixed point then has $f(p) = p + n$. Thus the purely combinatorial partition statement also holds for every decorated permutation, by applying the proposition to that lift. Fixed-point decorations do not alter the directed edge set or its advances. This observation does not extend the critical measurement theorem to looped cells; its domain remains the one in Theorem 1.

4.2 A combinatorial proof of partition existence

This subsection proves Proposition 15 from the refined endpoint order alone, by a finite induction. It uses no strand diagram and no plabic-graph citation.

Let

$$\mathbb{Z}_n = \{0,\dots,n-1\}$$

with its cyclic order, and let σ be a permutation of $\mathbb{Z}_n$. Put $s = \sigma^{-1}$. At each boundary position i retain two distinct refined points i^-, i^+, in that clockwise order. For $p \in \mathbb{Z}_n$, the chord C_p is directed from $s(p)^+$ to p^-. For distinct p, q, write $p \to q$ when

$$s(p)^+,\ s(q)^+,\ p^-,\ q^- \tag{26}$$

occur in clockwise cyclic order. Denote the resulting directed edge set by $E(\sigma)$, and put

$$\mathrm{adv}(p,q) = (q-p) \bmod n \in \{1,\dots,n-1\}.$$

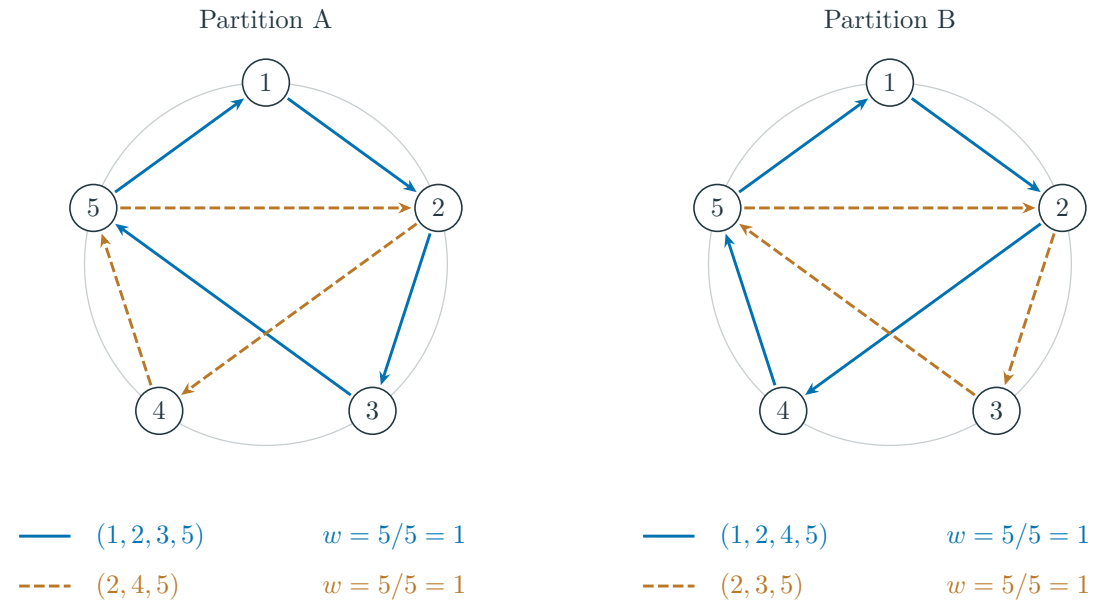


Figure 12: Two winding-one partitions for $f = (3, 5, 6, 7, 9)$. In each panel the blue solid cycle and the ochre dashed cycle together use every directed crossing edge exactly once. Each cycle has total cyclic advance 5, hence winding $w = 5/5 = 1$. The partitions differ in how the edges through labels 3 and 4 are assigned to cycles.

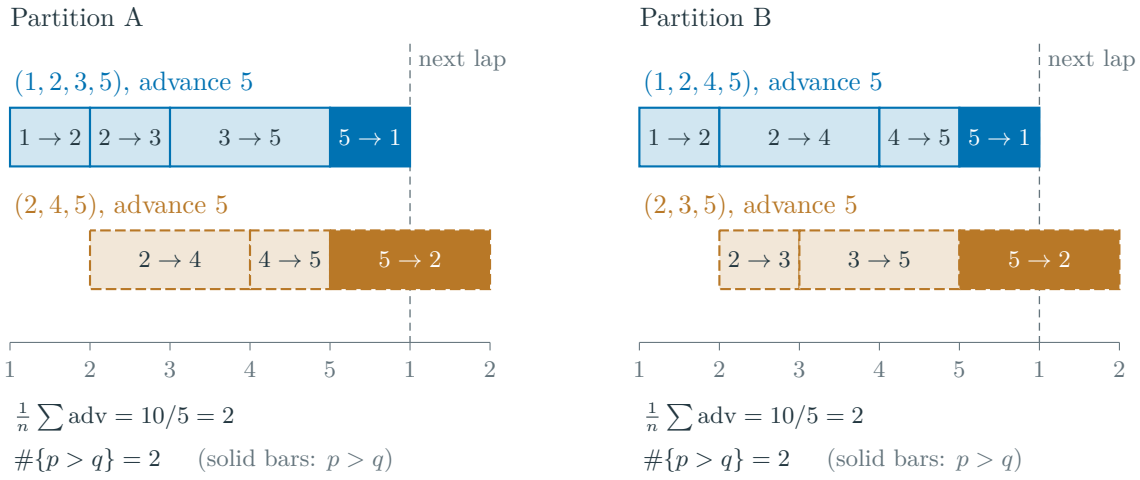


Figure 13: The count (25) for the two partitions of Figure 12, $f = (3, 5, 6, 7, 9)$. A cycle is a bar of length its total advance 5, cut into the advances $\mathrm{adv}(p, q)$ of its edges; solid bars are the edges with $p > q$, and the positions run on past label 5 into the next lap. Both partitions have total advance $10 = 5 \cdot 2$ and two edges with $p > q$, $(5, 1)$ and $(5, 2)$. The advances are counted directly.

The winding of a directed closed walk is its total advance divided by n.

To bridge conventions, identify the 0-based permutation σ of $\mathbb{Z}_n$ used here with the paper's induced permutation $\bar{f}$ (under $x \mapsto x+1$), so that $s = \sigma^{-1}$ becomes $s_x = \bar{f}^{-1}(x)$. Since f is loopless, the fixed points of σ are its coloops. The directed edge set $E(\sigma)$ is then exactly the directed crossing-edge set of G_f^x: both are read from the same ε-resolved endpoint order, in which each exit i^- occurs immediately before the entry i^+ at boundary position i, and the relation $p \to q$ of (26) is precisely Galashin's clockwise crossing order $b^+_{s_p}, b^+_{s_q}, b^-_p, b^-_q$. Establishing that $E(\sigma)$ is an edge-disjoint union of vertex-simple winding-one cycles, each of length at least 3, therefore proves Proposition 15.

We first record a direct consequence of the refined endpoint convention.

Proposition 16 (Direct degree balance)**.** *For every $p \in \mathbb{Z}_n$,*

$$\operatorname{outdeg}_{E(\sigma)}(p) = \operatorname{indeg}_{E(\sigma)}(p).$$

Proof. Fix p, and let

$$U_p = (s(p)^+, p^-)$$

be the open clockwise refined interval. This interval contains whole pairs r^-, r^+: after leaving $s(p)^+$, every intermediate boundary position is encountered first at r^- and then at r^+, while the two endpoints of U_p themselves are excluded. Consequently U_p contains equally many entry points and exit points.

For $q \neq p$, the cyclic-order definition gives

$$\begin{aligned} p \to q \quad &\Longleftrightarrow \quad s(q)^+ \in U_p \ \text{ and } \ q^- \notin U_p, \\ q \to p \quad &\Longleftrightarrow \quad q^- \in U_p \ \text{ and } \ s(q)^+ \notin U_p. \end{aligned}$$

If both endpoints of C_q lie in U_p, or both lie outside it, the two chords do not cross. Hence

$$\begin{aligned} \operatorname{outdeg}(p) - \operatorname{indeg}(p) &= \sum_{q \in \mathbb{Z}_n} \left(\mathbf{1}_{\{s(q)^+ \in U_p\}} - \mathbf{1}_{\{q^- \in U_p\}} \right) \\ &= \#\{\text{entry points in } U_p\} - \#\{\text{exit points in } U_p\} = 0. \end{aligned}$$

□

We now enlarge the finite state space used in the induction. A *decorated bounded window* is an integer tuple

$$w = (w_0, \ldots, w_{n-1})$$

such that the residues

$$\bar{w}_i := w_i \bmod n \in \mathbb{Z}_n$$

form a permutation and

$$i \le w_i \le i + n \qquad (0 \le i < n). \tag{27}$$

A position i is *fixed* if $\bar{w}_i = i$, and *active* otherwise. At a fixed position one has either $w_i = i$ or $w_i = i + n$. The residue permutation determines an inverse s_w, and we define $E(w)$ by the same rule as in (26), with s_w in place of s. The two decorations of a fixed position give the same refined chord: its endpoints i^-, i^+ are adjacent, so it crosses no other chord.

The original permutation gives the bounded window

$$w_i^\sigma = \begin{cases} \sigma(i), & \sigma(i) > i, \\ \sigma(i) + n, & \sigma(i) \le i, \end{cases} \tag{28}$$

which is the window of f in 0-based labels; a coloop i gets the decoration $w_i^\sigma = i + n$. Then $i < w_i^\sigma \le i + n$, its residue permutation is σ, and

$$E(w^\sigma) = E(\sigma). \tag{29}$$

Lemma 17 (Active-set restriction)**.** *The residue permutation $i \mapsto \bar{w}_i$ restricts to a permutation of the active positions.*

Proof. If i is fixed, then the residue i already occurs at position i. Since all residues are distinct, no other position can map to i. Thus the fixed positions map bijectively to themselves, and their complement does as well. □

Lemma 18 (Adjacent active ascent)**.** *If w has an active position, there are adjacent positions $a < b$ in the increasing list of active positions such that*

$$A := w_a < B := w_b.$$

Every r with $a < r < b$ is fixed, and

$$b \le A < B \le a + n. \tag{30}$$

If g is obtained by swapping A and B at positions a, b, then g is again a decorated bounded window.

Proof. Write the active positions as

$$i_1 < \cdots < i_m.$$

If $m > 0$, then $m \ge 2$, since a permutation cannot move exactly one element. By Lemma 17, the residue at i_1 is another active position and therefore is strictly larger than i_1; similarly, the residue at i_m is strictly smaller than i_m. Boundedness forces

$$w_{i_1} = \bar{w}_{i_1} \le i_m < i_1 + n \le \bar{w}_{i_m} + n = w_{i_m}.$$

Thus the sequence $w_{i_1}, \ldots, w_{i_m}$ cannot be strictly decreasing, so some adjacent pair in it is an ascent. Choose such a pair and call its positions $a < b$. Their adjacency in the active list says precisely that every $a < r < b$ is fixed.

Put

$$\alpha = A \bmod n, \qquad \beta = B \bmod n.$$

The label α is active, so $\alpha \ne a$. If $\alpha < a$, boundedness gives $A = \alpha + n \ge b$; if $\alpha > a$, adjacency of a, b in the active list gives $\alpha \ge b$, and boundedness then forces $A = \alpha$. Thus $b \le A$. Similarly, if $\beta < b$, then $\beta \le a$ and $B = \beta + n \le a + n$; if $\beta > b$, then $B = \beta \le n - 1 < a + n$. This proves (30).

After the swap, the value at a is B, and

$$a < B \le a + n.$$

The value at b is A, and

$$b \le A < B \le a + n < b + n.$$

The residues are unchanged as a set, so the swapped tuple is again a decorated bounded window. □

Define

$$\mathrm{inv}(w) = \#\{(i, j) : 0 \le i < j < n,\ w_i > w_j\}, \qquad \mu(w) = \binom{n}{2} - \mathrm{inv}(w). \tag{31}$$

Lemma 19 (Strict measure decrease)**.** *For the swap $w \mapsto g$ in Lemma 18,*

$$\mathrm{inv}(g) = \mathrm{inv}(w) + 1, \qquad \mu(g) = \mu(w) - 1. \tag{32}$$

Proof. For $a < r < b$, fixedness gives either $w_r = r$ or $w_r = r + n$. In the first case,

$$w_r = r < b \le A;$$

in the second,

$$w_r = r + n > a + n \ge B.$$

Thus no intermediate value belongs to $[A, B]$.

The pair (a, b) changes from the noninversion $A < B$ to an inversion. For an intermediate position r, the total inversion contribution of (a, r) and (r, b) is unchanged: its value lies either below both A, B or above both. A position outside $[a, b]$ sees the same two-element multiset $\{A, B\}$ on the same side. Hence exactly one inversion is added, proving (32). □

We next give the exact change in the directed crossing graph. For an integer lift r, define

$$\kappa(r^-) = 2r, \qquad \kappa(r^+) = 2r + 1. \tag{33}$$

Thus

$$\kappa((r + n)^\varepsilon) = \kappa(r^\varepsilon) + 2n, \qquad \varepsilon \in \{-, +\}.$$

Write $x \prec y$ when the chosen integer lifts satisfy $\kappa(x) < \kappa(y)$. A chain of four such inequalities contained in one interval of κ-length $2n$ records their clockwise cyclic order.

When displaying endpoint orders, the integer before the subscript records the chosen lift position, while the subscript records the chord label. Thus r_p^+ denotes the entry point of C_p represented at integer position r, so $r \equiv s_w(p) \pmod n$, and r_p^- denotes the exit point of C_p represented at integer position r, so $r \equiv p \pmod n$. The subscript does not enter the order coordinate:

$$\kappa(r_p^\varepsilon) = \kappa(r^\varepsilon), \qquad \varepsilon \in \{-, +\}.$$

The label attached to a fixed refined position is always read in the state w or g currently under discussion.

Figure 14 draws the lemma below. Cut the circle just before a^+ and lift one turn: the swap moves only the entries of α and β between a^+ and b^+, the open swept arc between them holds b^- and the isolated chords of fixed positions, and the three regimes are the three places of the entry S^+ of the chord C_b relative to the exits A^- and B^-.

Lemma 20 (Exact local transition)**.** *Use the notation of Lemma 18. If $A = b$, then*

$$E(w) = E(g). \tag{34}$$

Assume $A > b$. Let $c = s_w(b)$, and define

$$S = \begin{cases} c + n, & c < a, \\ c, & c > b. \end{cases} \tag{35}$$

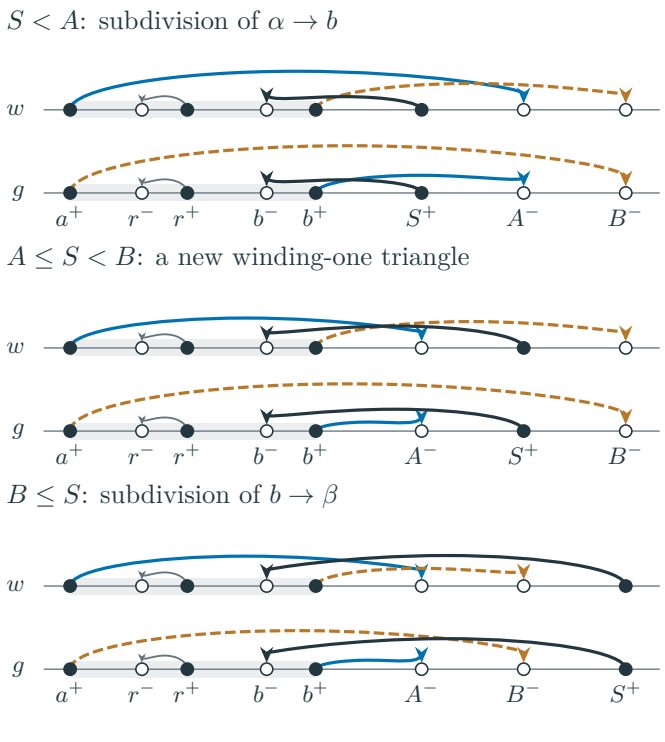
Then $b < S < a + n$, and exactly one of the following holds:

$$S < A: \quad E(w) = \big(E(g) \setminus \{\alpha \to b\}\big) \cup \{\alpha \to \beta, \beta \to b\}, \tag{36}$$

$$A \le S < B: \quad E(w) = E(g) \cup \{b \to \alpha, \alpha \to \beta, \beta \to b\}, \tag{37}$$

$$B \le S: \quad E(w) = \big(E(g) \setminus \{b \to \beta\}\big) \cup \{b \to \alpha, \alpha \to \beta\}. \tag{38}$$

Every displayed added edge is absent from $E(g)$, and every displayed removed edge is absent from $E(w)$.



$E(w) = E(g) - \{\alpha \to b\} + \{\alpha \to \beta,\ \beta \to b\}$

$\mathrm{adv}(\alpha, b) = \mathrm{adv}(\alpha, \beta) + \mathrm{adv}(\beta, b)$

The shaded band is the open swept arc from a^+ to b^+: it contains b^- and fixed positions only, whose chords cross nothing.

$E(w) = E(g) + \{b \to \alpha,\ \alpha \to \beta,\ \beta \to b\}$

advances $A - b$, $B - A$, $b + n - B$ sum to n

One lifted turn from a^+. Only the entries of α and β move, between a^+ and b^+; the regime is the place of the entry S^+ of the chord of b.

$E(w) = E(g) - \{b \to \beta\} + \{b \to \alpha,\ \alpha \to \beta\}$

$\mathrm{adv}(b, \beta) = \mathrm{adv}(b, \alpha) + \mathrm{adv}(\alpha, \beta)$

Chords run from entry to exit (arrowhead at the exit) and cross exactly when their endpoints interlace. Upper line: w; lower line: g. Chord of α blue, of β ochre dashed, of b black.

Figure 14: The three regimes of Lemma 20, on one lifted turn starting at a^+. Chords run from entry to exit; two chords cross exactly when their endpoints interlace. In each regime the upper line carries the chords of w and the lower line those of g; only the entries of α and β differ. The shaded band is the open swept arc from a^+ to b^+. Schematic; the positions are illustrative.

Proof. Only the entries of the labels α, β move:

	$s(\alpha)$	$s(\beta)$
w	a	b
g	b	a.

(39)

We first justify that no unlisted label can affect the transition.

The open refined arc from a^+ to b^+ contains b^-, but it does not contain a^-: at each position the order is r^-, r^+. It also contains both endpoints r^-, r^+ of every strictly intermediate position $a < r < b$. Such a position is fixed, and its adjacent-endpoint chord is isolated in both $E(w)$ and $E(g)$.

Now let q be active and

$$q \notin \{b, \alpha, \beta\}.$$

By Lemma 17, the exit label q is itself an active position. Since a, b are adjacent active positions, q^- is not in the open arc from a^+ to b^+; the possibility $q = a$ causes no exception, because a^- precedes a^+. Likewise, $s_w(q)$ is an active position. It is neither a nor b, since the entries there belong to α, β, and it cannot lie strictly between them. Thus $s_w(q)^+$ is outside the swept arc. The same is true in g, since the inverse position of q does not change.

For a pair involving α and such a q, the three stationary endpoints are $\alpha^-, s(q)^+, q^-$. Except when $\alpha = b$, which is the separate case $A = b$, none lies in the swept arc. The analogous assertion holds for β and q; if $\beta = a$, its exit is a^-, again outside the arc. In a finite cyclic order, deleting

three stationary points partitions the remaining points into three cyclic intervals. The points a^+ and b^+ lie in the same such interval whenever the open arc between them contains none of the three stationary points. Replacing one by the other therefore leaves the labeled cyclic order of the four endpoints unchanged. Hence only pairs contained in

$$\{b, \alpha, \beta\}$$

can change. In particular, the label a is never an additional special label: its exit a^- is outside the swept arc, and if $a = \beta$ it is already included in the displayed set.

Suppose first that $A = b$, so $\alpha = b$. Since $A < B \le a + n$, the four endpoints of the only special pair $\{b, \beta\}$ occur in w as

$$a_b^+ \prec b_b^- \prec b_\beta^+ \prec B_\beta^-. \tag{40}$$

The two endpoints of b are consecutive in this four-point order, as are the two endpoints of β, so the pair is noncrossing. In g, their order is

$$a_\beta^+ \prec b_b^- \prec b_b^+ \prec B_\beta^-. \tag{41}$$

which is again noncrossing. This proves (34).

Assume henceforth that $A > b$. Then α, β, b are distinct. The entry position $c = s_w(b)$ is active. It is not a, because $\alpha \neq b$, and it is not b, because b is active. It cannot lie strictly between a, b, so (35) is well defined and gives

$$b < S < a + n. \tag{42}$$

The relevant exit lifts satisfy

$$b < A < B < b + n. \tag{43}$$

For the pair $\{\alpha, \beta\}$, independently of the position of S, the order in w is

$$a_\alpha^+ \prec b_\beta^+ \prec A_\alpha^- \prec B_\beta^-. \tag{44}$$

This is exactly the defining order for $\alpha \to \beta$. In g, the order is

$$a_\beta^+ \prec b_\alpha^+ \prec A_\alpha^- \prec B_\beta^-. \tag{45}$$

The two α-endpoints are consecutive in this four-point order, so the pair is noncrossing. Thus $\alpha \to \beta$ belongs to $E(w) \setminus E(g)$ in all three regimes.

We now inspect the two remaining pairs in each regime.

Case $S < A$. For $\{\alpha, b\}$, the endpoint order in w is

$$b_b^- \prec S_b^+ \prec A_\alpha^- \prec (a + n)_\alpha^+. \tag{46}$$

The two b-endpoints and the two α-endpoints are each consecutive, so the pair is noncrossing. In g, the order is

$$b_\alpha^+ \prec S_b^+ \prec A_\alpha^- \prec (b + n)_b^-. \tag{47}$$

which is the defining order for $\alpha \to b$.

For $\{\beta, b\}$, the order in w is

$$b_\beta^+ \prec S_b^+ \prec B_\beta^- \prec (b + n)_b^-. \tag{48}$$

so $\beta \to b$. In g, the order is

$$S_b^+ \prec B_\beta^- \prec (a + n)_\beta^+ \prec (b + n)_b^-. \tag{49}$$

Here the two β-endpoints are consecutive, so the pair is noncrossing. Together with (44)–(45), these orders prove (36).

Case $A \le S < B$. For $\{\alpha, b\}$, the endpoint order in w is

$$S_b^+ \prec (a+n)_\alpha^+ \prec (b+n)_b^- \prec (A+n)_\alpha^-. \tag{50}$$

which is the defining order for $b \to \alpha$. In g, the order is

$$b_\alpha^+ \prec A_\alpha^- \prec S_b^+ \prec (b+n)_b^-. \tag{51}$$

The two α-endpoints are consecutive, so the pair is noncrossing. When $S = A$, the strict middle inequality in (51) is still valid because $A^- \prec A^+$.

For $\{\beta, b\}$, the endpoint order in w is

$$b_\beta^+ \prec S_b^+ \prec B_\beta^- \prec (b+n)_b^-. \tag{52}$$

so $\beta \to b$. In g, the order is

$$S_b^+ \prec B_\beta^- \prec (a+n)_\beta^+ \prec (b+n)_b^-. \tag{53}$$

which is noncrossing because the two β-endpoints are consecutive. Together with (44)–(45), these orders prove (37).

Case $B \le S$. For $\{\alpha, b\}$, the endpoint order in w is again

$$S_b^+ \prec (a+n)_\alpha^+ \prec (b+n)_b^- \prec (A+n)_\alpha^-. \tag{54}$$

so $b \to \alpha$. In g, the order is

$$b_\alpha^+ \prec A_\alpha^- \prec S_b^+ \prec (b+n)_b^-. \tag{55}$$

so the pair is noncrossing.

For $\{\beta, b\}$, the endpoint order in w is

$$b_\beta^+ \prec B_\beta^- \prec S_b^+ \prec (b+n)_b^-. \tag{56}$$

The two β-endpoints are consecutive, so the pair is noncrossing. In g, the order is

$$S_b^+ \prec (a+n)_\beta^+ \prec (b+n)_b^- \prec (B+n)_\beta^-. \tag{57}$$

which is the defining order for $b \to \beta$. When $S = B$, the exit B^- precedes the entry B^+, so the first β-exit encountered after $S_b^+ = B_b^+$ is $(B+n)_\beta^-$; this is why equality belongs to the third regime. Together with (44)–(45), these orders prove (38). □

Lemma 21 (Winding arithmetic)**.** *In the triangle case* (37)*,*

$$b \to \alpha \to \beta \to b$$

has winding one. In each subdivision case, the two-edge replacement has the same total advance as the removed edge.

Proof. By (43),

$$\begin{aligned} \mathrm{adv}(b, \alpha) &= A - b, \\ \mathrm{adv}(\alpha, \beta) &= B - A, \\ \mathrm{adv}(\beta, b) &= b + n - B. \end{aligned}$$

Their sum is n, proving the triangle assertion. In the left subdivision,

$$\begin{aligned}\operatorname{adv}(\alpha, b) &= b + n - A \\ &= (B - A) + (b + n - B) \\ &= \operatorname{adv}(\alpha, \beta) + \operatorname{adv}(\beta, b).\end{aligned}$$

In the right subdivision,

$$\begin{aligned}\operatorname{adv}(b, \beta) &= B - b \\ &= (A - b) + (B - A) \\ &= \operatorname{adv}(b, \alpha) + \operatorname{adv}(\alpha, \beta).\end{aligned}$$

□

Proof of Proposition 15. The induction establishes the stronger statement for every decorated bounded window w: $E(w)$ is an edge-disjoint union of winding-one directed cycles. It is a strong induction on the nonnegative integer $\mu(w)$ from (31).

If w has no active position, its residue permutation is the identity. Thus each w_i is either i or $i + n$, every chord has adjacent endpoints i^-, i^+, and

$$E(w) = \varnothing.$$

Conversely, a state with an active position cannot be terminal: Lemma 18 supplies an adjacent active ascent. Thus the all-fixed windows are exactly the terminal states of the reduction.

Suppose w is not terminal. Choose a, b as in Lemma 18, and let g be the swapped window. Lemma 19 gives $\mu(g) < \mu(w)$, so the induction hypothesis supplies an edge-disjoint partition of $E(g)$ into winding-one directed cycles.

In the identity case (34), retain this partition. In the triangle case (37), add

$$b \to \alpha \to \beta \to b$$

as a new part; it winds once by Lemma 21.

Consider the left subdivision (36). The edge $\alpha \to b$ has a unique owner cycle in the partition of $E(g)$. Replace that occurrence by

$$\alpha \to \beta \to b.$$

The new edges belong to $E(w) \setminus E(g)$, so they occur nowhere else in the old partition, and Lemma 21 preserves the total advance. The labels α, β, b are distinct. The substituted closed walk has total advance n, so it is vertex-simple by Lemma 14; in particular the inserted vertex β does not already occur elsewhere in the owner cycle.

The right subdivision (38) is identical: replace the uniquely owned edge $b \to \beta$ by

$$b \to \alpha \to \beta.$$

The inserted label α is distinct from b, β, the two new edges are absent from $E(g)$, the advance is preserved, and Lemma 14 again shows that the new owner cycle is vertex-simple.

In every case the resulting cycles are edge-disjoint and exhaust $E(w)$. This completes the induction. Applying it to w^σ and using (29) proves the required decomposition of $E(\sigma)$.

Next, a vertex-simple directed cycle of winding one is a rotation of the increasing order of its support. Indeed, for representatives $v_i \in \{0, \dots, n-1\}$,

$$\sum_i \operatorname{adv}(v_i, v_{i+1}) = n \cdot \#\{i : v_{i+1} < v_i\},$$

because the ordinary differences telescope. Winding one therefore means that the cycle has exactly one descent; rotating to begin immediately after that descent makes its labels strictly increasing. No resulting cycle has length 1, since the crossing graph is irreflexive, or length 2, since it is antisymmetric: for distinct p, q the four refined points $s(p)^+, s(q)^+, p^-, q^-$ are all distinct, and the clockwise pattern witnessing $q \to p$ is not a cyclic rotation of the one witnessing $p \to q$, so the two cannot hold at once. Thus every support has cardinality at least 3. Finally, summing $\mathrm{adv}(p, q) = q - p + n\mathbf{1}_{p>q}$ around each cycle gives exactly one wrapped edge per cycle, proving the count in (25) as well. □

Remark 22. The degree-balance proposition (Proposition 16) proves in- and out-degree balance of the directed crossing graph directly from the refined endpoint order, without the cycle decomposition; it is in this form that the gradient identity uses it (Proposition 12).

Remark 23 (Machine verification)**.** Three components of the proof have Lean formalizations: the cotangent cycle inequality (Lemma 25), which is analytic, and two combinatorial statements over every loopless f, coloops included: the row-difference identity (Proposition 10) and the winding-one partition (Proposition 15). The row-difference identity is formalized in full: crossing columns, the isolation, necklace-membership and zero-row facts for coloops, and, through `Conj43.pairExponent_nonCross`, `Conj43.full_pair_columns_zero` and `Conj43.rowdiff_full_pairs`, noncrossing-column vanishing and the full unordered-pair identity. The winding partition is formalized via the window induction of this subsection, together with both forms of the cycle count (25) and the extension to decorated permutations; the strand-diagram proof is not formalized. The auxiliary cover theorem of that proof (*Existence of a winding-one partition* in Appendix A) is a further named Lean result, not a further mathematical component.

The analytic steps are formalized as well: the lifted cycle sum (10), the hinge identity (Lemma 3), the assembled Hessian with its positivity and kernel (Proposition 26), the potential's gradient $B\ell$ and Hessian $B\,\mathrm{diag}(\cot\delta_e)B^{\mathsf{T}}$, strict convexity on the gauge slice, an injective gradient that is an open embedding into $\mathrm{im}\,B$, the variational inverse (2), and the computation that turns the logarithm of (22) on necklace rows into G_f (Proposition 12). These compose into a *conditional* form of Theorem 1: if a map on the gauge slice is continuous and onto a set Crit, and a continuous map N on Crit satisfies, along it, the logarithmic necklace-row form of (22), then the first map is a homeomorphism, with inverse $G_f^{-1} \circ P_f D \circ N$. Appendix A states these hypotheses in mathematical notation and names the results of [1] and [4] that they stand for. The interface with the definitions of [1] is formalized as well: against Lean transcriptions of those definitions, the encoding used here is a bijection onto his loopless bounded affine permutations, its necklace rows are given by his affine formula, its crossings and misalignment classes are his, its exponent table is that of (22), and Lemma 9 holds on his refined boundary. Lean constructs neither the measurement map nor the twist, so it certifies components, their interface and a conditional composition, not Theorem 1 itself; whether the transcribed definitions and the hypotheses say what [1] and [4] say is for the reader to check. Two later results are formalized, the first in full. Of the sharp modulus, Lean has Theorem 28 with its extremal pair, the bounds of Lemma 27 on $\kappa^\star_m(\eta)$, the two-sided bound (69), and the inverse-Lipschitz bound (70). Of Corollary 2, it has the combinatorial half: $f^\downarrow$ is a bounded affine permutation of rank $k-1$, looplessness is exactly what keeps it at or above the identity, and $(f^{\mathrm{elec}}_\tau)^\downarrow = f^{\mathrm{Ising}}_\tau$. The homeomorphism Φ_f, which rests on Theorem 1, is not formalized.

The current formal source and the historical certificate archive, with its checksum, are described in Appendix A.

4.3 A second proof, from strand diagrams

The following proof of Proposition 15 is geometric and independent of §4.2; it is not formalized, and Remark 24 lists what it imports.

Second proof of Proposition 15. Suppress fixed strands, which are isolated in the crossing graph, and split each boundary position into distinct points i^-, i^+ in clockwise order. A nonfixed strand S_p runs from the entry s_p^+ to the exit p^-. Write $\mathrm{adv}(p,q) = (q-p) \bmod n \in \{1,\dots,n-1\}$.

Black-clique cycles. Choose a reduced alternating strand diagram with strand permutation f, which exists by Postnikov's monotone triple-diagram construction [5, Lem. 14.6, Cor. 14.7 with Cor. 14.2]. At a boundary vertex merged from the pair b_i, b_i' in the passage from triple diagrams to alternating strand diagrams [5, the map $T \mapsto D$ after Def. 14.3, and Lem. 14.4], use the black boundary resolution, constructed in the boundary collars below. It is not the order the merge itself gives, which puts each entry before its exit (Figure 15(d), not used): a black boundary corner becomes one small transverse crossing, while the other germ order becomes none. Every transverse intersection is then a corner of exactly one counterclockwise strand face, hence of the corresponding black vertex of the reduced plabic graph.

Read the exit labels at that black vertex in clockwise order. They form a black clique

$$L \setminus \{b_1\}, \dots, L \setminus \{b_m\},$$

with $b_1, \dots, b_m$ in cyclic order. This clockwise orientation is the one fixed in [7, Rem. 9.6, with Thm. 9.12 and the two remarks after it]: the plabic-tiling boundary vectors are clockwise, and the black-clique vertices $v_L - v_{b_i}$ retain that orientation. Thus the black vertex supplies the directed cycle

$$C_F : b_1 \longrightarrow b_2 \longrightarrow \cdots \longrightarrow b_m \longrightarrow b_1.$$

Its advances telescope to n, so it winds once. Bivalent black vertices give honest winding-one bigons. Taking all black vertices gives a multiset of winding-one cycles, possibly with both orientations of the same unordered edge.

Signed excess, including the boundary. Use target labels: S_p ends at p, and an ordinary graph face contains p in its label precisely when it lies to the left of S_p. Contract same-color interior edges before taking the strand diagram; the two strand segments along such an edge are parallel, and contraction preserves the face labels [7, §6]. The correspondence between the graph's cyclic face order and the plabic tiling is supplied by [7, Thm. 9.12 and the following two remarks]; Remark 9.6 there fixes its clockwise orientation. Degree-two vertices are treated as two-corner cycles, even though their two-element cliques are represented by single edges in the tiling.

Let $m(p,q)$ be the multiplicity of $p \to q$ among the clockwise black-clique cycles. In an ordinary Cartesian chart, define the local clockwise intersection sign by

$$\varepsilon_c(p,q) = -\operatorname{sgn}\det(t_p, t_q),$$

where t_p, t_q are the oriented strand tangents. Put the black vertex west of an interior crossing and its white neighbor east. Label the rays, in clockwise order starting northwest, as

$$p_{\mathrm{out}}, \quad q_{\mathrm{out}}, \quad p_{\mathrm{in}}, \quad q_{\mathrm{in}}.$$

We may take $t_p = (-1,1)$ and $t_q = (1,1)$, so $\varepsilon_c(p,q) = +1$. The black, counterclockwise strand face occupies the west sector: its directed boundary enters on S_q and leaves on S_p. The ordinary graph faces north and south of the crossing have labels

$$K \cup \{q\} = L \setminus \{p\}, \qquad K \cup \{p\} = L \setminus \{q\}, \qquad L = K \cup \{p,q\}.$$

Indeed, the north face is left of S_q and right of S_p, and the south face has the opposite memberships. Clockwise around the black vertex, north precedes south. The corresponding clique step is therefore $p \to q$: its contribution to $m(p,q) - m(q,p)$ is $+1$, exactly the intersection sign. For (q,p) both contributions are -1; for every other pair they are zero. Orientation-preserving changes of chart give all interior corners.

Here is the boundary construction explicitly (Figure 15). Write $r = \bar{f}(i) \neq i$; S_i exits at i and S_r enters there. In a small collar with the disk to the west, clockwise along the boundary is downward. Place the exit i^- above the entry i^+. At a black neighbor, the exit germ approaches from southwest and the entry germ returns northwest. Joining these to the upper exit and lower entry makes one transverse crossing with the ray configuration just described, with $(p,q) = (r,i)$. It contributes the black-clique step $r \to i$ and sign $+1$. At a white neighbor, the exit germ is northwest and the entry germ is southwest; the two connections to the same ordered endpoints are disjoint. Disjoint collars implement these choices simultaneously, without meeting any other strand. Fixed strands have already been suppressed. This specifies the refinement after Postnikov's boundary merge; it is an additional local construction, not a statement of the merge itself [5, Def. 14.3 and the following map; cf. the construction preceding Cor. 14.2].

Every black corner is now counted once, and every transverse crossing has exactly one black owner. A bivalent black vertex has clique steps $p \to q$ and $q \to p$, so its two local signs cancel. Summing over all corners gives

$$m(p,q) - m(q,p) = \sum_c \varepsilon_c(p,q) = I_{\mathrm{cw}}(S_p, S_q). \tag{58}$$

The four refined endpoints of two distinct strands are distinct. Homotoping the two strands to chords through proper arcs with these endpoints fixed preserves algebraic intersection: generic interior tangencies create or delete opposite signs, and no intersection can cross the boundary. The homotopy need not remain alternating. In the clockwise endpoint order s_p^+, s_q^+, p^-, q^- the ordered chord tangents have negative Cartesian determinant. Thus

$$I_{\mathrm{cw}}(S_p, S_q) = \begin{cases} 1, & s_p^+, s_q^+, p^-, q^- \text{ occur clockwise,} \\ -1, & s_q^+, s_p^+, q^-, p^- \text{ occur clockwise,} \\ 0, & \text{otherwise.} \end{cases}$$

Consequently the signed excess is exactly the directed crossing-edge vector of G_f^x.

Integral cancellation. First delete every bigon, cancelling its two opposite occurrences. Suppose occurrences of $p \to q$ and $q \to p$ remain. They have distinct owner cycles: a winding-one cycle is vertex-simple by Lemma 14, and a vertex-simple cycle containing both orientations is precisely the already deleted bigon.

Remove $p \to q$ from its owner C_1 and $q \to p$ from its owner C_2. The remainders are paths

$$P_1 : q \rightsquigarrow p, \qquad P_2 : p \rightsquigarrow q.$$

If $d = \mathrm{adv}(p,q)$, their advances are $n-d$ and d. Consequently P_1P_2 is a closed walk of total advance n, hence a vertex-simple winding-one cycle by Lemma 14. Replace C_1, C_2 by this cycle. Each operation removes two edge occurrences, preserves every signed multiplicity in (58), and therefore terminates.

At termination only one orientation of each unordered pair remains. Equation (58) then forces exactly one occurrence of every positive crossing edge and no occurrence of a noncrossing edge. The surviving winding-one cycles are edge-disjoint and exhaust G_f^x. No bigon survives, so every cycle

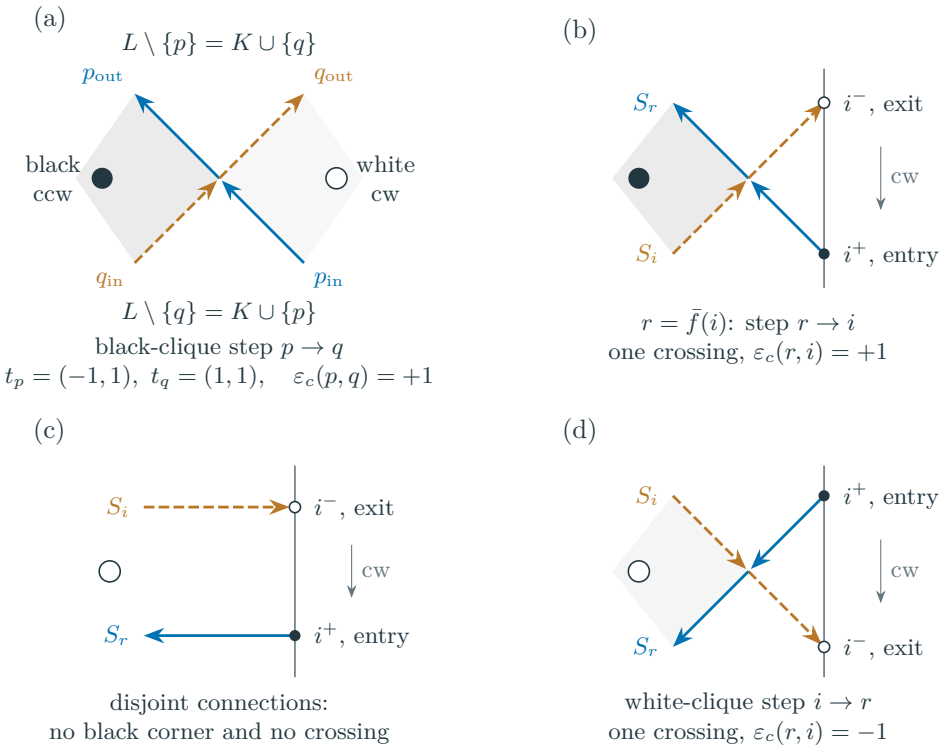


Figure 15: The local sign count behind (58). (a) An interior crossing at a black corner, with the black, counterclockwise strand face to the west: the rays in clockwise order are $p_{\text{out}}, q_{\text{out}}, p_{\text{in}}, q_{\text{in}}$, the faces north and south are labelled $L \setminus \{p\}$ and $L \setminus \{q\}$, and the clique step $p \to q$ has sign $\varepsilon_c(p, q) = +1$. (b) The black resolution at a boundary position i next to a black vertex, with the exit i^- before the entry i^+ clockwise: one crossing, the step $r \to i$, $r = \bar{f}(i)$. (c) Next to a white vertex: no crossing. (d) The other refinement, which is not used: it puts the entry first and gives the white-clique step $i \to r$ with the opposite sign. Schematic.

has length at least three. For the count, $\mathrm{adv}(p,q) = q - p + n\mathbf{1}_{p>q}$. Summing around a winding-one cycle cancels the raw differences, so that cycle contains exactly one wrapped edge. Summing over the partition gives (25). □

Remark 24 (what the strand-diagram proof imports, and why nothing depends on it)**.** The proof just given rests on three imported facts — the existence of a reduced alternating strand diagram with prescribed strand permutation and the merge of the boundary vertex pairs [5, Lem. 14.6, Cor. 14.7 with Cor. 14.2; the map $T \mapsto D$ following Def. 14.3, and Lem. 14.4], and the clockwise orientation of the black-clique vertices [7, Rem. 9.6, with Thm. 9.12 and the two remarks after it] — and on the local sign bookkeeping (58), which is derived above ray by ray, at interior corners and in the boundary collars. The pointers into [5] have been checked against arXiv:math/0609764 (single version, no journal counterpart); the black boundary resolution used above is not stated by that merge, whose map puts each entry before its exit; it is the additional local construction given in the boundary collars of the proof. The [7] pointers have been checked. A reader who prefers not to take those on trust can rely on the proof in §4.2: it proves the same statement from the refined endpoint order alone, uses none of these citations, and is the version that is machine checked (Remark 23). Nothing later in the paper depends on which of the two proofs is used.

4.4 The cotangent cycle inequality

Lemma 25. *Let $\alpha_1, \dots, \alpha_m$ be positive angles with $m \geq 3$ and $\sum \alpha_i = \pi$, and let $d_1, \dots, d_m \in \mathbb{R}$ with $\sum d_i = 0$. Then $\sum_i \cot(\alpha_i)\, d_i^2 \geq 0$, with equality iff all $d_i = 0$.*

Proof. At most one α_i exceeds $\pi/2$ (two would sum to $> \pi$). Suppose first that all $\alpha_i \leq \pi/2$. At most one of them equals $\pi/2$: two such angles would already sum to π, leaving the remaining $m - 2 \geq 1$ positive angles a sum of 0. Every other cotangent is positive, so the form is ≥ 0, and it vanishes only if $d_i = 0$ whenever $\alpha_i < \pi/2$; the condition $\sum d_i = 0$ then forces the remaining d_i, if there is one, to vanish as well. Otherwise let $\alpha_m > \pi/2$ and $S = \pi - \alpha_m = \sum_{i<m} \alpha_i < \pi/2$. Weighted Cauchy–Schwarz gives $\sum_{i<m} \cot(\alpha_i) d_i^2 \geq (\sum_{i<m} d_i)^2 / \sum_{i<m} \tan(\alpha_i)$, and strict superadditivity of tan on $(0, \pi/2)$ (with all partial sums $\leq S < \pi/2$) gives $\sum_{i<m} \tan \alpha_i < \tan S$. Using $\sum_{i<m} d_i = -d_m$ and $\cot \alpha_m = -\cot S$,

$$\sum_i \cot(\alpha_i) d_i^2 > \frac{d_m^2}{\tan S} + \cot(\alpha_m) d_m^2 = \cot(S) d_m^2 - \cot(S) d_m^2 = 0$$

when $d_m \neq 0$; if $d_m = 0$ the positive terms force all $d_i = 0$. □

4.5 Positive semidefiniteness

Figure 16 sketches the argument of this subsection.

Proposition 26. *For every $\theta \in \Theta_f^{\mathrm{adm}}$, $H_f(\theta) \succeq 0$ and*

$$\ker H_f(\theta) = \mathrm{Span}\{\mathbf{1}_C : C \text{ a component of } G_f^x\}.$$

Proof. By Proposition 15 the edge set partitions into winding-one cycles; each edge appears in exactly one, so (4) gives, for every h,

$$h^{\mathsf{T}} H_f h = \sum_{\text{cycles } Z} \sum_{e=(p,q)\in Z} \cot(\delta_e(\theta))\, (h_q - h_p)^2.$$

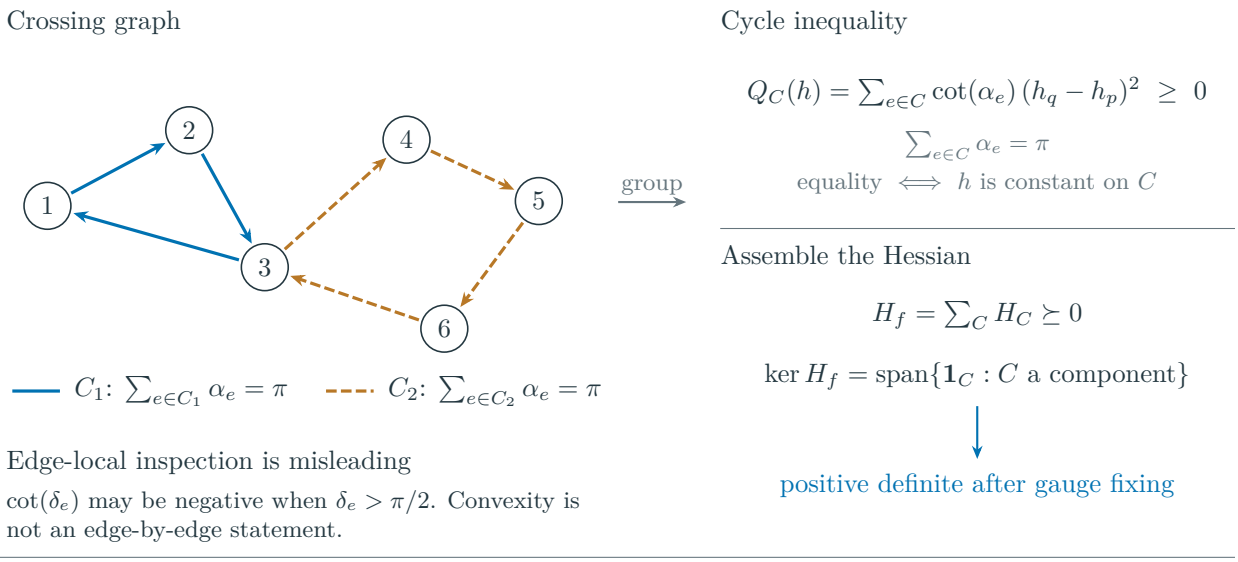


Figure 16: The directed crossing edges partition into edge-disjoint winding-one cycles. Individual cotangent weights may be negative, but each cycle quadratic form is positive semidefinite; their sum has exactly the crossing-component shifts as its kernel.

The angles δ_e are the lifted ones of (7), positive on Θ_f^{adm} by (8), and around each winding-one cycle they sum to π by (10). The h_q-h_p also sum to 0 around the cycle. Thus Lemma 25 makes each inner sum ≥ 0, with equality iff h is constant along every edge of Z. Hence $H_f\succeq 0$, and $h\in\ker H_f$ iff h is constant along every crossing edge, i.e. constant on each connected component. □

Figure 17 shows, at the worked example, the form $h^{\mathsf{T}}H_f h$ on a plane tangent to the gauge slice and Ψ_f along a segment of the slice.

4.6 The sharp convexity modulus away from the angular walls

The kernel theorem admits a sharp quantitative strengthening. For $m\geq 3$ and $0<\eta\leq\pi/m$, define

$$\kappa_m^\star(\eta):=\frac{\cot\eta-(m-1)\cot((m-1)\eta)}{m}. \tag{59}$$

This number is positive. Indeed, if $(m-1)\eta\geq\pi/2$, the second cotangent is nonpositive; otherwise strict superadditivity of tangent gives $(m-1)\tan\eta<\tan((m-1)\eta)$.

We first record a diagonal-form criterion. If real numbers $a_1,\dots,a_m$ have at most one nonpositive entry, then

$$\sum_i a_i d_i^2\geq 0\qquad\text{whenever}\ \sum_i d_i=0 \tag{60}$$

if and only if either every $a_i\geq 0$, or exactly one entry, say a_m, is negative, the others are positive,

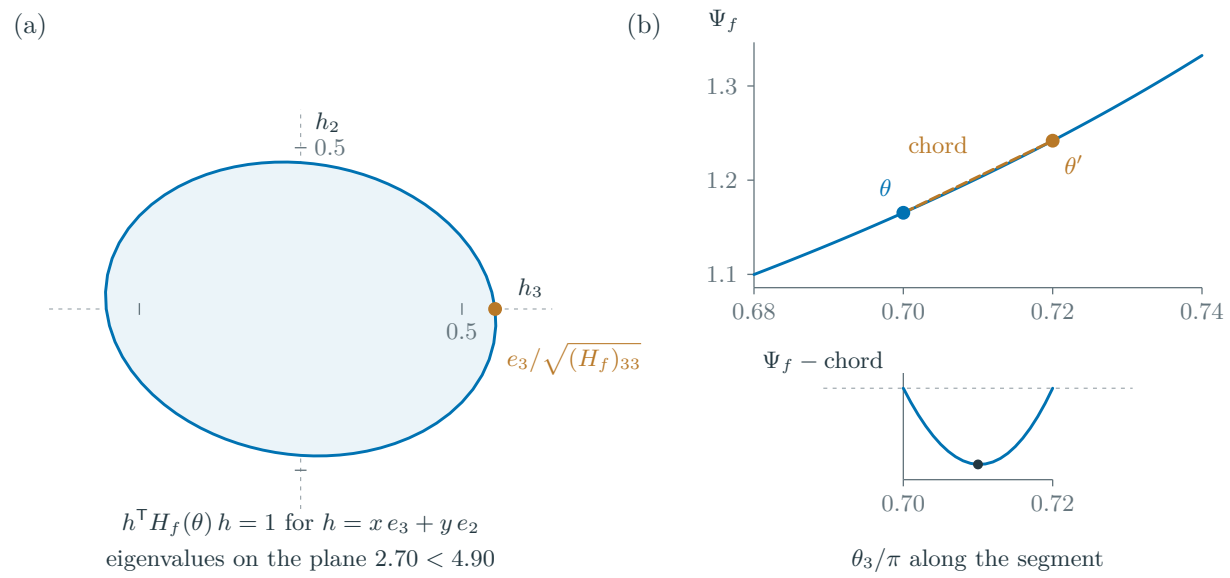


Figure 17: Worked example $\bar{f} = (3, 5, 1, 2, 4)$, $\theta/\pi = (0, 0.1, 0.7, 0.3, 0.8)$. (a) The curve $h^{\mathsf{T}} H_f(\theta) h = 1$ for $h = x\, e_3 + y\, e_2$, an ellipse: $H_f(\theta)$ is positive definite on directions with $h_1 = 0$ (Proposition 26). (b) Ψ_f, with $L(u) = \int_{\pi/2}^{u} \log \sin t \, dt$, along $\theta + \pi s\, e_3$ with its chord from θ to $\theta' = \theta + 0.02\pi\, e_3$, and below it Ψ_f minus the chord. Numerical values, in floating point.

and

$$\sum_i \frac{1}{a_i} \le 0. \tag{61}$$

For in the latter case, putting $T = \sum_{i<m} a_i^{-1}$, weighted Cauchy–Schwarz gives

$$\sum_{i<m} a_i d_i^2 \ge \frac{d_m^2}{T}.$$

Condition (61) says $T \le -1/a_m$, so $1/T + a_m \ge 0$. Conversely, if (61) fails, take $d_m = -1$ and $d_i = (1/a_i)/T$ for $i < m$. Then $\sum_i d_i = 0$ and the form equals $1/T + a_m < 0$.

Lemma 27 (Localization of the shifted sign change)**.** *Let $m \ge 3$ and $0 < \eta < \pi/m$, and put $c = \kappa_m^\star(\eta)$. Then*

$$0 < c < \tan\left(\frac{(m-2)\eta}{2}\right). \tag{62}$$

The upper bound fails at the excluded endpoint: $\kappa_m^\star(\pi/m) = \cot(\pi/m) = \tan\big((m-2)(\pi/m)/2\big)$, so there the inequality is an equality. This is harmless, because Theorem 28 disposes of $\eta = \pi/m$ in its first sentence and invokes the lemma only for $\eta < \pi/m$.

Proof. Only the upper bound remains after the positivity observation above. Put

$$a = \frac{m\eta}{2}, \qquad u = \frac{(m-2)\eta}{2}, \qquad g(t) = t \cot t.$$

Then $0 < u < a < \pi/2$, $\eta = a - u$, $(m-1)\eta = a + u$, and

$$c = \frac{g(a-u) - g(a+u)}{2a}. \tag{63}$$

Product-to-sum gives

$$g(a-u)-g(a+u)=\frac{a\sin(2u)-u\sin(2a)}{\sin^2 a-\sin^2 u}.$$

The function $\tan t/t$ is strictly increasing on $(0,\pi/2)$, since its derivative has the sign of $t-\sin t\cos t$, whose derivative is $2\sin^2 t$. Hence $u\sin a\cos u-a\sin u\cos a>0$, and a common-denominator calculation yields

$$2a\tan u-[g(a-u)-g(a+u)]=\frac{2\cos a}{\cos u(\sin^2 a-\sin^2 u)}\cdot(u\sin a\cos u-a\sin u\cos a)>0.$$

Now use (63). □

Theorem 28 (Sharp cotangent cycle inequality)**.** *Let $m\ge 3$, $0<\eta\le\pi/m$, and $\alpha_1,\dots,\alpha_m\ge\eta$ with $\sum_i\alpha_i=\pi$. If $\sum_i d_i=0$, then*

$$\sum_i \cot(\alpha_i)d_i^2\ge\kappa_m^\star(\eta)\sum_i d_i^2. \tag{64}$$

The constant is best possible: the angle–velocity pair

$$\begin{aligned}(\alpha_1,\dots,\alpha_m)&=(\eta,\dots,\eta,\pi-(m-1)\eta),\\(d_1,\dots,d_m)&=(1,\dots,1,-(m-1))\end{aligned} \tag{65}$$

attains equality, as do simultaneous permutations and nonzero scalar multiples of the displayed velocity.

Figure 18 shows the extremal pair (65) for two choices of (η,m).

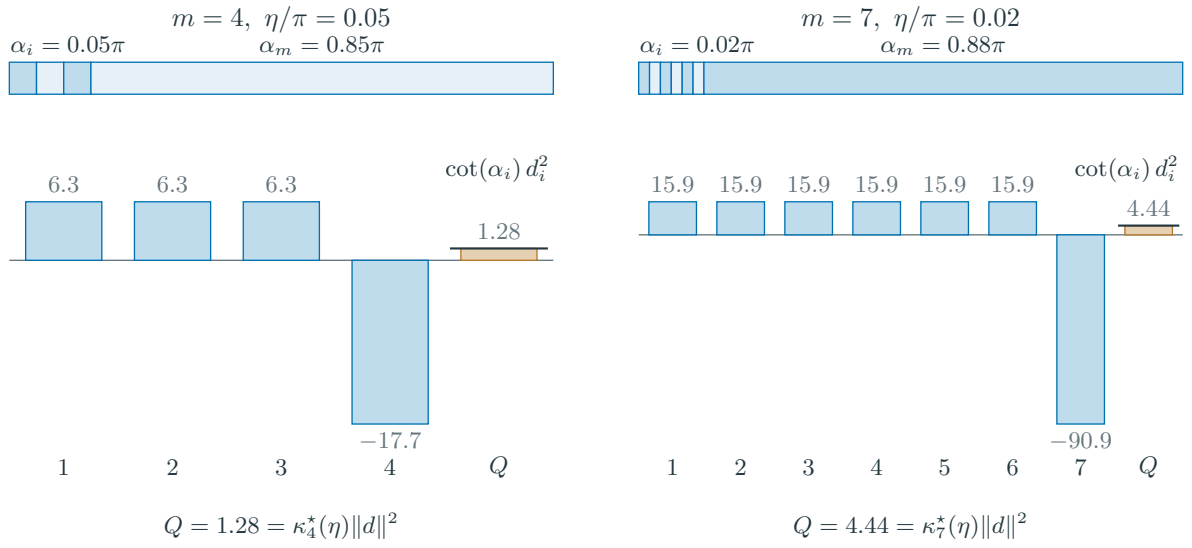


Figure 18: The extremal pair (65) of Theorem 28 for $(\eta/\pi,m)=(0.05,4)$ (left) and $(0.02,7)$ (right). Above, the angles α_i stacked to π; below, the terms $\cot(\alpha_i)\,d_i^2$ and their sum Q, which equals $\kappa_m^\star(\eta)\sum_i d_i^2$ (the line on the last bar). Numerical values, in floating point.

Proof. If $\eta = \pi/m$, the angle constraints force every $\alpha_i = \eta$, while (59) reduces to $\kappa_m^\star(\eta) = \cot\eta$; the assertion and sharpness are immediate. Assume henceforth that $0 < \eta < \pi/m$.

Put $c = \kappa_m^\star(\eta)$, $\phi = \arctan c$, and $\vartheta = \pi/2 - \phi$, so that $\cot\vartheta = c$. By Lemma 27,

$$0 < \phi < \frac{(m-2)\eta}{2}, \qquad \vartheta > \frac{\pi - (m-2)\eta}{2}. \tag{66}$$

The shifted coefficient $A_i := \cot\alpha_i - c$ is nonpositive exactly when $\alpha_i \geq \vartheta$. There cannot be two such angles, since then

$$\sum_i \alpha_i \geq 2\vartheta + (m-2)\eta > \pi.$$

If every $A_i \geq 0$, the result follows; this includes the case in which an angle equals ϑ (there is at most one with $\alpha_i \geq \vartheta$), so that its $A_i = 0$. Suppose therefore that, after relabeling, $A_m < 0$ and $A_i > 0$ for $i < m$. Then $\alpha_m > \vartheta$, hence $\cot\alpha_m - c \neq 0$ and $h(\pi - S)$ below is defined throughout the region considered.

Write $r = m - 1$, $\beta_i = \alpha_i$ for $i < m$, $S = \sum_{i<m}\beta_i$, and define

$$h(t) = \frac{1}{\cot t - c}, \qquad F(\beta_1, \ldots, \beta_r) = \sum_{i<m} h(\beta_i) + h(\pi - S).$$

Thus $F = \sum_i A_i^{-1}$. We show that F is nonincreasing in each β_i throughout the one-negative region. Since

$$h'(t) = \frac{1}{(\cos t - c\sin t)^2} = \frac{\sec^2(t+\phi)}{1+c^2},$$

one has

$$\frac{\partial F}{\partial \beta_i} = h'(\beta_i) - h'(\alpha_m). \tag{67}$$

Both $\beta_i + \phi$ and $S - \phi = \pi - (\alpha_m + \phi)$ lie in $(0, \pi/2)$. Moreover,

$$(S - \phi) - (\beta_i + \phi) = \sum_{j \neq i, m} \beta_j - 2\phi \geq (m-2)\eta - 2\phi > 0.$$

Since $\sec^2$ increases on $(0, \pi/2)$ and $\sec^2(\alpha_m + \phi) = \sec^2(S - \phi)$, (67) is negative.

Set $P = \cot\eta$ and $Q = \cot((m-1)\eta)$. At the boundary configuration,

$$c = \frac{P - (m-1)Q}{m}, \qquad P - c = \frac{(m-1)(P+Q)}{m}, \qquad -Q - c = -\frac{P+Q}{m}.$$

Here

$$P + Q = \frac{\sin(m\eta)}{\sin\eta \sin((m-1)\eta)} > 0,$$

so the boundary point indeed lies in the one-negative region, and

$$F(\eta, \ldots, \eta) = \frac{m}{P+Q} - \frac{m}{P+Q} = 0.$$

Starting at this boundary point and increasing the coordinates one at a time stays in the same one-negative region, with no change of the distinguished index m. At every intermediate point, $\eta \leq \beta_i \leq \beta_i^{\text{final}} < \vartheta$, while $\alpha_m = \pi - \sum_{i<m}\beta_i$ decreases monotonically but remains at least its final value $\alpha_m^{\text{final}} > \vartheta$. Thus $A_m < 0$ and every $A_i > 0$ for $i < m$ throughout the path. Therefore

$$F(\beta_1, \ldots, \beta_r) \leq F(\eta, \ldots, \eta) = 0. \tag{68}$$

Thus (61) holds. Applying the diagonal criterion to $A_i = \cot \alpha_i - c$ proves (64).

For the data (65), the shifted quadratic form equals

$$(m-1)\frac{(m-1)(P+Q)}{m} + (m-1)^2\left(-\frac{P+Q}{m}\right) = 0.$$

Hence the Rayleigh quotient is exactly c, proving sharpness. □

Fix a winding-one partition $\mathcal{Z}_f$ and set

$$\kappa^\star_{f,\mathcal{Z}}(\eta) := \min_{Z\in\mathcal{Z}_f} \kappa^\star_{|Z|}(\eta), \qquad L_f^{\mathrm{graph}} := BB^{\mathsf{T}}.$$

Corollary 29 (Sharp cyclewise Laplacian sandwich)**.** *Suppose $E_f \neq \varnothing$. If $\delta_e(\theta) \geq \eta$ for every crossing edge and $\eta \leq \pi/|Z|$ for every $Z \in \mathcal{Z}_f$, then*

$$\kappa^\star_{f,\mathcal{Z}}(\eta) L_f^{\mathrm{graph}} \preceq H_f(\theta) \preceq \cot(\eta) L_f^{\mathrm{graph}}. \tag{69}$$

If λ_f^+ is the smallest positive eigenvalue of L_f^{graph}, then on any convex set of admissible angles on which every crossing angle is at least η the inverse of the component-shift quotient of G_f satisfies

$$\|[\theta] - [\theta']\|_{\mathrm{quot}} \leq \frac{\|G_f(\theta) - G_f(\theta')\|_2}{\kappa^\star_{f,\mathcal{Z}}(\eta)\lambda_f^+}. \tag{70}$$

Here the target norm is the Euclidean norm in the explicit logarithmic necklace-gradient coordinates, and $\|[x]\|_{\mathrm{quot}}$ is the Euclidean distance from x to the space of component shifts.

Proof. On each winding-one cycle the lifted angles sum to π and the edge differences sum to zero. Sum Theorem 28 over the edge-disjoint partition. The upper bound follows from

$$\cot(\eta) L_f^{\mathrm{graph}} - H_f = B\,\mathrm{diag}(\cot\eta - \cot\delta_e)B^{\mathsf{T}} \succeq 0.$$

For the inverse estimate, choose the minimum-norm representative of $[\theta'] - [\theta]$, integrate the Hessian lower bound along the joining segment, and apply Cauchy–Schwarz. This gives strong monotonicity with constant $\kappa^\star_{f,\mathcal{Z}}(\eta)\lambda_f^+$ and hence (70). □

Remark 30 (Meaning of sharpness)**.** Theorem 28 is sharp for a single cycle of fixed length. The corollary takes the minimum over the cycles in one chosen winding-one partition. It does not claim that this minimum is the optimal coercivity constant of the full Hessian when cycles share vertices or when a different winding-one partition is chosen. The metric statement remains confined to the explicit logarithmic necklace-gradient coordinates; it is not a raw-Plücker or intrinsic-Grassmannian bi-Lipschitz theorem.

5 The homeomorphism

Proof of Theorem 1. Let f be loopless; no fixed-point-free hypothesis is needed, since Propositions 10, 12, 15 and 26 are proved for every loopless f (Remark 7). Readers who prefer the component route may instead invoke Lemma 5, whose commuting square (17) reduces the statement to the fixed-point-free factors; the argument below does not use it.

By (8) the ambient admissible chamber is $\Theta_f^{\mathrm{adm}} = \{0 < \delta_e < \pi\}$, an open convex polyhedron (each δ_e is an *affine* function of θ by (7), with gradient (9)); it is unbounded, being invariant under

the global shift. The gauge slice $\Theta_f^{>0} \subseteq \Theta_f^{\mathrm{adm}}$ (one angle fixed per crossing component) is an open convex polytope of dimension $d_f = n - c_f$: within a component any two labels are joined by a path of crossing edges, and on each winding-one cycle the δ_e are positive and sum to π by (10), which bounds them; the angles of a singleton component are fixed outright by the gauge. Openness is meant in the affine hull, and it is what the inverse-function and invariance-of-domain steps below need: Θ_f^{adm} is open in $\mathbb{R}^n$, being cut out by finitely many strict affine inequalities, so $\Theta_f^{>0} = \Theta_f^{\mathrm{adm}} \cap V$, with $V = \{\theta : \theta_{p_1} = \cdots = \theta_{p_{c_f}} = 0\}$ an affine subspace of dimension $n - c_f$, is open in V; and $\Theta_f^{>0}$ is nonempty, because $\theta_j = \pi j/n$ satisfies every inequality $\theta_p < \theta_q < \theta_p + \pi$ with $p < q$ and component shifts move it into the slice. So V is the affine hull of $\Theta_f^{>0}$ (Galashin records the same fact after [1, eq. (4.3)], identifying $\Theta_f^{>0}$ with the interior of a d_f-dimensional polytope). By Proposition 26, $\ker H_f = \mathrm{Span}\{\mathbf{1}_C\}$ is exactly the gauge directions, so $H_f|_{\Theta_f^{>0}} \succ 0$ everywhere on $\Theta_f^{>0}$: $\Psi_f|_{\Theta_f^{>0}}$ is strictly convex.

Surjectivity. By (1), $\mathrm{Meas}_f(\Theta_f^{>0}) = \mathrm{Meas}_f(\Theta_f^{\mathrm{adm}}) = \mathrm{Crit}_f^{>0}$: every admissible θ is carried into the slice by component shifts, which change neither the δ_e (both labels of a crossing lie in one component of G_f^x, so no crossing joins distinct components and Θ_f^{adm} is shift-invariant) nor the value of Meas_f [1, Prop. 4.2 and the paragraph following it]. Hence $\mathrm{Meas}_f|_{\Theta_f^{>0}}$ is onto $\mathrm{Crit}_f^{>0}$. This is the one place where the theorem uses an input beyond the machinery of §3–§4: the present paper's coordinates give gauge-invariance of G_f, which does not by itself give gauge-invariance of Meas_f, because R_f is not claimed injective.

Injectivity. For $\theta, \theta' \in \Theta_f^{>0}$ with $h = \theta' - \theta$, $h^{\mathsf{T}}(G_f(\theta') - G_f(\theta)) = \int_0^1 h^{\mathsf{T}} H_f(\theta + th) h \, dt > 0$ unless $h = 0$; so $G_f|_{\Theta_f^{>0}}$ is injective. Since $R_f \circ \mathrm{Meas}_f = G_f$, equal measurements have equal G_f, hence equal θ in $\Theta_f^{>0}$: Meas_f is injective on $\Theta_f^{>0}$.

Homeomorphism. $\Psi_f|_{\Theta_f^{>0}}$ strictly convex with nonsingular Hessian makes $G_f|_{\Theta_f^{>0}}$ a local diffeomorphism (inverse function theorem) between the Euclidean spaces $\Theta_f^{>0}$ and $\mathrm{im}(B)$ ($\dim \Theta_f^{>0} = \dim \mathrm{im}(B) = d_f$, and $\mathrm{im}\, H_f = \mathrm{im}(B)$ because $\ker H_f = \ker B^{\mathsf{T}}$); being globally injective, it is a homeomorphism onto its open image, with continuous inverse G_f^{-1}. The invariance-of-domain step is applied *only* between the known Euclidean spaces $\Theta_f^{>0}$ and $\mathrm{im}(B)$, never to the critical image. Now R_f is continuous on $\mathrm{Crit}_f^{>0}$ and $R_f(\mathrm{Crit}_f^{>0}) = \mathrm{im}(G_f|_{\Theta_f^{>0}}) \subseteq \mathrm{dom}(G_f^{-1})$, so

$$\mathrm{Meas}_f^{-1} = G_f^{-1} \circ R_f$$

is a continuous inverse to $\mathrm{Meas}_f|_{\Theta_f^{>0}}$, which is a continuous bijection onto $\mathrm{Crit}_f^{>0}$ by the injectivity and surjectivity steps. Hence $\mathrm{Meas}_f \colon \Theta_f^{>0} \to \mathrm{Crit}_f^{>0}$ is a homeomorphism, and the inverse is the displayed explicit composition.

Figure 19 shows G_f on one plane of the gauge slice at the worked example.

Variational inverse. Write T for the vector space parallel to the gauge slice, and fix $X \in \mathrm{Crit}_f^{>0}$ and $y = R_f(X)$. The preimage $\theta_* \in \Theta_f^{>0}$ just obtained satisfies $G_f(\theta_*) = y$, so it is a stationary point of $F_y(\theta) = \Psi_f(\theta) - \langle y, \theta \rangle$ on the slice. Strict convexity makes it the unique global minimizer. Conversely, at a minimizer the first variation gives $G_f(\theta) - y \in T^{\perp}$. This vector also belongs to $\mathrm{im}\, B$, and $T^{\perp} \cap \mathrm{im}\, B = \{0\}$: a vector normal to the slice is supported on the one fixed label of each component, whereas a vector in $\mathrm{im}\, B$ has sum zero on each component. Thus $G_f(\theta) = y$, proving (2). Existence here uses $y = R_f(X)$ with $X \in \mathrm{Crit}_f^{>0}$; no existence claim for arbitrary $y \in \mathrm{im}\, B$, or convergence guarantee for a numerical minimization algorithm, is needed. By (5), the same minimizer uniquely maximizes $\sum_e \Lambda(\delta_e(\theta)) + \langle y, \theta \rangle$, the total cycle volume plus the linear term.

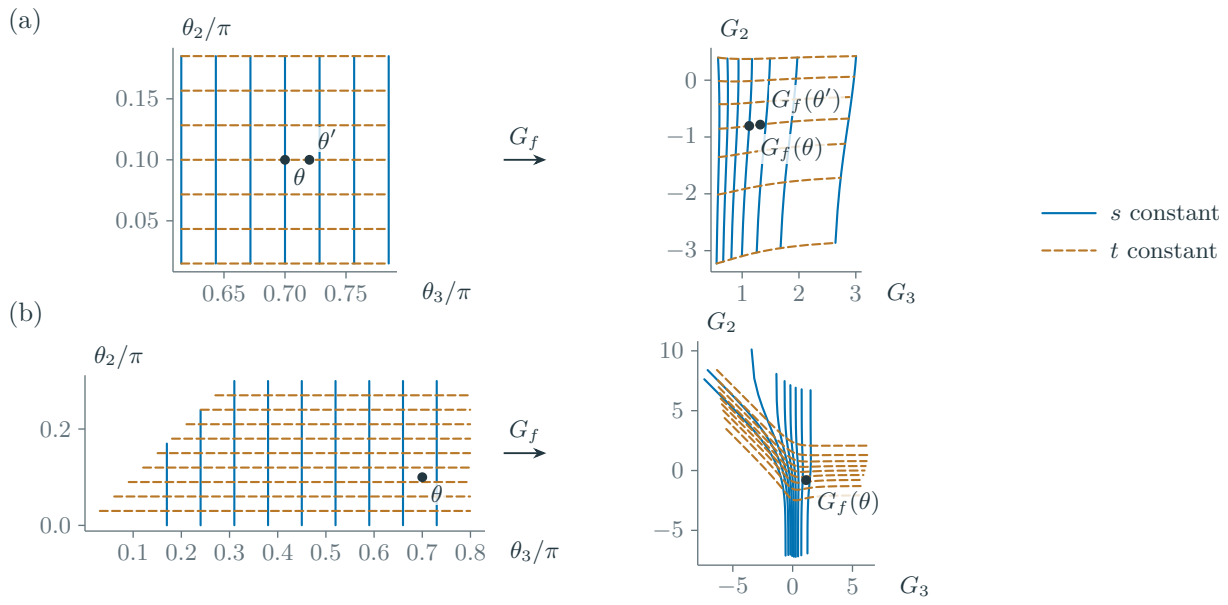


Figure 19: The plane $\theta + \pi(s\, e_3 + t\, e_2)$ of the gauge slice and its image under G_f, drawn in the components (G_3, G_2), which is the gradient of the restriction of Ψ_f to the plane, for $\bar{f} = (3, 5, 1, 2, 4)$ and $\theta/\pi = (0, 0.1, 0.7, 0.3, 0.8)$. (a) A grid near θ and the points θ, $\theta' = \theta + 0.02\pi\, e_3$ with their images; the image grid does not fold. (b) Lines across the whole admissible part of the plane; their images run off where a gap δ_e reaches 0 or π. Numerical values, in floating point.

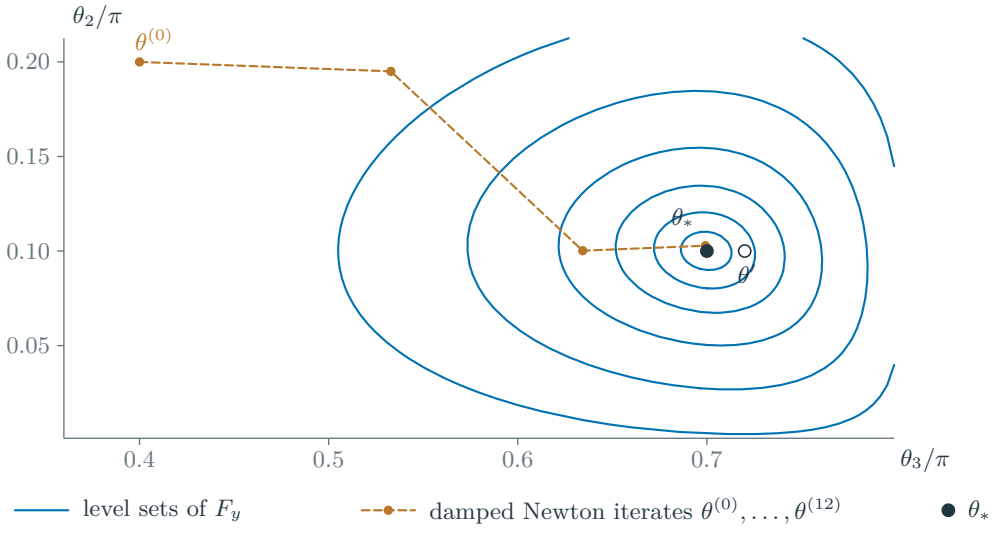


Figure 20: Level sets of $F_y = \Psi_f - \langle y, \cdot \rangle$ on the plane $\theta_* + \pi(s\, e_3 + t\, e_2)$, for $\bar{f} = (3, 5, 1, 2, 4)$, $\theta_*/\pi = (0, 0.1, 0.7, 0.3, 0.8)$ and $y = G_f(\theta_*)$; here $L(u) = \int_{\pi/2}^{u} \log \sin t \, dt$. Marked: the minimizer θ_*, a competitor $\theta' = \theta_* + 0.02\pi\, e_3$ with $F_y(\theta') > F_y(\theta_*)$, and twelve damped Newton steps in the gauge slice from $\theta^{(0)}/\pi = (0, 0.2, 0.4, 0.6, 0.8)$, projected to the plane, as an illustration with no convergence claim. Numerical values, in floating point.

Figure 20 shows the level sets of F_y on one plane of the slice at the worked example. □

Corollary 31 (inverting G_f is minimizing)**.** *For every $y \in G_f(\Theta_f^{>0})$ there is exactly one $\theta \in \Theta_f^{>0}$ with $G_f(\theta) = y$, and it is the unique minimizer of $F_y = \Psi_f - \langle y, \cdot \rangle$ on $\Theta_f^{>0}$.*

Proof. Write $y = G_f(\theta_0)$ with $\theta_0 \in \Theta_f^{>0}$. As in the paragraph *Variational inverse* above, θ_0 is a stationary point of F_y on the gauge slice, so by strict convexity it is the unique minimizer of F_y on $\Theta_f^{>0}$. Any $\theta \in \Theta_f^{>0}$ with $G_f(\theta) = y$ equals θ_0, because G_f is injective on the gauge slice, and any minimizer equals θ_0 by uniqueness. □

The corollary restates the proof of Theorem 1 from the side of G_f: computing the inverse of the gradient at a point of its image is the same as minimizing F_y.

Corollary 32 (the necklace coordinates determine the critical point)**.** *For every loopless f the map*

$$N_f \colon \operatorname{Crit}_f^{>0} \to \mathbb{P}^{n-1}_{>0}, \qquad X \mapsto [\Delta_{I_1}(X) : \cdots : \Delta_{I_n}(X)],$$

is a topological embedding, with inverse $N \mapsto \operatorname{Meas}_f(G_f^{-1}(-P_f D \log N))$ on its image, where log *is taken coordinatewise on any positive representative.*

Proof. The necklace minors are continuous and positive on $\operatorname{Crit}_f^{>0} \subseteq \Pi_f^{>0}$, so N_f is continuous, and $-P_f D \log N_f(X) = R_f(X)$ by (24), independently of the representative. If $N_f(X) = N_f(X')$, then $R_f(X) = R_f(X')$, and $X = X'$ because $\operatorname{Meas}_f^{-1} = G_f^{-1} \circ R_f$ by Theorem 1. On the image of N_f the displayed inverse is the composition of continuous maps $-P_f D \log$, G_f^{-1} (continuous on the open image of the slice, as in the proof of Theorem 1) and Meas_f. □

Corollary 33 (Real-analytic critical coordinates)**.** *For every loopless bounded affine permutation f, including those with coloops, the critical cell $\operatorname{Crit}_f^{>0}$ is a real-analytic embedded submanifold of $\Pi_f^\circ(\mathbb{R})$, and hence of $\operatorname{Gr}(k,n)(\mathbb{R})$, of dimension $d_f = n - c_f$. The measurement map*

$$\operatorname{Meas}_f : \Theta_f^{>0} \longrightarrow \operatorname{Crit}_f^{>0}$$

is a real-analytic diffeomorphism, with inverse $G_f^{-1} \circ R_f$. Its differential is injective on the tangent space $T = \{h : h_{p_1} = \cdots = h_{p_{c_f}} = 0\}$ at every point of $\Theta_f^{>0}$. The ordinary necklace map N_f of Corollary 32 is also a real-analytic embedding into $\mathbb{P}^{n-1}_{>0}$. All these assertions include admissible collisions of noncrossing labels.

Proof. We use the standard real-analytic structure on the real Grassmannian: on the chart where a k-column minor is nonzero, a plane is represented uniquely by a matrix whose corresponding columns form the identity matrix, and the remaining entries are coordinates. Its transition maps are rational with nonzero denominators. The open positroid variety is a smooth locally closed algebraic subvariety defined over $\mathbb{R}$ [6, Thm. 5.9]; its real locus has the induced real-analytic submanifold structure. We do not assert smoothness of its closure or of the complex critical variety.

The gauge slice is open in T, and has dimension d_f, as in the proof of Theorem 1. The analytic map $\theta \mapsto (e^{i\theta_p})_p$ of (14) takes the whole slice into Θ_f° of (13). Composing with the regular map of [1, Prop. 5.1] is real-analytic, and [1, Lem. 5.13] and (15) identify this composition with Meas_f. Its values are real. In the Grassmannian charts just described, its coordinate functions are therefore real-valued real-analytic functions, giving analyticity into the real Grassmannian. Since the values lie in the smooth real locus of Π_f°, the same assertion holds with that target.

The left twist is regular on Π_f° [1, §5.4], and its necklace minors do not vanish there [1, eq. (2.4)]. After fixing projective scale locally, $\log|\Delta_{I_r}(\widetilde{\tau}(X))|$ is real-analytic on the real locus. Applying $P_f D$ removes the common scale. This gives a real-analytic $\mathbb{R}^n$-valued extension of R_f to a neighbourhood of $\mathrm{Crit}_f^{>0}$. Equivalently, use the ordinary-minor formula of Remark 13. Only its restriction to $\mathrm{Crit}_f^{>0}$ needs to take values in $W := \mathrm{im}(B)$.

By Proposition 12, $G_f = B \log\sin\delta$ is real-analytic on the admissible chamber, and the identity $R_f \circ \mathrm{Meas}_f = G_f$ holds on the whole slice. Thus, for $h \in T$,

$$dR_f|_{\mathrm{Meas}_f(\theta)}\, d\,\mathrm{Meas}_f\,|_\theta(h) = H_f(\theta)h.$$

Here the differential of R_f can be taken using the ambient extension. Proposition 26 says that the kernel of H_f consists exactly of component shifts. Its intersection with T is zero, so $d\,\mathrm{Meas}_f\,|_\theta$ is injective on T. Theorem 1 gives that Meas_f is a homeomorphism onto its image with the subspace topology. The analytic immersion theorem now makes this image an embedded real-analytic submanifold of dimension d_f.

For the inverse, $H_f(\theta)$ has image W, and $H_f(\theta)|_T : T \to W$ is an isomorphism: it is injective and both spaces have dimension d_f. This is an isomorphism between two possibly different subspaces of $\mathbb{R}^n$, not an assertion that H_f preserves T. The analytic inverse function theorem therefore applies to $G_f : \Theta_f^{>0} \to W$. Its injectivity, established in the proof of Theorem 1, makes the local analytic inverses agree on overlaps. Consequently G_f^{-1} is analytic on the open set $G_f(\Theta_f^{>0}) \subset W$. The restriction of R_f to the embedded critical cell is analytic as a W-valued map: one may compose its ambient extension with any linear projection $\mathbb{R}^n \to W$. Its values belong to $G_f(\Theta_f^{>0})$, so $\mathrm{Meas}_f^{-1} = G_f^{-1} \circ R_f$ is real-analytic.

Finally, N_f is analytic into positive projective space. The well-defined analytic map $\mathcal{R}([z]) = -P_f D \log z$, for any positive representative z, satisfies $\mathcal{R} \circ N_f \circ \mathrm{Meas}_f = G_f$. Differentiating shows that $N_f \circ \mathrm{Meas}_f$ has injective differential, and therefore so does N_f on $\mathrm{Crit}_f^{>0}$. Its topological embedding is Corollary 32, so it is a real-analytic embedding. Its inverse on its image is the analytic map $N \mapsto \mathrm{Meas}_f(G_f^{-1}(-P_f D \log N))$.

There is no deleted collision locus in this argument. Equation (5.2) of [1] excludes $t_p = \pm t_q$ only for crossing pairs; [1, Prop. 5.1 and Lem. 5.13] apply at every admissible specialization. The last paragraph of Proposition 12 establishes the identity there directly. Noncrossing collisions are interior points of the slice; crossing collisions are inadmissible and are outside the statement. If $d_f = 0$, the slice and critical cell are single points, and the conclusions have their usual zero-dimensional meaning. □

Remark 34 (Independent measurement controls)**.** Two finite numerical cross-checks of $R_f \circ \mathrm{Meas}_f = B\ell$ — one against Galashin's independent top-cell reconstruction [1, §9], and one on every non-top loopless cell with $2 \le n \le 7$ — are recorded in Appendix A (*Finite checks*). They are not premises.

6 The all-rank critical Gr–OG shift

For $f = (3, 5, 6, 7, 9)$, Figure 21 shows f beside its shift $f^\downarrow(p) = f(p-1)$.

Proof of Corollary 2. Galashin observes that the reduced strand diagram of a loopless f is the dual reduced strand diagram of $f^\downarrow$; hence the f-admissible and dual $f^\downarrow$-admissible angle chambers are identified by $\theta \mapsto \theta$ [1, §8.4]. Proposition 8.9 of [1] says that, if Conjecture 4.3 holds for f, the dashed arrow in its Figure 24(b) is the composition of the three homeomorphisms

$$\mathrm{Crit}_f^{>0} \xrightarrow{\mathrm{Meas}_f^{-1}} \Theta_f^{>0} \xrightarrow{\mathrm{id}} \widehat{\Theta}_{f^\downarrow}^{>0} \xrightarrow{\widehat{\mathrm{Meas}}_{f^\downarrow}} \widehat{\mathrm{Crit}}_{f^\downarrow}^{>0}.$$

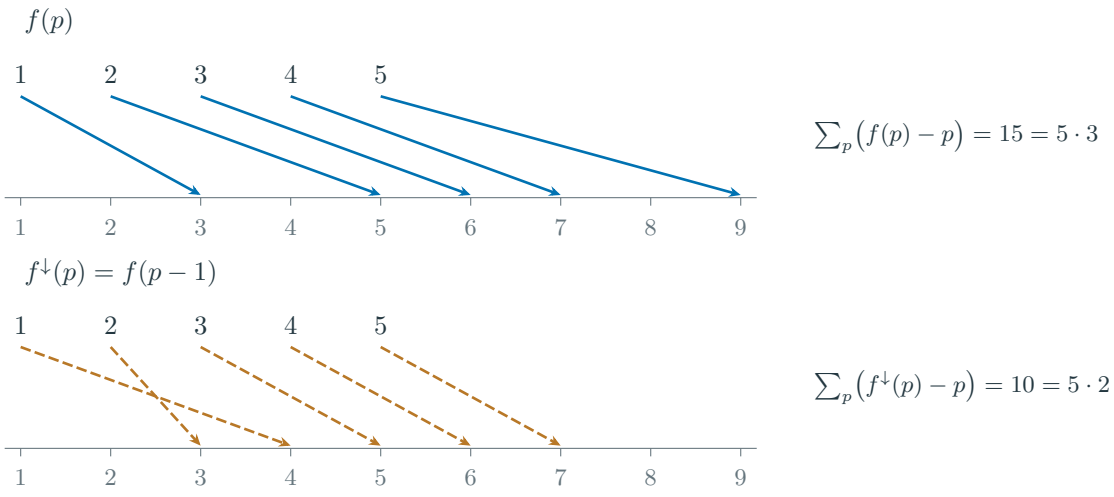


Figure 21: The shift of the bounded affine permutation $f = (3, 5, 6, 7, 9)$, $\bar{f} = (3, 5, 1, 2, 4)$, $n = 5$. Top: $p \mapsto f(p)$. Bottom: $p \mapsto f^{\downarrow}(p) = f(p-1)$, where $f(0) = f(5) - 5 = 4$. The displacement sums are $15 = 5 \cdot 3$ and $10 = 5 \cdot 2$, so the rank drops from 3 to 2. Evaluated by the formal definitions of Appendix A.

Theorem 1 supplies the hypothesis for every loopless f and gives $\mathrm{Meas}_f^{-1} = G_f^{-1} \circ R_f$, proving the first assertion and its displayed constructive formula. For $k = 1$ the statement is degenerate but true: $f^{\downarrow} \in \mathcal{B}(0, n)$ and $\mathrm{Gr}(0, n)$ is a point, so both sides are points and Φ_f is the unique map between them.

Now let τ be a matching on $[2N]$, and put $f = f_\tau^{\mathrm{elec}}$ and $g = f_\tau^{\mathrm{Ising}}$. The residue of $f^{\downarrow}(p) = f(p-1)$ is $\tau(p)$ [1, §7]. Moreover,

$$p \le f^{\downarrow}(p) = f(p-1) \le p + 2N - 1,$$

so the shift is coloopless; it is also loopless because $\tau(p) \neq p$. Uniqueness of the loopless bounded lift gives $f^{\downarrow} = g$.

We use the regular dual measurement of [1, Remark 5.3]: raw graph coordinates from (3.6) may have a common vanishing factor. This map is invariant under translations on dual crossing components. Indeed, put $h = \hat{g}$ and, for a dual g-admissible tuple φ, set $\widetilde{\psi} = \widetilde{\varphi} \circ h^{-1}$, using affine angle extensions. Proposition 3.4 of [1] gives

$$\widehat{\mathrm{Meas}}_g(\varphi) = \mathrm{alt}^{\perp}\big(\mathrm{Meas}_h(\psi)\big).$$

The relabeling in that proposition sends a dual g-crossing component C to the ordinary h-crossing component $\bar{h}(C)$. Translating φ on C therefore translates ψ on $\bar{h}(C)$, which preserves Meas_h by [1, Proposition 4.2]. The identity holds for the regular maps also where a raw graph vector vanishes.

Let θ be τ-admissible and let θ^0 be its normalization in the chosen gauge slice $\Theta_f^{>0}$. Proposition 7.2 of [1] makes θ both f- and g-admissible, and §8.4 identifies the ordinary f-crossing components with the dual g-crossing components. Thus ordinary gauge invariance and the dual invariance just established give

$$\Phi_f\big(\mathrm{Meas}_f(\theta)\big) = \widehat{\mathrm{Meas}}_g(\theta^0) = \widehat{\mathrm{Meas}}_g(\theta) = \mathrm{Meas}_g(\theta).$$

For the last equality, a boundary edge at b_p of a black-boundary representative of g has strand label $\{p, \tau(p)\}$. Its dual weight is $\sin(\pi/2) = 1$ by τ-isotropy, so (3.6) and (1.2) agree on every edge.

Theorem 7.3 of [1] identifies the two ordinary measurements with the displayed embedded response and correlation matrices. Consequently the restriction is to the image of the τ-admissible tuples in $\mathrm{Crit}_f^{>0}$; the gauge representative θ^0 need not itself be isotropic. □

Remark 35 (what Corollary 2 is not)**.** Corollary 2 is a statement about *critical* loci only. It is not an answer to Question 9.2 of [3], which asks for a homeomorphism between the full electrical and Ising spaces: for the top matching with $N \geq 3$ the open electrical network cell has dimension $\binom{N}{2}$, while the connected critical family has dimension $N-1$, and Proposition 8.9 of [1] supplies nothing in the missing noncritical directions.

A note on process

Learned Response Labs is a research collaboration between the author and AI systems. The proofs, text, software, atlases, and other artifacts of this program were primarily generated by AI coding harnesses (OpenAI Codex and Anthropic Claude Code) operating adversarially: results produced by one system were frozen, independently rederived, and hostilely audited by the other before internal acceptance, with machine-checked Lean 4/Mathlib components where noted. The author designed and directed the research program — the choice of problems, the harness and adversarial-audit architecture, and the cross-domain references that shaped it.

Before release, each paper is reviewed internally in a *reading edition*, which sets every formal statement beside what checks it (a frozen Lean statement or a pinned exact certificate) and every step without one beside its source; a release goes out only when every item has been confirmed to say what the paper says. External review is pending, and reviews received will be noted with the paper's public reading edition.

The decision to circulate this document, and the judgment that the process behind it makes the results worth a referee's time, are the author's; the mathematics itself awaits external verification, which is exactly what this document requests. Accordingly, the headline results are offered as *proof candidates pending external review*, and no claim in this document should be treated as established until that review has occurred. This external-review preprint is prepared for public circulation; it is not yet an arXiv version.

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A Formal statements and their geometric interface

The formal source for this paper is written in Lean `4.33.0-rc1` with Mathlib; the exact Mathlib revision is recorded in its dependency lockfile, `lake-manifest.json`. The coloop-inclusive source supplement (*public source-supplement DOI pending*) includes a source manifest, that lockfile, the exact declarations and their proof modules. Every theorem named in this appendix, except the auxiliary lemmas marked as such, is also stated, without its proof, in one of fifteen frozen statement files kept apart from the proofs: thirteen for the `Conj43.` statements, and two for the `TP.` statements on loops and on decorated permutations, which come from a companion formal library for totally positive combinatorics and are included in the supplement. The proofs are checked against those frozen statements by the Comparator tool for Lean, which compares each proved statement with its frozen copy, replays the proofs in the Lean kernel, and admits only the axioms `propext`, `Classical.choice` and `Quot.sound`. The displays below render the formal statements in mathematical notation. Lean does not construct the measurement map: the conditional theorem at the end of this appendix takes the properties of the measurement map that the proof of Theorem 1 uses as hypotheses, and the paragraph after it says what each hypothesis stands for.

Domain and crossing convention. `Conj43.LooplessBAP n` contains a permutation σ of $\{0,\dots,n-1\}$. For $n>0$, its bounded lift is $f(i)=i+1+((\sigma(\bar{i})-i-1) \bmod n)$, with the periodic extension understood. Thus a fixed point of σ represents a coloop $f(i)=i+n$; it never represents a loop. Add one to the Lean labels to obtain the paper's labels. Writing $s_p=\sigma^{-1}(p)$, the resolved circle has modulus $5n$, entry position $5s_p+4$ and exit position $5p+1$. The predicate `Conj43.CrossEdge F p q` requires strict clockwise order

$$(5s_p+4,\ 5s_q+4,\ 5p+1,\ 5q+1) \pmod{5n}.$$

Here clockwise means that the four positive cyclic increments sum to $5n$. This is the endpoint order s_p^+, s_q^+, p^-, q^- used in §2, including the exit-before-entry convention at a shared label (Lemma 9(b)). The two Mis classes are read from the same positions, clockwise and counterclockwise. The crossing list enumerates this predicate on all ordered pairs, without duplicates. The first necklace term is $\{x : x \le \sigma^{-1}(x)\}$; subsequent terms remove the current label and insert its image. In particular a coloop belongs to every necklace term. Lean row r is the paper's row I_{r+1}.

The interface with Galashin's definitions. Lean also transcribes the definitions of [1], in his labels $1,\dots,n$: loopless bounded affine permutations, the affine necklace formula $I_r=\{f(a) \bmod n : a<r\le f(a)\}$, the refined boundary with its exit and entry points x^-, x^+ [1, Def. 1.5], affine f-crossings and alignments, the classes $\mathrm{Mis}^{\uparrow\downarrow}$ and $\mathrm{Mis}^{\downarrow\uparrow}$ [1, §5.4], and the exponent of (22). The affine f-crossing is transcribed with $t=f^{-1}(q)$; the definition before [1, eq. (3.2)] prints $t:=f^{-1}(p)$, repeating the first argument. Against these transcriptions:

- the bounded lift, with labels shifted by one, is a loopless bounded affine permutation of rank between 1 and n whose residues are σ shifted by one (`Conj43.galashin_isLooplessBAP`, `Conj43.galashin_rank_range`, `Conj43.galashin_residues`); the encoding is injective and onto the loopless bounded affine permutations of every rank (`Conj43.galashin_injective`, `Conj43.galashin_surjective`), never has a loop (`Conj43.galashin_no_loops`), and makes a fixed point of σ exactly a coloop (`Conj43.galashin_coloop`);
- Lean's necklace row r is his I_{r+1} (`Conj43.galashin_necklace`);
- the positions $5s_p+4$ and $5p+1$ order any four refined points as his boundary does (`Conj43.encode_clockwise4`), parts (a)–(c) of Lemma 9 hold on his refined boundary (`Conj43.halfopen_a`, `Conj43.halfopen_b`, `Conj43.halfopen_L1`, `Conj43.halfopen_L2`, `Conj43.halfopen_L3`), and so does (21) (`Conj43.necklace_iff_refinedArc`);
- `Conj43.CrossEdge F p q` holds exactly when representatives of $p+1$ and $q+1$ form his affine f-crossing (`Conj43.crossEdge_iff_affine`); two distinct labels are aligned in his sense exactly when they are in no crossing and in neither Mis class (`Conj43.alignment_iff_affine`); the two Mis predicates are his classes (`Conj43.misUD_iff_galashin`, `Conj43.misDU_iff_galashin`); two strands cross in some direction exactly when their chords interleave on the circle (`Conj43.crossEdge_iff_interleave`); and no label crosses itself, while two distinct labels have four distinct refined endpoints (`Conj43.crossEdge_irrefl`, `Conj43.chord_endpoints_distinct`);
- `Conj43.pairExponent` is the exponent of the pair in (22) on the face label that his necklace formula gives (`Conj43.pairExponent_eq_galashin`).

Cotangent inequality. `Conj43.cot_cycle_inequality` states, for $m \geq 3$ and $\alpha, d : \{0, \dots, m-1\} \to \mathbb{R}$,

$$(\forall i,\ \alpha_i > 0)\,, \quad \sum_i \alpha_i = \pi, \quad \sum_i d_i = 0 \quad \Longrightarrow \quad \begin{cases} \sum_i \cot(\alpha_i) d_i^2 \geq 0, \\ \sum_i \cot(\alpha_i) d_i^2 = 0 \iff \forall i,\ d_i = 0. \end{cases}$$

Row difference. `Conj43.rowdiff` states, for $n > 0$ and F : `LooplessBAP n`,

$$\texttt{Pmat}\ F\ \texttt{Dmat}\ n\ \texttt{Mmat}\ F = \texttt{Bmat}\ F,$$

an identity of integer matrices whose columns are the directed crossing edges, with $B_{x,(p,q)} = \mathbf{1}_{x=q} - \mathbf{1}_{x=p}$, $(Dz)_r = z_{r+1} - z_r$ and $(Pz)_x = z_{s_x}$. The full form is stated separately, with columns indexed, as in §2.3, by the unordered pairs $p < q$. There $(M_N)_{r,\{p,q\}}$ is the exponent of the pair on necklace row r read from (22) (`Conj43.pairExponent`: $+1$ for a $\mathrm{Mis}^{\uparrow\downarrow}$ pair on a row containing both labels, $+1$ for a $\mathrm{Mis}^{\downarrow\uparrow}$ pair on a row containing neither, -1 for a crossing $p \to q$ on a row containing p but not q, and 0 otherwise), and the column of B is $e_q - e_p$ if $p \to q$, $e_p - e_q$ if $q \to p$, and 0 if the pair carries no crossing. For such a pair, `Conj43.pairExponent_nonCross` states that its exponent is 0 on every necklace row and `Conj43.full_pair_columns_zero` that its column vanishes in M_N, in $P_f D M_N$ and in B; `Conj43.rowdiff_full_pairs` states

$$P_f\, D\, M_N = B$$

over all unordered pairs, which is Proposition 10 as stated. The coloop facts of its proof (a coloop crosses nothing, lies in every necklace row, and has zero rows in B and in $P_f D M_N$) are `Conj43.not_crossEdge_of_coloop_fst`, `Conj43.coloop_mem_necklace`, `Conj43.Bmat_coloop_row` and `Conj43.PDM_coloop_row`. In the encoding of the previous paragraph, part (c) of Lemma 9 for labels that are not coloops is proved as the auxiliary Lean lemmas `Conj43.arc_L1`, `Conj43.arc_L2` and `Conj43.arc_L3`, and equation (21) by induction along the necklace recursion. The proof of `Conj43.rowdiff` uses `Conj43.arc_L3` and equation (21) for the crossing columns and `Conj43.inArcN_indicator_diff`, the boundary of a half-open arc, for their row differences; the vanishing of the Mis columns uses `Conj43.arc_L1` and `Conj43.arc_L2`. None of these auxiliary lemmas is among the frozen statements.

Existence of a winding-one partition. `Conj43.winding_one_partition` states, for F : `LooplessBAP n`, that there is a list of vertex lists $\mathcal{C}$ such that

$$\forall c \in \mathcal{C}, \quad |c| \geq 3, \quad c \text{ has no repeated vertex}, \quad \mathrm{winding}(c) = n,$$
$$\operatorname*{concat}_{c \in \mathcal{C}} \mathrm{cycleEdges}(c) \text{ is a permutation of } \mathrm{crossEdges}(F).$$

Here $\mathrm{cycleEdges}(c)$ pairs consecutive entries, including the last with the first, and winding is the sum of $(q - p) \bmod n$ over those edges. The permutation assertion gives both edge exhaustion and disjointness. The auxiliary theorem `Conj43.crossEdges_subset_cover` supplies the cover inclusion used in its proof. `Conj43.partition_card` states that every such list $\mathcal{C}$ has as many entries as there are wrapped crossing edges (p, q), those with $q < p$, which is the second form of (25); it rests on `Conj43.winding_eq_wrapped`: a vertex list with no repeated vertex winds n times its number of wrapped edges. For the first form, `Conj43.partition_advance` states that $n|\mathcal{C}|$ is the total advance $\sum_{(p,q) \in E_f} \mathrm{adv}(p, q)$ of the crossing edges, and `Conj43.crossEdges_advance` that this total is n times the number of wrapped edges. For a decorated permutation, loops allowed, `TP.decCrossEdge_iff` and `TP.decCrossEdges_eq` state that its crossing relation and its crossing edges are those of the loopless lift of its underlying permutation, and `TP.dec_winding_one_partition` gives it the same partition.

Notation for the analytic statements. Write E_f for the directed crossing edges, $\delta_{pq}(\theta) = \theta_q - \theta_p + \pi\,\mathbf{1}[q<p]$ for $(p,q)\in E_f$ (equation (7)), $\Theta_f^{\mathrm{adm}} = \{\theta\in\mathbb{R}^n : 0<\delta_e(\theta)<\pi \text{ for all } e\in E_f\}$, B the real incidence matrix with a column for each $e\in E_f$, $\ell_e = \log\sin\delta_e$, $G = B\ell$,

$$Q_\theta(h) = \sum_{(p,q)\in E_f} \cot\delta_{pq}(\theta)\,(h_q-h_p)^2, \qquad \Psi(\theta) = \sum_{e\in E_f}\int_{\pi/2}^{\delta_e(\theta)} \log\sin t\,dt;$$

this Ψ satisfies (23) and differs from any other antiderivative choice, such as the normalization $L(0)=0$ of §1, by a constant. Two labels lie in the same component when a path of crossing edges, taken in either direction, joins them. For $a\in\mathbb{R}^n$ the formal gauge slice is

$$S_a = \{\theta\in\Theta_f^{\mathrm{adm}} :\ \theta_p = a_p \text{ whenever } p \text{ is the least label of its component}\}.$$

With $a=0$ this is Galashin's slice $\Theta_f^{>0}$ [1, eq. (4.3)] for the representatives $p_C = \min C$; another choice of representatives gives a slice carried onto this one by component shifts, which change neither Meas_f nor G_f.

Cycle sums, the assembled Hessian and the hinge. For a vertex list c of length at least 3 with $\mathrm{winding}(c) = n$, `Conj43.cycle_wrap_sum` states that exactly one edge of c is wrapped, and, if c has no repeated vertex, `Conj43.delta_cycle_sum` states $\sum_{e\in c}\delta_e(\theta) = \pi$ for every $\theta\in\mathbb{R}^n$, which is (10). `Conj43.hessian_matrix_form` states $h^{\mathsf T}B\,\mathrm{diag}(\cot\delta_e)B^{\mathsf T}h = Q_\theta(h)$, and for $\theta\in\Theta_f^{\mathrm{adm}}$, `Conj43.hessian_psd`, `Conj43.hessian_kernel_components` and `Conj43.hessian_matrix_kernel` state

$$Q_\theta(h)\ge 0, \qquad Q_\theta(h)=0 \iff B\,\mathrm{diag}(\cot\delta_e)B^{\mathsf T}h = 0 \iff h \text{ is constant on every component},$$

which is Proposition 26. For $\theta\in\Theta_f^{\mathrm{adm}}$, $(p,q)\in E_f$ and $t_j = e^{i\theta_j}$, `Conj43.hinge_identity` states $|[\![t_q,t_p]\!]_+| = 2\sin\delta_{pq}(\theta)$, with the bracket of (12), which is Lemma 3.

The potential, its gradient, and convexity on the slice. `Conj43.logSinPrimitive_derivative` states $\frac{d}{du}\int_{\pi/2}^{u}\log\sin t\,dt = \log\sin u$ for $0<u<\pi$, and `Conj43.admissible_isOpen` that Θ_f^{adm} is open in $\mathbb{R}^n$. At every $\theta\in\Theta_f^{\mathrm{adm}}$, `Conj43.potential_gradient` states that Ψ is differentiable with gradient $G(\theta)$, and `Conj43.gradient_hessian` that G is differentiable with derivative $B\,\mathrm{diag}(\cot\delta_e(\theta))B^{\mathsf T}$, both also as strict derivatives; this is the calculus of Proposition 12. `Conj43.incidence_degree_balance` states $B\mathbf{1}=0$, `Conj43.gradient_edge_constant` states $B(\ell+c\mathbf{1}) = B\ell$, and `Conj43.gradient_component_shifts` states $G(\theta+s)=G(\theta)$ for every s constant on components. For $\theta\in\Theta_f^{\mathrm{adm}}$ and $h\neq 0$ vanishing at the least label of every component, `Conj43.hessian_gauge_positive` states $Q_\theta(h)>0$. For every $a\in\mathbb{R}^n$: Ψ is strictly convex on S_a (`Conj43.potential_slice_strictConvex`); $\langle G(y)-G(x),y-x\rangle>0$ for distinct $x,y\in S_a$ (`Conj43.gradient_slice_strictMonotone`), so G is injective on S_a (`Conj43.gradient_slice_injective`); $G|_{S_a}$ is a topological embedding into $\mathbb{R}^n$ (`Conj43.gradient_slice_embedding`) and an open embedding into $\mathrm{im}\,B$ with its subspace topology (`Conj43.gradient_incidence_openEmbedding`); each $x\in S_a$ is the unique minimizer on S_a of $z\mapsto\Psi(z)-\langle G(x),z\rangle$ (`Conj43.potential_variational_inverse`); and for every $y\in G(S_a)$ the function $z\mapsto\Psi(z)-\langle y,z\rangle$ has exactly one minimizer on S_a (`Conj43.variational_image_minimizer`); and that minimizer is the gradient preimage: y has exactly one preimage in S_a, and it is the minimizer (`Conj43.variational_gradient_preimage`, a corollary of the two; with $a=0$ it is Corollary 31). These are the convexity, injectivity, homeomorphism and variational steps of the proof of Theorem 1. An earlier composition theorem, `Conj43.measurement_conditional_homeomorphism`, assumes the assembled identity $R\circ\mathrm{Meas} = G$; the theorem below derives that identity instead.

Necklace logarithms. Put $\lambda_{pq}(\theta) = \log|[\![t_q, t_p]\!]_+|$ for $p < q$, $t_j = e^{i\theta_j}$ (Lean's logarithm is 0 at 0; every pair whose bracket can vanish on Θ_f^{adm} has coefficient 0 below). For every $\theta \in \mathbb{R}^n$, `Conj43.necklace_log_crossing` states that $M_N\lambda(\theta)$, summed over all unordered pairs, equals the same sum over the crossing edges alone, so the Mis columns vanish on necklace rows; `Conj43.necklace_recovery_constant` states $P_f D(z + c\mathbf{1}) = P_f D z$. For $\theta \in \Theta_f^{\mathrm{adm}}$, `Conj43.necklace_log_hinge` states $\lambda_e(\theta) = \ell_e(\theta) + \log 2$ on crossing edges, and `Conj43.necklace_log_gradient` states $P_f D\, M_N\lambda(\theta) = G(\theta)$. This is the computation in the proof of Proposition 12, from the logarithm of (22) to G_f.

The conditional theorem. `Conj43.gal512_measurement_homeomorphism` states the following. Let $n \geq 1$ and $a \in \mathbb{R}^n$, let X be a topological space and $\mathrm{Crit} \subseteq X$, and let $\mathrm{Meas}\colon S_a \to \mathrm{Crit}$ and $N\colon \mathrm{Crit} \to \mathbb{R}^n$ be maps such that

(H_1) Meas is continuous; (H_2) Meas is surjective; (H_3) N is continuous;

(H_4) for every $\theta \in S_a$ there is $c(\theta) \in \mathbb{R}$ such that, for every row r,

$$N(\mathrm{Meas}\,\theta)_r = \sum_{1 \leq p < q \leq n} (M_N)_{r,\{p,q\}}\, \lambda_{pq}(\theta) + c(\theta).$$

Put $R = P_f D \circ N$. Then $G|_{S_a}$ is a homeomorphism onto $G(S_a)$, Meas is a homeomorphism $S_a \to \mathrm{Crit}$, R maps Crit into $G(S_a)$, and

$$\mathrm{Meas}^{-1} = \left(G|_{S_a}\right)^{-1} \circ R.$$

On the way, `Conj43.gal512_gradient` derives $R \circ \mathrm{Meas} = G$ on S_a from (H_4) alone, and `Conj43.gal512_recovery_incidence` derives $R(\mathrm{Crit}) \subseteq \mathrm{im}\, B$ from (H_2) and (H_4). `Conj43.gauge_slice_nonempty` states $S_a \neq \varnothing$ for every a, and `Conj43.gal512_hypotheses_nonvacuous` exhibits, for every f and a, a nonempty Crit with maps satisfying (H_1)–(H_4) (the image of the logarithmic necklace map, with N the inclusion). So the hypotheses are consistent; that Galashin's objects satisfy them is what they stand for, not something Lean shows.

What the hypotheses stand for. Take $X = \mathrm{Gr}(k, n)$, $\mathrm{Crit} = \mathrm{Crit}_f^{>0}$, $a = 0$, so that $S_0 = \Theta_f^{>0}$, $\mathrm{Meas} = \mathrm{Meas}_f|_{\Theta_f^{>0}}$, and $N(X) = \left(\log|\Delta_{I_r}(\widetilde{\tau}(X))| - \log|\Delta_{I_1}(\widetilde{\tau}(X))|\right)_r$; dividing by the first necklace coordinate is one continuous choice of the projective scale, which D removes, so $R = P_f D \circ N$ is the map R_f of Proposition 12.

- (H_1), continuity of the measurement map on the slice: [1, Prop. 5.1] makes Meas_f a regular map on the complex slice Θ_f° of (13), and by [1, Lem. 5.13] its value at $t = e^{i\theta}$, $\theta \in \Theta_f^{>0}$, is $\mathrm{Meas}_f(\theta)$.
- (H_2), surjectivity onto the critical cell: equation (1), from [1, Prop. 4.2 and the paragraph following it] and the sentence after [1, eq. (4.3)].
- (H_3), continuity of the logarithmic twisted-necklace coordinates on $\mathrm{Crit}_f^{>0}$: the left twist is a regular automorphism of Π_f° ([4]; [1, §5.4]); $\mathrm{Crit}_f^{>0} \subseteq \Pi_f^{>0} \subseteq \Pi_f^\circ$, because every interior edge weight is positive at an admissible tuple [1, Prop. 4.2, Def. 1.8 and eq. (2.6)]; and the necklace Plücker coordinates do not vanish on Π_f° [1, eq. (2.4)].
- (H_4), Proposition 5.12 of [1]: equation (22) at the necklace face labels $I = I_r$, at $t = e^{i\theta}$ for θ in the slice (so $t \in \Theta_f^\circ$), after taking $\log|\cdot|$, the common scalar becoming $c(\theta)$. It is assumed on

the whole slice, including admissible collisions of noncrossing labels; the proof of Proposition 12 explains why the identity holds there. The exponent table M_N is the transcription of (22) in the proof of Proposition 10.

The sharp modulus and the shift. `Conj43.kappaStar` is $\kappa^\star_m(\eta)$ of (59). For $m \geq 3$ and $0 < \eta \leq \pi/m$, `Conj43.kappaStar_pos` states $\kappa^\star_m(\eta) > 0$; for $0 < \eta < \pi/m$, `Conj43.kappaStar_lt_tan` states $\kappa^\star_m(\eta) < \tan((m-2)\eta/2)$, as in Lemma 27; and `Conj43.sharp_cot_cycle` states, for $\alpha, d : \{0, \dots, m-1\} \to \mathbb{R}$,

$$(\forall i,\ \alpha_i \geq \eta)\,, \quad \sum_i \alpha_i = \pi, \quad \sum_i d_i = 0 \quad \Longrightarrow \quad \sum_i \cot(\alpha_i) d_i^2 \geq \kappa^\star_m(\eta) \sum_i d_i^2,$$

which is Theorem 28; `Conj43.sharp_cot_extremizer` states that the pair (65) satisfies these hypotheses with equality. For a list $\mathcal{C}$ as in the partition paragraph and θ with $\delta_e(\theta) \geq \eta > 0$ on every crossing edge and $\eta \leq \pi/|c|$ for every $c \in \mathcal{C}$, `Conj43.sharp_laplacian_sandwich` states (69) as inequalities of quadratic forms, with $L_f^{\mathrm{graph}} = BB^{\mathsf{T}}$. Under the same hypotheses on $\mathcal{C}$ and η, let C be a convex set on which every crossing angle is at least η, and let $\lambda > 0$ be the least positive element of the spectrum of L_f^{graph}. Write $\|\cdot\|_2$ for the Euclidean norm and $\|[x]\|_{\mathrm{quot}}$ for the infimum of $\|x - c\|_2$ over the component shifts c. Then for $\theta, \theta' \in C$, `Conj43.sharp_strong_monotone` states $\kappa^\star_{f,\mathcal{Z}}(\eta)\lambda \|[\theta - \theta']\|^2_{\mathrm{quot}} \leq \langle G_f(\theta) - G_f(\theta'), \theta - \theta' \rangle$, and `Conj43.sharp_inverse_lipschitz` states (70) with $\lambda_f^+ = \lambda$. For the shift, let f be the affine lift of F : `LooplessBAP n` and k its rank. `Conj43.rank_sum` states $\sum_{i<n}(f(i) - i) = kn$. With $f^\downarrow(p) = f(p-1)$, `Conj43.shift_bounds`, `Conj43.shift_periodic`, `Conj43.shift_bijective` and `Conj43.shift_rank` state that $p \leq f^\downarrow(p) \leq p + n$ (for $n > 0$), that $f^\downarrow(p+n) = f^\downarrow(p) + n$, that $f^\downarrow$ is a bijection of $\mathbb{Z}$, and that $\sum_{i<n}(f^\downarrow(i) - i) = (k-1)n$. `Conj43.shift_self_le_iff` states that for $g : \mathbb{Z} \to \mathbb{Z}$ with $g \geq \mathrm{id}$, $g(p-1) \geq p$ for every p exactly when g has no fixed point. For a matching τ on $2N \geq 2$ labels, `Conj43.elec_shift_eq_ising` states $(f_\tau^{\mathrm{elec}})^\downarrow = f_\tau^{\mathrm{Ising}}$, the two permutations having residues $p \mapsto \tau(p+1)$ and $p \mapsto \tau(p)$ as in §6.

What is not formalized. Lean does not construct the critical measurement map, Galashin's curve, the boundary-measurement graphs, the left twist or the Grassmannian; Propositions 4.2, 5.1 and 5.12 and Lemma 5.13 of [1] and the twist of [4] enter only through (H_1)–(H_4). Also not formalized are the strand-diagram proof of Proposition 15 (only the window induction of §4.2 is), the component reduction (Lemma 5), the homeomorphism Φ_f of Corollary 2, the untwisted recovery (Remark 13 and Corollary 32), and the hyperbolic-volume reading of the introduction. Nor is the real-analytic strengthening of Corollary 33.

Artifact scope, and what remains for the reader. The historical archive 10.5281/zenodo.21664478 uses the same Lean and Mathlib versions, but its `LooplessBAP` additionally requires $\sigma(i) \neq i$ for every label. Its combinatorial statements therefore have fixed-point-free scope, and it contains none of the statements added after it: the full-pair row difference, the analytic statements, the conditional theorem and the interface. Its SHA-256 is

`3fcfbaad43f5196c5304bdb57f407ea394830cd476b7780595882a3ef9a45529`

and its DOI does not identify the newer source. The source supplement above admits coloops; it is the snapshot of the formal source that contains every statement named in this appendix. Its retained cross-validation report compares the definitions with an exact-integer mirror on 5,912 loopless cells with $2 \leq n \leq 7$, including 3,737 with coloops. This is finite interface corroboration, not

a proof of the geometric imports or of the composed theorem. Given the formal statements, what remains for the reader is a reading of definitions and hypotheses, not of proofs. The encodings — the positions $5s_p+4$ and $5p+1$, the necklace recursion and `Conj43.pairExponent` — are proved equal to Galashin's definitions as Lean transcribes them, so what the reader checks is that the transcriptions listed in the interface paragraph say what [1] defines. The half-open reading (Lemma 9), on which the whole gradient identity rests, is proved three times: in the encoding, inside the formal proofs of `Conj43.rowdiff` and `Conj43.rowdiff_full_pairs`; on Galashin's refined boundary, by the interface statements; and by hand in §3. The hypotheses: that (H_1)–(H_4) say what the cited results say, as listed above. And the statements that are not formalized, whose proofs are the written ones.

Finite checks. These checks are cited in the main text in Remarks 11 and 34. None is a premise.

The row-difference identity. An independent pure-integer reimplementation, sharing no code with the earlier checks or with their half-open-arc routine, reconstructed the necklace rows and all three pair classes directly from the source definitions. It checked all 458,733 unordered-pair exponent columns (169,490 directed-crossing columns) on all 17,007 fixed-point-free cells with $3 \le n \le 8$ and found zero row-difference mismatches. Two later reimplementations, sharing no code with that one or with each other, reproduced both counts and verified $P_f D M_N = B$ on all 3,737 loopless cells with at least one coloop and $n \le 7$, again with zero mismatches. An exhaustive run over those same cells, and one of those two reimplementations independently, confirm the step's four assertions — a coloop lies in every necklace row, is in no crossing, admits no $\mathrm{Mis}^{\uparrow\downarrow}$ partner in either orientation, and has identically zero M_N columns — with no exceptions. Separately, the Lean definitions of the window, rank, crossing edges, necklace, and the matrices $B, M_N, P_f D M_N$ were bit-identical to an exact rational computation on all twelve fixed-point-free cells with $n \le 4$; those cells include fifteen crossing edges with the coincidence $s_p = q$. These counts are corroboration, not premises.

Measurement controls. On top-cell measurements, the historical archive includes a two-route check against Galashin's independent reconstruction in [1, §9]. Starting from the Fourier representative of $\mathrm{Meas}_{k,n}(\theta)$, it reads R_f from the left twist and inverts G_f from a regular-polygon initial point. Separately, it extracts the bridge ratios of [1, eq. (9.4)] from raw Plücker coordinates and the square cross-ratios of [1, eq. (9.3)] from left-twist Plücker coordinates, then applies [1, Lem. 9.3] and the remainder of the proof of [1, Thm. 4.4]. On eighteen deterministic points with $3 \le k \le n-2$ and $n \le 8$, every Plücker-to-ratio identity holds exactly and the two reconstructed angle tuples agree to at most 6.0×10^{-15}; a one-percent corrupted bridge ratio is detected. This is a finite numerical cross-check, not a premise or a proof for general f.

The source supplement accompanying this revision also contains a check on all 5,885 non-top loopless cells with $2 \le n \le 7$, at two distinct-parameter configurations per cell. It evaluates Galashin's curve, solves the left-twist equations, and takes necklace minors, without constructing the exponent matrix. The resulting $R_f(\mathrm{Meas}_f(\theta))$ agrees with $B\ell(\theta)$ to at most 2.4×10^{-13} across 11,770 points, including configurations with coloops, reordered noncrossing labels, and obtuse crossing angles. Of these points, 10,546 have crossing edges: at each, reversing one wrapped incidence column is detected, and omitting its affine lift violates admissibility. The other 1,224 points have no crossing edges and supply no orientation test. The declared numerical tolerance is 10^{-8}. This finite convention check does not test collisions or prove the identity beyond its sampled parameters.

Two differences between the text and the formal source. The termination measure of the window induction in §4.2 is $\mu(w) = \binom{n}{2} - \mathrm{inv}(w)$ (Lemma 19). The formal proof (`Conj43.WindowAscent`) instead uses the position-weighted lift sum $\texttt{Window.measure}(w) = \sum_{i=0}^{n-1} i\, w_i$. An (a,b)-swap with $a < b$ and $w_a < w_b$ lowers it by $(b-a)(w_b - w_a) > 0$, without counting intermediate

inversions; the induction is otherwise the same. Likewise the formal proof represents a winding-one cycle by its sorted vertex set, with edges `consecutiveEdges` (defined in module `Conj43.Defs` and used in `Conj43.SeedCover`). Separately, in the frozen statement file the docstring of `Conj43.gal512_measurement_homeomorphism` describes its hypothesis `hMeas` as "Galashin Proposition 5.1 and Theorem 6.9". The mention of Theorem 6.9 is historical and is not used: (H_1) rests on Proposition 5.1 and Lemma 5.13, as stated above, and the frozen file is left unchanged, as the fixed reference for the statements. The supplement contains the retained cross-validation report but not the scripts that produced it, which read an external exact enumeration of cells; it contains its own standalone checks.

B A worked example

Take the loopless window

$$f = (3, 5, 6, 7, 9), \qquad \bar{f} = (3, 5, 1, 2, 4), \qquad (s_1, s_2, s_3, s_4, s_5) = (3, 4, 1, 5, 2).$$

Here $n = 5$ and $k = 3$. The displayed values were evaluated by the formal definitions of Appendix A. The strand S_p runs from $b^+_{s_p}$ to b^-_p, with each exit b^-_j immediately before b^+_j in clockwise boundary order (Figure 22). The necklace is

$$(I_1, I_2, I_3, I_4, I_5) = (\{1, 2, 4\}, \{2, 3, 4\}, \{3, 4, 5\}, \{1, 4, 5\}, \{1, 2, 5\}).$$

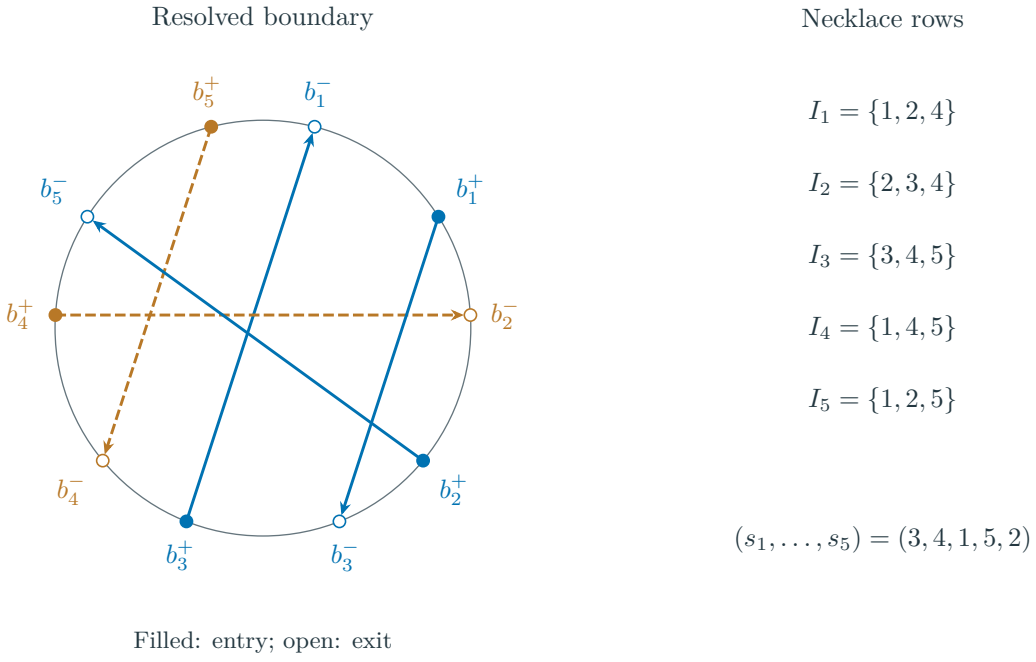


Figure 22: The resolved boundary and necklace of $f = (3, 5, 6, 7, 9)$. Chords point from entry to exit; their intersections depict endpoint interlacing. Filled entries and open exits remain distinct when they share an integer label. The necklace rows on the right record membership on the half-open arcs $(s_p, p]$.

Pair classes. The seven directed crossing edges are

$$E_f = \{(1, 2), (2, 3), (2, 4), (3, 5), (4, 5), (5, 1), (5, 2)\}.$$

Exactly $(2, 4)$, $(4, 5)$ and $(5, 2)$ have $s_p = q$. The remaining unordered pairs are

$$\mathrm{Mis}_f^{\uparrow\downarrow} = \{\{1, 3\}\}, \qquad \mathrm{Mis}_f^{\downarrow\uparrow} = \{\{1, 4\}\}, \qquad \text{alignment: } \{3, 4\}.$$

No necklace row contains both 1 and 3, and every row contains 1 or 4. Thus both misalignment columns vanish; the alignment is absent from the exponent formula. Figure 9 shows one representative of each class.

Matrices. Number the crossing columns in the displayed order of E_f, from 1 to 7. Figure 23 gives M_N, $P_f D M_N$ and B in that order. The three other unordered-pair columns are zero in all three matrices and are omitted from the display. For example, column 2, the edge $2 \to 3$, reads

$$m = (-1, 0, 0, 0, -1)^{\mathsf{T}}, \qquad P_f D m = (0, -1, 1, 0, 0)^{\mathsf{T}} = e_3 - e_2.$$

Columns: directed crossing edges

	$1 \to 2$	$2 \to 3$	$2 \to 4$	$3 \to 5$	$4 \to 5$	$5 \to 1$	$5 \to 2$
	1	2	3	4	5	6	7
1	0	−1	0	0	−1	0	0
2	0	0	0	−1	−1	0	0
3	0	0	0	0	0	−1	−1
4	−1	0	0	0	0	0	−1
5	0	−1	−1	0	0	0	0

M_N

	1	2	3	4	5	6	7
1	−1	0	0	0	0	1	0
2	1	−1	−1	0	0	0	1
3	0	1	0	−1	0	0	0
4	0	0	1	0	−1	0	0
5	0	0	0	1	1	−1	−1

$P_f D M_N$

	1	2	3	4	5	6	7
1	−1	0	0	0	0	1	0
2	1	−1	−1	0	0	0	1
3	0	1	0	−1	0	0	0
4	0	0	1	0	−1	0	0
5	0	0	0	1	1	−1	−1

B

Rows: r for M_N; x for $P_f D M_N$ and B

Figure 23: The seven crossing columns for $f = (3, 5, 6, 7, 9)$, in the order shown above the matrices. Necklace positions index the rows of M_N; exit labels index the rows of $P_f D M_N$ and B. The two lower matrices agree entry by entry. All three omitted noncrossing columns are zero.

Figure 24 follows two columns of Figure 23 through D and P_f.

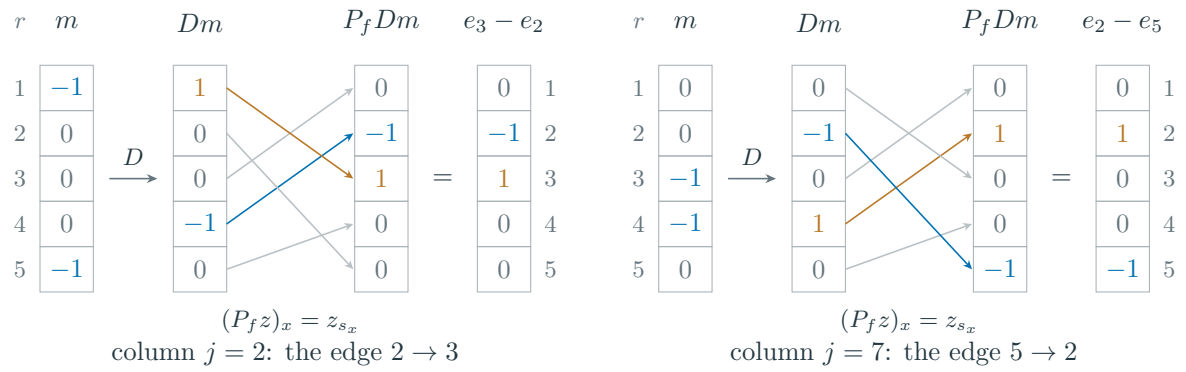


Figure 24: Columns $j = 2$ (the edge $2 \to 3$) and $j = 7$ (the edge $5 \to 2$) of Figure 23, followed from the column m of M_N through D and P_f to the column of B, for $\bar{f} = (3, 5, 1, 2, 4)$ and $s = (3, 4, 1, 5, 2)$. Row x of $P_f Dm$ is row s_x of Dm; blue marks -1 and ochre $+1$. Evaluated by the formal definitions of Appendix A.

Half-open columns. For a column indexed by $p \to q$, the entry of M_N on row r is $-\mathbf{1}_{r\in(s_p,s_q]}$. In column order, the seven arcs are

$$(3,4],\ (4,1],\ (4,5],\ (1,2],\ (5,2],\ (2,3],\ (2,4],$$

and their row sets are, respectively,

$$\{4\},\ \{1,5\},\ \{5\},\ \{2\},\ \{1,2\},\ \{3\},\ \{3,4\}.$$

In particular, the wrapped arc $(4,1]$ contains rows 5 and 1, while its initial row 4 is excluded. Figure 25 draws the seven arcs.

j	$p \to q$	$(s_p, s_q]$	Rows in the arc					
1	$1 \to 2$	$(3,4]$	$\{4\}$	1	2	3	**4**	5
2	$2 \to 3$	$(4,1]$	$\{1,5\}$	**1**	2	3	4	**5**
3	$2 \to 4$	$(4,5]$	$\{5\}$	1	2	3	4	**5**
4	$3 \to 5$	$(1,2]$	$\{2\}$	1	**2**	3	4	5
5	$4 \to 5$	$(5,2]$	$\{1,2\}$	**1**	**2**	3	4	5
6	$5 \to 1$	$(2,3]$	$\{3\}$	1	2	**3**	4	5
7	$5 \to 2$	$(2,4]$	$\{3,4\}$	1	2	**3**	**4**	5

Each column is minus the indicator of its arc.

Figure 25: Every crossing column as a half-open arc of necklace rows. Double solid circles mark rows in the arc; dashed circles mark the other rows. The exponent is -1 on the marked rows and 0 elsewhere. The cyclic order makes $(4,1] = \{5,1\}$ and $(5,2] = \{1,2\}$.

Partitions. The two winding-one partitions in Figure 12 are

$$\{(1,2,3,5),(2,4,5)\}, \qquad \{(1,2,4,5),(2,3,5)\}.$$

Both have two cycles, one containing the wrapped edge $(5,1)$ and the other $(5,2)$. Each cycle advances a total of 5 around the labels, so its winding is one. Either choice therefore gives the cycle decomposition used in the Hessian argument.